

The Sentiment Channel of Fiscal Policy[‡]

Marco Bellifemine

LSE

Adrien Couturier

UCL

Jacopo Tozzo

Bank of Italy

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Abstract

How does government spending stimulate the economy? We uncover a new transmission channel—the sentiment channel—by showing that a rise in government spending makes firms overoptimistic about their future demand, thereby stimulating investment and output. We assemble a novel dataset linking microdata on Italian firms’ sales, sales forecasts, and public procurement contracts. Using a natural experiment that shifted public spending across municipalities, we find that an increase in government demand makes firms systematically overoptimistic about their future sales. This overoptimism is pervasive, as firms also raise their expectations about export sales. To interpret these findings, we develop a theory of expectations in which shocks to total sales make a firm overoptimistic about both its public and private sales. We embed this model of expectations in a heterogeneous-firm New Keynesian model disciplined with our causal estimates. Government spending boosts firms’ optimism which prompts them to invest, crowding-in investment in general equilibrium, thereby producing a 75% increase in the government spending multiplier. The sentiment channel is state-dependent: tighter financing conditions limit firms’ ability to act on improved expectations and weaken its contribution to the fiscal multiplier.

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Bellifemine: London School of Economics. Email: m.bellifemine@lse.ac.uk.

Couturier: University College London. Email: adrien.couturier@ucl.ac.uk.

Tozzo: Bank of Italy. Email: jacopo.tozzo@bancaditalia.it. The views expressed do not represent those of the Bank of Italy or the Eurosystem.

1 Introduction

How does government spending stimulate economic activity in a recession? There is a long-standing view, common in the press and in policy discussions, that fiscal policy operates not only through direct demand effects, but also by boosting business optimism and hence investment. This idea dates back to Keynes, who, at the height of the Great Depression, famously argued that government expenditures can move beliefs and sentiment just as much as fundamentals: *“Large schemes of work being undertaken [by the government] would give an immediate fillip to the industry of the country”* (Keynes and Henderson, 1929).

Yet, incorporating this channel in standard macroeconomic models has proven challenging. As Mankiw (2009) and Cochrane (2009) noted after the Global Financial Crisis, the lack of direct empirical evidence on how government spending affects private sector beliefs has made it difficult to discipline models that feature such mechanisms. The result has been, as Mankiw puts it, that *“until we figure it out, it is best to be suspicious of any policy whose benefits are supposed to work through the amorphous channel of confidence”*. Hence, the literature has largely resorted to models with rational expectations to understand the effects of fiscal stimulus. In this context, it is not possible to define a notion of optimism or assess the quantitative relevance of such a channel for fiscal policy.

This paper makes progress by formalizing empirically, theoretically and quantitatively what we call the sentiment channel of fiscal policy. Using firm-level data, we show that receiving a public procurement contract boosts firms’ optimism about their future total sales. Incorporating this sentiment channel in a heterogeneous-firm New Keynesian model has important policy implications. The firm-level optimism induced by an aggregate government spending shock overturns the crowding-out of investment predicted by standard models and leads to larger and empirically plausible multipliers.

First, to provide causal evidence for the sentiment channel, we assemble a novel dataset linking firms’ sales forecasts with their realizations, and match it to the universe of public procurement contracts in Italy. Leveraging a natural experiment that induced plausibly exogenous variation in municipal spending, we find that positive government demand shocks make firms systematically overoptimistic: their sales forecasts rise by more than their actual sales. Importantly, this optimism is not limited to public revenues and spills over to other domains like private (export) sales.

To rationalize these findings, we develop a new theory of expectation formation in which a positive sales shock generates systematic optimism, distorting beliefs across domains of demand. This bias is particularly relevant for investment decisions, which hinge on expectations of future demand. In this context, financial frictions play a central role,

as they affect the ability of firms to make intertemporal choices. We therefore incorporate our model of expectation formation into a standard q -theory of investment with financial frictions, and show analytically that financial constraints are a key mechanism shaping the transmission of optimism to investment: investment is increasing in optimism about future demand, but the effect is dampened for financially constrained firms.

Finally, we embed this behavioral theory of investment in a heterogeneous-firm New Keynesian model to quantify the sentiment channel of fiscal policy and run policy counterfactuals. We calibrate the behavioral bias to our empirical estimates and find that the sentiment channel is quantitatively large: the optimism induced by government spending leads to a crowding-in of firm investment, raising the five-year cumulative government multiplier by 75% relative to a benchmark with no sentiment, from 0.72 to 1.26. Therefore our channel allows to reach empirically plausible multipliers and offers an alternative mechanism to Heterogeneous Agents New Keynesian models.

The sentiment channel is state-dependent, owing to the central role of financial frictions highlighted in our theory. When financing constraints tighten, firms are less able to translate improved expectations into investment, reducing the contribution of sentiment to the fiscal multiplier. These findings suggest that incorporating empirically disciplined models of expectations in standard theories has important consequences for policy analysis.

Empirics Estimating a sentiment channel of fiscal policy is challenging, as it requires data on the expectations and realizations of firms' outcomes, together with exogenous variation in government spending at the firm level. Our first contribution consists in assembling a novel dataset combining the quasi-universe of public procurement contracts in Italy with a large survey of firms' expectations and with administrative data on income statements.¹ This allows us to track firm-specific procurement revenues and link them to both forward-looking beliefs and realized outcomes.

Our second empirical contribution is a new source of identification: we exploit a 2015 reform to Italy's municipal fiscal rules that modified the formula determining municipal surplus targets. Before the reform, targets were based on average municipal expenses over 2009-2011. The reform revised the base years to 2010-2012, effectively loosening targets for municipalities that had high expenditures in 2009 relative to 2012. Since the details of the reform were not publicly debated prior to its announcement, the change in base years was plausibly unanticipated by both municipalities and firms, generating exogenous variation in surplus goals across municipalities. We employ our contract-level procurement data to

¹Public procurements are sizable, amounting to 36% of government consumption.

construct a measure of firms' exposure to different municipalities. We then rely on a shift-share event study combining firm exposures with the municipal variation in fiscal targets.

This enables us to provide the first causal estimate of how government spending shocks affect the firms' expectations. To accomplish this, we use a natural experiment to study how firms' beliefs respond to public sales shocks, thus complementing existing work that has primarily relied on unconditional correlations (Barrero, 2022, Ma et al., 2024). These methods pool together different sources of shocks, and are hence silent about the underlying drivers of behavioral biases. This matters, as recent evidence shows that expectations respond differently to different types of shocks (Born et al., 2023). Our findings thus help reconcile the mixed evidence on over- and under-reaction in the literature.²

A rise in government procurement spending makes firms overly optimistic. A procurement shock that increases firm revenues by 10% leads sales forecasts to exceed realizations by 6.5% in the subsequent year (while forecast errors are zero on average). This optimism extends beyond public revenues, as firms also raise their expectations about private sales, which we proxy with export sales. Together, these findings provide new causal evidence for a sentiment channel of fiscal policy, in which government spending affects economic activity through belief distortions.

A theory of cross-domain extrapolation Our empirical results show not only that firms overreact to procurement shocks, but also that they extrapolate these shocks into domains that should be unaffected, such as export sales. Standard models of overreaction, like diagnostic expectations, can account for the first finding but not for the latter. Models of imperfect information, on the other hand, are at odds with the fact that firms can precisely observe their private and public revenues. In order to rationalize our empirical findings we therefore build a model of expectation formation in which recent news about total sales shapes beliefs about its private and public sub-components. Our mechanism is a form of cross-domain extrapolation that combines two central ideas in the behavioral literature: *representativeness* and *salience* (Tversky and Kahneman, 1983, Bordalo et al., 2023b). *Representativeness* implies that managers overweight the probability of events that have become more likely in light of recent news. *Salience* governs which variable captures their attention when applying this heuristic. In particular, we posit that total sales are salient, meaning that they stand out in the managers' minds more than its public and private sub-components.

In this setting, any positive surprise to total sales—whether driven by public or by

²An alternative strategy in the literature is to rely on survey experiments (Coibion et al., 2022). To our knowledge, this approach has not been used to study the response of firms' expectations to government spending.

private revenues—cues memories of both high private *and* public sales. *Representativeness* then leads managers to overweight these states when forming expectations, making them overly optimistic about both components. The effect is symmetric for negative news. In this way, shocks to one domain (e.g. public sales) spill over into beliefs about the other domain (e.g. private sales).

Building on [Maxted \(2023\)](#) continuous time formulation of diagnostic expectations, we prove that this model of expectation formation can be conveniently represented in recursive form with a variable, *sentiment*, driving latent optimism. This variable summarizes recent sales news and is defined as an exponentially weighted memory of past sales shocks. Thus, good sales news originating from private and/or public demand endogenously raises sentiment, tilting beliefs about future public, private, and total sales upward, consistent with our empirical results.

A q -theory of investment with sentiment Expectations matter for real activity only insofar as they affect firms’ choices. Among these, investment is the natural decision for which beliefs are important, as it is inherently forward looking. At the same time, firm investment is constrained by financing capacity, making financial frictions central to how sentiment translates into real activity. To study how sentiment shapes firms’ investment choices, we therefore embed our theory of expectation formation in a canonical q -theory ([Hayashi, 1982](#), [Abel and Eberly, 1994](#)) with financial frictions ([Fazzari et al., 1988](#)). This setting yields an analytical mapping between firms’ potentially biased expectations about future demand and investment decisions.

We analytically decompose the optimal investment choice in the frictionless, rational expectations benchmark augmented by two wedges: one reflecting financial frictions and another capturing belief distortions about future profitability driven by sentiment.

A key prediction of our theory is that the relationship between investment and sentiment is weaker for financially constrained firms, as financing constraints limit firms’ ability to make intertemporal decisions. Because sentiment enters our framework only through forecast errors about future profits, we can test this prediction directly in our survey data. In line with the theory, financial constraints dampen the sensitivity of investment to expectations in the data.

General equilibrium model Finally, to assess the macroeconomic relevance of the sentiment channel, we embed our behavioral q -theory of investment in a general equilibrium heterogeneous-firm New Keynesian model calibrated to our microdata and reduced-form estimates. We cast the economy as a mean field game featuring a “rationality wedge”,

reflecting the gap between firms’ perceived and actual economic dynamics. Our methodology is portable and can be applied to a wide range of heterogeneous-agent models exhibiting deviations from rational expectations, see for example [Bellifemine et al. \(2025\)](#). This allows us to solve for the general equilibrium response of the economy to a government spending shock. Consistent with our empirical setting, we model government spending shocks as correlated shocks to individual firms’ public procurement.

We calibrate the expectation formation process to our reduced-form empirical estimates and find that the sentiment channel is quantitatively important. The optimism induced by government spending more than offsets the negative impact of higher interest rates on investment, overturning the standard crowding-out result: investment is crowded-in in general equilibrium. This is consistent with the empirical literature that finds a crowding-in effect of government expenses on private investment ([Edelberg et al., 1999](#), [Auerbach and Gorodnichenko, 2012](#)). This effect produces a 75% increase in the five-year cumulative multiplier, raising it to 1.26 from a baseline of 0.72.

The strength of the sentiment channel depends on firms’ financing conditions. When financing constraints tighten, the sensitivity of firm investment to expectations is muted. In our illustrative experiment, a severe tightening of financing conditions reduces the sentiment channel’s contribution to the five-year cumulative fiscal multiplier by 25% relative to normal financing conditions.

Related literature “Animal spirits” have long been invoked as drivers of aggregate fluctuations ([Keynes, 1936](#), [Akerlof and Shiller, 2009](#)). A large literature documents that beliefs overreact ([Afrouzi et al., 2023](#)), and in particular to domain-specific news in asset prices ([Bordalo et al., 2019, 2024](#)), in professional forecasts ([Bordalo et al., 2020](#), [Afrouzi et al., 2023](#)), and in managers’ forecasts of their own firms ([Gennaioli et al., 2016](#), [Bordalo et al., 2026a](#)). Managerial forecasts, which are the focus of this paper, have been documented to display persistent biases ([Barrero, 2022](#), [Ma et al., 2024](#), [Born et al., 2023](#)). Evidence that experiences in one domain affect beliefs in another comes mostly from households ([Binder and Makridis, 2022](#), [Bordalo et al., 2023a](#), [Taubinsky et al., 2026](#), [Cenzon, 2025](#)). [Bordalo et al. \(2025, 2026b\)](#) develop and test models in which contextual cues shape which experiences are recalled and used to imagine future outcomes, allowing experiences from other domains to influence beliefs through perceived similarity. We show that such contagion also operates for firms and, importantly, can be caused by a policy shock: government procurement makes firms overoptimistic about total sales and, through it, about unrelated private sales. Our theory of expectations builds on the representativeness and salience heuristics ([Tversky and Kahneman, 1983](#), [Bordalo et al., 2023a](#)), which diagnostic expecta-

tions turn into a model of overreaction to news about the forecast variable itself (Bordalo et al., 2018). We leverage the continuous-time formulation of diagnostic expectations in Maxted (2023) by generalising it to a multivariate setting, allowing for a tractable model of cross-domain extrapolation that can be readily embedded in dynamic macroeconomic models.

This paper also relates to a new and growing strand of literature that emphasizes the role of expectations in driving aggregate demand fluctuations. Angeletos and Lian (2022) develop a theory in which household expectations depart from full information and introduce a “confidence multiplier”; Bordalo et al. (2026b) instead derive one from memory, where one agent’s spending cues optimism in others. Bianchi et al. (2024) show that diagnostic expectations on endogenous variables can generate boom–bust dynamics through a feedback between beliefs and realized outcomes, without any contamination across variables. More directly related to our fiscal application, L’Huillier et al. (2024) embed diagnostic expectations in an estimated New Keynesian model and show that they amplify nominal frictions and strengthen the effectiveness of fiscal policy. Relative to all four, we provide firm-level causal evidence of a sentiment-based multiplier following a fiscal shock and a general-equilibrium quantification of its implications. Methodologically, our quantitative model is close to Bordalo et al. (2026a), who incorporate diagnostic expectations in a heterogeneous-firm model to study boom–bust credit demand cycles; we instead propose deviations from rational expectations as a channel through which fiscal policy affects aggregate demand via aggregate investment.

Finally, we add to a large body of work studying the transmission of fiscal policy to aggregate economic activity, which mostly focuses on the response of real outcomes, more recently using granular procurement data (di Giovanni et al., 2025, Cox et al., 2024); we instead study a channel operating through the response of firms’ beliefs. Our quantitative framework relates to heterogeneous-agent models of fiscal transmission, which mostly study household heterogeneity (Auclert et al., 2025); we focus on firm heterogeneity, underexplored by comparison (Winberry et al., 2025), and propose a new transmission channel operating through biased expectations, consistent with evidence on the crowding-in of private investment (Ramey and Shapiro, 1998, Edelberg et al., 1999, Auerbach and Gorodnichenko, 2012, McClure and Yding, 2025a). This differs from the news-shocks literature, where fluctuations are driven by news about future fundamentals (Beaudry and Portier, 2004, 2007, Jaimovich and Rebelo, 2009, Lorenzoni, 2009, Chahrour and Jurado, 2018, Lagerborg et al., 2023): we instead study endogenous expectation fluctuations caused by a policy shock. In this sense, we are closer to Bachmann and Sims (2012), who use a structural VAR to study how survey measures of confidence respond to govern-

ment spending shocks at the aggregate level; we instead measure the response of firms' expectations directly and trace its consequences in general equilibrium. Our investment model builds on q -theory with financial frictions (Tobin, 1969, Hayashi, 1982, Abel and Eberly, 1994, Fazzari et al., 1988), extended to allow biased expectations, connecting to heterogeneous-firm models of idiosyncratic shocks (Khan and Thomas, 2008, Winberry, 2021, Ottonello and Winberry, 2020, Koby and Wolf, 2020) and to evidence that managers' expectations predict investment beyond Tobin's q (Gennaioli et al., 2016).

Outline The rest of the paper is organized as follows. Section 2 describes the data and presents our empirical findings. Section 3 develops our theory of cross-domain extrapolation and its recursive formulation. In Section 4 we embed this theory in a model of firm investment and analyze the role of sentiment for investment. Section 5 describes our calibration strategy, quantifies the sentiment channel of fiscal policy, and examines its state-dependence. Section 6 concludes.

2 Empirical Evidence

We begin by studying how government spending affects firms' expectations. To this end, we assemble a novel dataset that links the quasi-universe of Italian public procurement contracts to survey data on firms' expectations and administrative balance sheet and income statement records. Our identification strategy exploits a reform to municipal budget rules that generates plausibly exogenous variation in public spending across Italian municipalities. We leverage this variation to estimate the causal effects of government spending on firms' sales forecast errors.

2.1 Identifying a sentiment channel of fiscal policy

To motivate our empirical strategy, we first describe the ideal specification for testing a sentiment channel of fiscal policy: namely regressing a firm's forecast error on shocks to its public sales. Forecast errors are a natural outcome variable, as they provide a direct way to test for deviations from full information rational expectations (FIRE). Under the null hypothesis of rationality, forecast errors should be orthogonal to any variable in the forecaster's information set; otherwise, it could be possible to improve the forecast with available information. Under the assumption that firms observe their public sales shock and understand the stochastic process for public sales, being able to forecast a firm's forecast error with past sales shocks therefore constitutes a rejection of the null.

Importantly, testing for deviations from FIRE does not itself require the public sales shock to be exogenous. This point is illustrated by recent work documenting systematic biases in managerial expectations from *unconditional* forecast error correlations (Barrero, 2022, Ma et al., 2024). Beyond establishing the presence of biases; a causal design might be more informative about the underlying form of deviation from FIRE. Indeed, recent evidence shows that expectations respond differently to different types of shocks (Born et al., 2023). By contrast, to establish that the deviations from FIRE are caused by fiscal policy—that is, to identify an expectation channel of fiscal policy—requires exogenous shocks to government spending.³

Taken together, these considerations underscore the challenge of identifying a sentiment channel of fiscal policy: such an endeavor requires firm-level data that jointly span (i) expectations, (ii) realizations, (iii) public sales, and (iv) exogenous fiscal shocks. We now describe how we assemble a dataset that meets these requirements.

2.2 Data sources

We construct our dataset by combining four data sources. First, we obtain firm-level sales forecasts from the Bank of Italy’s Survey of Industrial and Service Firms (INVIND). Second, we link forecasts to realized outcomes from administrative balance sheet and income statement records from the Company Accounts Data System (CADS). Third, we merge this firm-level panel with contract-level procurement records from the Italian Anti-Corruption Authority (ANAC). This combined dataset allows us to track firms’ expected and realized sales and investment, as well as their municipality-specific exposure to procurement revenues. Finally, to measure the public spending shock induced by the reform, we use fiscal accounts for all Italian municipalities from Orbis’ proprietary database Aida PA. We now describe each source and the construction of the merged dataset in detail.

Survey of Industrial and Service Firms Our data on firm-level sales expectations come from the Bank of Italy’s Survey of Industrial and Service Firms (INVIND), an annual survey conducted between February and May of each year since 1972. From 1995 onward, respondents have reported numerical forecasts for end-of-year sales, thus providing us with nearly three decades of expectations data. Relative to existing surveys eliciting quantitative forecasts, a key advantage of INVIND is its large sample size and strong panel dimen-

³A common approach in the literature to identify the drivers of behavioral biases is to use survey experiments. To our knowledge, this has not been applied in the context of government spending and firms. Moreover, survey experiments may suffer from limited external validity, as responses are elicited in low-stakes settings.

sion. In recent years, about 5,000 firms participate in each wave and respondents are observed on average for 11 years. The survey targets firms with at least 20 employees headquartered in Italy operating in industrial and non-financial private service sectors. For the construction sector, the reference universe comprises firms with at least 10 employees. To ensure representativeness, the sample design is stratified by sector of economic activity, geographical area, and firm size. Response rates are high, averaging around 70 percent in recent years.

Company Accounts Data System To compare firms' expectations with realized outcomes, we merge our INVIND data with the Company Accounts Data System (CADS), a proprietary database maintained by Cerved Group S.p.A. CADS aggregates administrative balance-sheet and income-statement filings from all Italian limited liability companies. Italian law mandates that firms report these data annually to local Chambers of Commerce. From CADS, we extract firms' annual revenues, total assets, investment, and a proprietary credit score measure based on Altman's Z-score. We also retrieve a number of firm characteristics that will be useful for our analysis, including the NACE classification of economic sector, location, and year of incorporation.

Public procurement contracts To link firms' forecasts to their revenues from the public sector, we collect contract-level procurement data from the Italian anti-corruption authority (ANAC). These data cover the universe of public procurement contracts exceeding €40,000 awarded between 2012 and 2024. Crucially, each contract record includes identifiers for both the awarding public body and the winning firm, as well as the contract value. This allows us to construct firm-municipality procurement histories and merge them with our matched INVIND-CADS panel.

Municipal income statements Finally, to construct a measure of the municipal public spending shock induced by the reform of interest, we obtain municipal income statements from Aida PA. We extract annual current expenditures and tax revenues for all Italian municipalities from 2009 to 2017, which form the basis for our shock measure.

Summary statistics In our empirical analysis we employ two different samples. To study the effect of government spending on realized outcomes, we construct a large panel by merging CADS and ANAC, covering about 25,000 firms annually. To study forecast errors, we merge our linked CADS-INVIND data with ANAC. This restricts the sample to approximately 800 firms per year but provides the crucial link between expectations, out-

comes, and public sales shocks. Detailed summary statistics for both samples are reported in Table A.1 in Section A.

2.3 Descriptive statistics

Before turning to our main empirical results, we present descriptive evidence on firms' expectations and on the structure of public procurement.

Procurement data We first describe our contract-level procurement data, which form the basis for our measure of firms' exposure to municipal spending. Public procurement constitutes a large share of government activity, representing 36.3% of government consumption over the period 2012-2022. Of particular relevance for our empirical analysis, the 7,896 Italian municipalities represent an important source of procurement. They account for 21.7% of awarded contracts (by number) and 8.3% (by value).

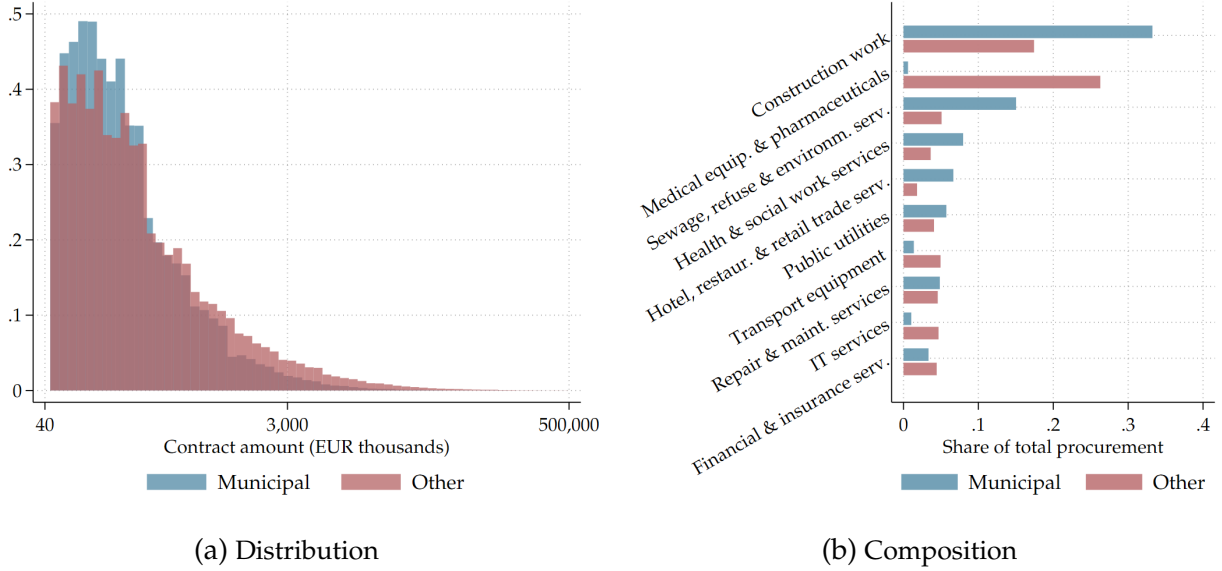
These aggregate patterns are mirrored at the firm level, where public procurement represents a substantial revenue source. Among firms with public contracts, public procurements account for 14.5% of total sales on average. Conditional on contracting with municipalities, municipal procurement alone represents 12.1% of firm sales.⁴ These magnitudes underscore why municipal procurement shocks are a meaningful source of demand variation for firms.

To further illustrate the salience of procurement as a revenue source, Figure 1a shows the distribution of contract values awarded by municipalities and other public bodies. The median contract is sizable—€123,000 for municipalities and €146,000 for other public bodies. Average contract values are even higher, driven by the presence of a few “mega-projects”. These are more common in non-municipal procurement, where the mean reaches €893,000 (compared to €396,000 for municipalities). Such contracts mainly reflect large infrastructure projects and centralized purchases by regional and national authorities, which are largely absent from municipal procurement.

Turning to the sectoral composition of public procurement spending, Figure 1b summarizes the breakdown of procurement contracts in our dataset, distinguishing between those awarded by municipal and non-municipal entities. Compared to other public bodies, municipal procurement is more concentrated in *Construction works*—which includes road paving and building repairs—and *Sewage, refuse, cleaning and environmental services*.

⁴Figure A.1 in Section A.1 shows the distribution of the share of total public procurement in firms' revenues, as well as the share specifically attributable to municipal contracts, conditional on receiving at least one procurement contract.

Figure 1: Municipal vs Non-Municipal Procurements



Note: Panel (a) shows the distribution of contract sizes for municipal and non-municipal contracts. We remove contracts below €40,000, since reporting is non-mandatory below this threshold. Panel (b) displays the distribution of procurement contracts across ten categories, distinguishing between those issued by municipalities and those issued by other public entities. For visibility, we only keep the ten most important sectors.

These categories account for 33% and 15% of all municipal procurement expenses, respectively, while they only represent 17% and 5% of contracts in the non-municipal sample. On the other hand, municipal procurement is under-represented in *Medical equipment and pharmaceutical products*, which is virtually absent from municipal procurement but constitutes nearly 26% of contracts in the rest of the sample. These patterns reflect the division of government responsibilities: municipalities focus on local infrastructure and sanitation while regional bodies handle healthcare provision. Outside these categories, however, municipal procurement is broadly similar to non-municipal procurement, with roughly half of spending distributed across other sectors such as *Hotel, restaurant and retail trade services*, *Repair & maintenance services*, and *Financial & insurance services*.

Finally, to inform the rest of our empirical analysis, it is useful to discuss the timing of contract execution and payments. Italian accounting standards require firms to record revenues on an accrual basis—that is, when goods or services are delivered rather than when payments are received. Importantly, municipal contracts typically begin shortly after award: the median (mean) lag between award and start dates is 11 (41) days, and between award and completion is 337 (460) days.⁵ Payments, however, follow with substantial delay: the median (mean) lag from contract start to first payment is 152 (196) days,

⁵For non-municipal contracts, the corresponding figures are 28 (60) and 265 (358), respectively.

and to the last payment 675 (856) days.⁶

Expectations data We first show that firms’ expectations are predictive of their decisions. Specifically, panel (a) in Figure 2 examines the relationship between firms’ expectations and their actions by plotting the investment rate (investment over capital) against expected future revenue growth. Crucially, we residualize both axes by *realized* future revenue growth and firm fixed effects. Hence, this specification isolates the component of expectations orthogonal to realized outcomes, capturing how optimism or pessimism relative to realizations correlates with actions. The relationship is strongly positive: a 10 percentage-point increase in expected revenue growth is associated with a 0.18 percentage-point higher investment rate, controlling for firm fixed effects and keeping realized revenue growth fixed. This effect is economically meaningful. Given that the average investment rate in our sample is roughly 6%, a one standard-deviation increase in expected growth corresponds to an investment increase of about 3% relative to the mean. While purely correlational, panel (a) of Figure 2 may be interpreted as providing motivating evidence for a central premise of our analysis: that firms’ investment choices are closely linked to their expectations.

Because the responses are confidential and released only in aggregate form, respondents do not face incentives for strategic misreporting which is typical in other settings, such as public earnings calls. Typically, CEOs or CFOs would answer these surveys and report their internal forecast. This explains not only the accuracy of the forecast but also the correlation it exhibits with actual firm decisions. While this unconditional accuracy lends credibility to the data, it does not imply that expectations are rational.

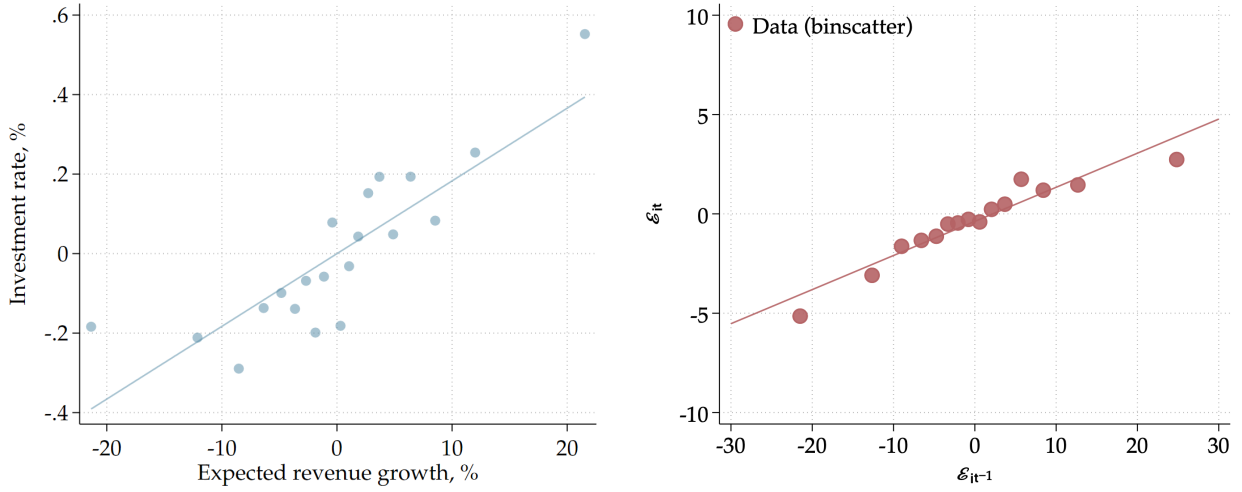
To make progress on this front, we construct our main variable of interest, the sales forecast error, using the INVIND survey. Each year, at the beginning of the calendar year, the survey asks managers to predict end-of-year sales. We therefore define the sales forecast error for firm i in year t , denoted $\mathcal{E}_{it}$, as the difference between log realized sales at the end of the year and the log of the sales forecast made at the beginning of the year:

$$\mathcal{E}_{it} = \log(\text{Sales}_{it}) - \log(\mathbb{F}_{it-1}\text{Sales}_{it}), \quad (1)$$

where $\mathbb{F}_{it-1}\text{Sales}_{it}$ is the forecast made at the start of the calendar year t for sales over the year t . Under this convention, a positive error ($\mathcal{E}_{it} > 0$) indicates that realized sales exceeded expectations, meaning the firm was overly pessimistic *ex post*. The first stylized

⁶The corresponding figures for non-municipal contracts are 180 (214) and 865 (1015), respectively.

Figure 2: Expectations



Note: Panel (a) plots a binned scatter plot of the investment rate (defined as investment over capital) against expected revenue growth, defined as $\log \mathbb{E}_{it-1} \text{Sales}_{it} - \log \text{Sales}_{it-1}$, residualizing for realized future revenue growth and firm fixed effects. Panel (b) plots the one-year-ahead autocorrelation of the forecast error within a firm.

fact about these forecast errors is that they are on average zero across the sample.⁷

Second, we replicate the findings in [Ma et al. \(2024\)](#): firms' forecast errors are autocorrelated, even within a firm. To show this, regress a firm's forecast error in period $t + 1$ on its forecast error in period t , with firms' fixed effects.⁸ In line with their findings, panel (b) on Figure 2 shows that forecast errors are autocorrelated, with an annual autocorrelation of 17%. This is an important moment which will bring discipline to our model, it is also suggestive of deviations from full information rational expectations. Indeed, if firms had full information, their past forecast error should not be predictive of future forecast errors: under rationality, a variable in an agent's information set should not be able to forecast their forecast errors. The next part of the paper will bring some causal interpretation to this forecast error.

Taken together, these patterns establish the economic relevance of our expectations data. Managers' forecasts are accurate on average and correlate with firms' choices. This motivates our next step: studying how these expectations respond to fiscal shocks as a key mechanism through which policy affects real activity.

⁷The mean is about -2%, which is reflective of the Jensen inequality term stemming from the way we construct the forecast errors.

⁸We use the [Arellano and Bover \(1995\)](#) procedure to estimate the coefficients using dynamic panel GMM.

2.4 Identifying procurement shocks to firms: a shift-share approach

Having assembled our dataset combining firms' forecasts, realizations, and public sales, we now proceed to isolate exogenous variation in government spending. We exploit a 2015 reform to Italian municipal budget laws as a natural experiment. We implement a shift-share design (in which identification comes from the shifts, as in [Borusyak et al. \(2022\)](#)) that provides plausibly exogenous shocks to firm-level public demand. The institutional context and the construction of the shift-share instrument are described below.

The shifts: 2015 Stability Pact Our source of exogenous variation in public spending comes from a 2015 reform of the Italian *Patto di Stabilità dei Comuni* (henceforth, Stability Pact), the fiscal framework governing municipal budget surpluses.⁹

The Stability Pact was introduced in 1999 as a domestic mechanism to align local finances with EU fiscal rules established in the 1992 Maastricht Treaty and the 1997 Stability and Growth Pact. Starting in 2007, the Stability Pact imposed budget surplus targets on municipalities. In 2013, the government announced the targets for 2014–2017 and set them at 14.07% of average municipal spending in 2009–2011.¹⁰ Because the targets were tied to past spending, surplus goals differed across municipalities.

Our natural experiment comes from a reform announced on October 15, 2014. The government revised the surplus targets for 2015–2018 in two ways: it lowered the required rate to 8.6%, and, importantly, shifted the reference years from 2009–2011 to 2010–2012.¹¹ The reform was designed to provide additional fiscal space to municipalities. While this aggregate objective was known in advance, the technical details (the change in reference years that forms the basis of our identification strategy) were not publicly debated, rendering them plausibly unanticipated by both municipalities and firms. Together with the mechanical link to past spending, this lack of anticipation makes the reform a source of plausibly exogenous variation in surplus requirements across municipalities.

To capture this variation, we define the shift for municipality m as the change in log municipal expenditures between 2009 and 2012:

$$g_m \equiv \log(\text{Exp}_{m2009}) - \log(\text{Exp}_{m2012}). \quad (2)$$

Positive values of g_m indicate that municipality m spent more in 2009 than in 2012, and

⁹[Coviello et al. \(2021\)](#) study the 2008 reform to the Stability Pact, which reduced municipal spending. That reform is not suitable for our setting because our procurement data from ANAC begin in 2012.

¹⁰See art. 31 of *Legge 12 Novembre 2011, n.183* as updated by art .1, comma 532 of *Legge 27 Dicembre 2013, n. 147* in Section [A.2](#).

¹¹See art. 1, comma 489 of *Legge 23 Dicembre 2014, n.190* in Section [A.2](#).

therefore experienced a larger reduction in its surplus target following the reform. Figure A.5 in Section A.4 documents the distribution of these shifts across municipalities. The average shift is -0.7% . Our identifying variation, however, comes from the cross-sectional dispersion, with a standard deviation of 3.85% .

The shift-share instrument Having defined the municipal-level shifts, we translate them into firm-level public demand shocks using a shift-share design. For each firm i , we define its exposure to municipality m as the share of municipal procurement revenues obtained from that municipality during the 2012-2014 pre-reform period:

$$s_{im} = \frac{\sum_{t=2012}^{2014} Sales_{imt}}{\sum_{t=2012}^{2014} \sum_{m'=1}^M Sales_{im't}}, \quad (3)$$

where $Sales_{imt}$ denotes firm i 's procurement revenues from municipality m in year t . The weights s_{im} capture firms' exposure to the reform through their municipal customer base; by construction, they sum to one for each firm. We then construct the firm-level shift-share instrument as the weighted sum of municipal shifts g_m , using the exposure weights s_{im} .¹²

$$z_i = \sum_{m=1}^M s_{im} g_m. \quad (4)$$

Following Borusyak et al. (2022), we assess the effective number of shocks underlying our shift-share instrument by computing the inverse Herfindahl–Hirschman index (HHI) of average firm-level exposures. Formally,

$$\text{inverse HHI} \equiv \frac{1}{\sum_m s_m^2} \quad \text{where} \quad s_m \equiv \frac{1}{N} \sum_{i=1}^N s_{im}. \quad (5)$$

Intuitively, the inverse HHI measures how dispersed exposures are across municipalities: larger values indicate identification comes from many small shocks rather than a few dominant ones. In our data, the inverse HHI equals 136.28 in the full CADS sample and 104.01 in the matched INVIND sample, suggesting that our identification rests on a large and reasonably diffuse set of shocks. These magnitudes are well above the heuristic threshold of 20 typically used in the literature.

Figure A.6 and Figure A.7 in Section A.4 show the distribution of the average municipal shares s_m and the shift-share instrument z_i across firms, respectively. Firms have an

¹²Throughout the analysis, we trim the bottom and top 0.5% of the shifts g_m when constructing our shift-share instrument z_i .

average exposure of 1 to 4% to the few largest Italian municipalities—most notably Milan and Rome—but, as already clear from the large inverse HHI, concentration of exposures is small. As for the distribution of z_i , it closely mirrors that of the underlying shifts, with a mean of -1.5% and a standard deviation of 3.8%.

Potential threats to identification Before presenting our main empirical results, we discuss some potential threats to identification.

A first concern is that the shifts did not affect municipalities only via the reform, as they may capture other persistent determinants of procurement, such as local political cycles or mean-reversion in spending. To address this, Figure A.9 in Section A.5 shows that municipalities with larger shifts g_m experienced a large increase in procurement expenses *after* the reform, while showing no differential behavior *before* the reform was enacted. To further support exogeneity, in Figure A.3 we also show that the shifts are uncorrelated with pre-reform municipal characteristics. Figure A.8a further shows that the extent to which the reform relaxed municipal budgets is geographically widespread, with no evidence of spatial clustering.

A related question is whether municipalities responded to the reform on margins other than procurement expenses. To assess this, Figure A.10 in Section A.5 plots the behavior of municipal tax revenues around the reform. Tax revenues show no economically or statistically significant differences across municipalities, regardless of the reform’s effect on their fiscal space. This suggests that municipalities absorbed the additional fiscal space through higher spending rather than tax cuts, consistent with institutional constraints and their limited autonomy over local tax rates.

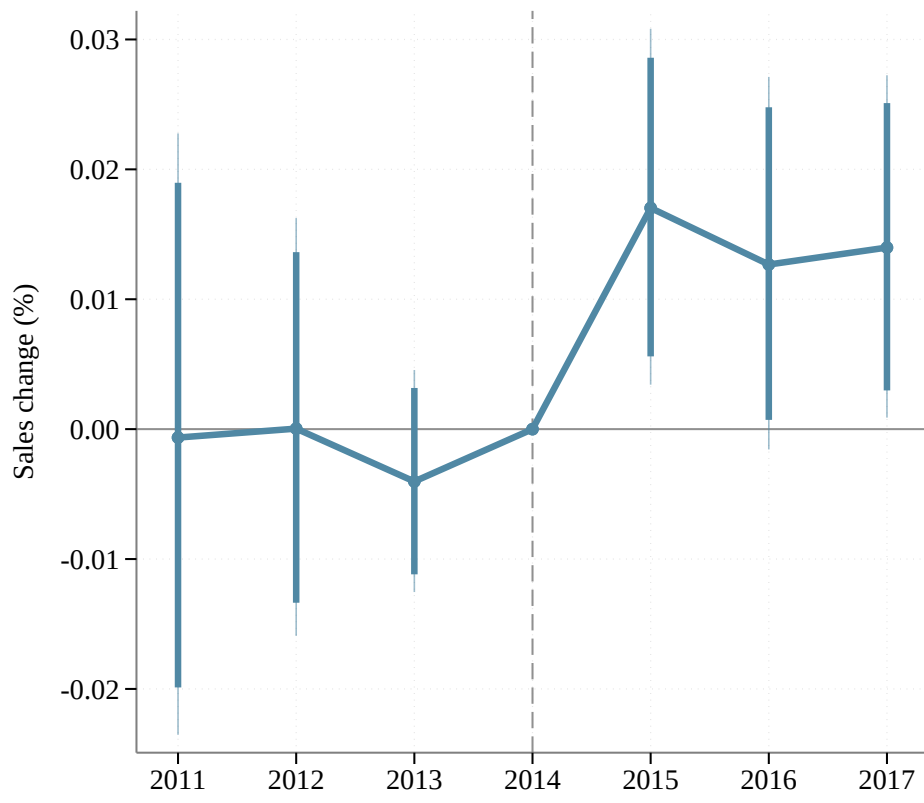
We next examine whether the firm-level shift-share instrument defined in (4) captures meaningful exposure to municipal spending. To do so, we estimate the following event study specification on the full CADS sample:

$$\log Sales_{it} = \alpha_i + \gamma_{s(i)t} + \sum_{h=2011, \neq 2014}^{2017} \beta_h (\mathbb{1}_{h=t} \times z_i) + \varepsilon_{it}, \quad (6)$$

where $\log Sales_{it}$ denotes log firm revenues, z_i is the shift-share instrument constructed above, $\gamma_{s(i)t}$ is a 1-digit NACE sector $\times$ time fixed effect, and α_i denotes a firm fixed effect. We use 2014 as the reference year, and construct exposure-robust standard errors following [Borusyak et al. \(2022\)](#).

Figure 3 plots the estimated coefficients $\hat{\beta}_h$, normalized to correspond to a one-standard-deviation increase in z_i . The coefficients are flat prior to 2015, consistent with firms with different exposures being on parallel trends. After the reform, revenues rise

Figure 3: Response of Firms' Revenues



Note: estimated $\hat{\beta}_h$ from the event study design in (6) based on the full CADS sample. The y-axis denotes the percentage change (relative to the 2014 reference year) in revenues. Wide and thin lines represent 90% and 95% confidence bands, respectively, based on exposure robust standard errors.

persistently for firms exposed to municipalities with greater fiscal space. A one-standard-deviation increase in z_i raises firm revenues by between 1% and 2% per year throughout 2015-2017. These results can be viewed as a first stage, indicating that exposure to the reform directly translated into higher revenues through procurement spending.

Another potential concern is whether firms anticipated the reform. As noted above, the technical details were not publicly debated in advance. In line with this, balance tests in Figure A.4 in Section A.3 show that the instrument is uncorrelated with pre-reform firm characteristics, including sales forecast errors, revenues, and profitability. To further rule out anticipation, we re-estimate (6) using firm assets rather than sales as the outcome. Figure A.11 in Section A.5 shows flat coefficients prior to 2015, indicating that exposed firms did not build capacity in advance and a substantial rise in investment following the reform.

2.5 The response of expectations to government spending

We now examine how government spending affects firms' expectations about future demand. To this end, we estimate a specification analogous to (6), this time on the matched INVIND-CADS sample, using sales forecast errors as the outcome:

$$\mathcal{E}_{it} = \alpha_i + \gamma_{s(i)t} + \sum_{h=2011, \neq 2014}^{2017} \beta_h (\mathbb{1}_{h=t} \times z_i) + \varepsilon_{it} \quad (7)$$

Where $\mathcal{E}_{it}$ denotes the sales forecast error defined in (1) and all other variables are defined as above.¹³ We weight our estimates using survey weights and construct exposure-robust standard errors as defined by [Borusyak et al. \(2022\)](#).

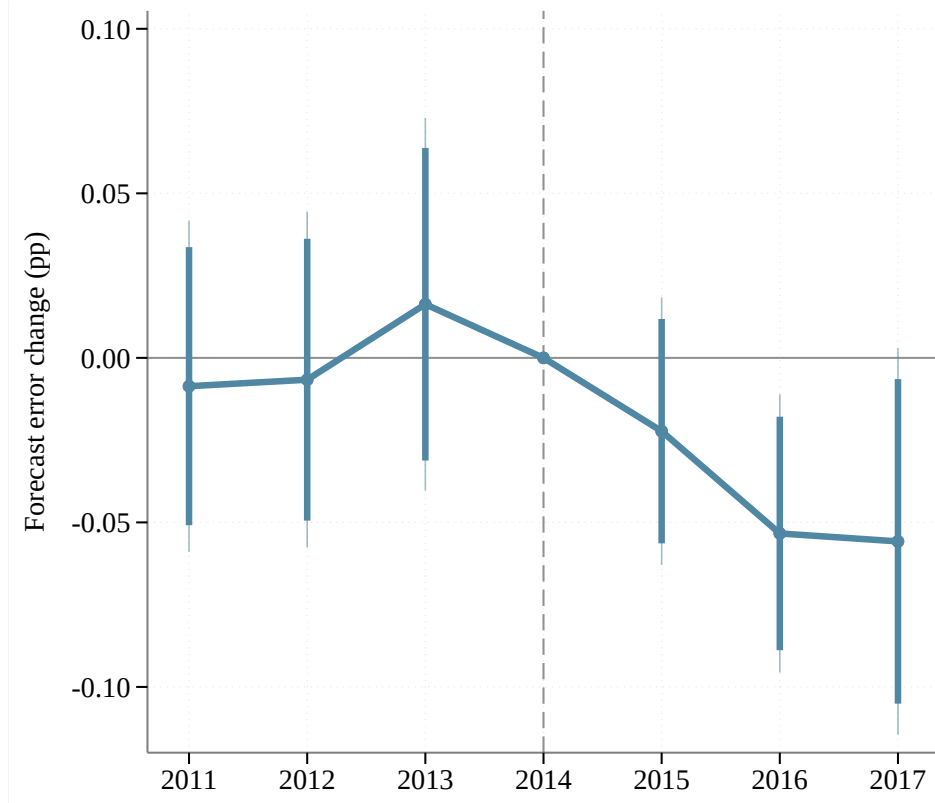
Figure 4 presents the results from this regression. Prior to 2015 forecast errors show no differential trends across firms with different exposure. After the reform, forecast errors turn increasingly negative for more exposed firms, reflecting excessive optimism about future sales.¹⁴ The effect is sizable: a one-standard-deviation increase in z_i raises expected sales relative to realizations by 6 percentage points in both 2016 and 2017. The 2015 estimates, although negative, are not statistically significant. This likely reflects survey timing: because responses were collected in the first half of the year, many exposed firms had not yet experienced higher procurement revenues. By 2016, once these revenues had materialized, the overoptimism is both large and precisely estimated.

Robustness In Figure A.12 in Section A.5, we report a series of robustness checks for the results in Figure 4. Most importantly, we show that our estimates are unchanged if we include province $\times$ year fixed effects, which suggests that the effects we identify are driven by the direct impact of municipal spending and not by local general equilibrium forces. We also show that our results are not only robust to using unwinsorized forecast errors and shift-share measures but also hold with finer 2-digit and 4-digit sector $\times$ year, or without the inclusion of sector $\times$ year fixed effects. Finally, in Table A.2 we decompose the result in Figure 4 into the response of price and quantity forecast errors and show that our results are mainly driven by the forecast error on quantities, and not by firms misperceiving the price dynamics of public procurement contracts.

¹³To mitigate the influence of extreme observations, we winsorize the forecast error below the 2.5th percentile and above the 97.5th percentile. In Figure A.12 in Section A.5 we show results are unchanged when using the raw, unwinsorized variable.

¹⁴Since the average forecast error in our sample is close to zero and slightly negative, the estimates point to excessive optimism rather than a reduction in pessimism.

Figure 4: Response of Forecast Errors



Note: estimated $\hat{\beta}_h$ from the event study design in (7). The y-axis denotes the change in sales forecast error (relative to the base year 2014) in percentage points. Wide and thin lines represent 90% and 95% confidence bands, respectively, based on exposure robust standard errors.

Instrumental variable estimates To quantify the extent of overreaction in economically meaningful terms, we estimate an instrumental variable regression in which past realized sales are instrumented with our shift-share instrument. Specifically, we estimate:

$$\mathcal{E}_{it} = \alpha_i + \gamma_{s(i)t} + \beta \widehat{\log Sales}_{it-1} + u_{it}, \quad (8)$$

where $\widehat{\log Sales}_{it-1}$ denotes log sales instrumented with the shift-share instrument interacted with a post-reform dummy, $z_i \times Post_t$.¹⁵ Table 1 shows the results from this regression. The exposure robust first-stage F-statistic exceeds the conventional critical value of 10, consistent with Figure 3. A 1% increase in sales, instrumented by the shift-share instrument, leads to a rise in optimism, making forecast errors more negative by about 0.65

¹⁵We lag realized sales by one survey wave to allow firms time to observe the effect of the reform on sales. We exclude year 2015 from the regression, since past sales were not yet affected by the reform in that year.

Table 1: Two-stage least squares regression

	First Stage	Reduced Form	IV
$Post_t \times z_i$	0.067*** (0.021)	-0.044*** (0.014)	
$\log Sales_{it-1}$			-0.652** (0.257)
Exposure-robust F-stat.			10.645
Observations	4,086	4,086	4,086
Firm FEs	✓	✓	✓
Sector-Year FEs	✓	✓	✓

Note: the first stage reports results from the regression of the endogenous regressor variable $\log(Sales_{it-1})$ on the shift-share instrument z_i interacted with post-reform dummy $Post_t$. The reduced form reports results from the regression of the sales forecast error $\mathcal{E}_{it}$ on the shift-share instrument z_i interacted with post-reform dummy $Post_t$. The IV column reports estimates for the instrumental variable regression in (8). Exposure robust standard errors in parentheses.

percentage points.¹⁶

2.6 Cross-domain extrapolation

Our main empirical result, presented above, is that shocks to *public* sales lead firms to overreact in their expectations of *total* sales. Because total sales reflect both public and private revenues, this raises the question of whether the optimism induced by public spending is confined to the directly affected domain or whether it also distorts expectations of private demand. This question relates to a recent literature documenting cross-domain extrapolation, where salient outcomes in one domain distort expectations in unrelated domains (Binder and Makridis, 2022, Bordalo et al., 2023b, 2025, Taubinsky et al., 2026, Cenzon, 2025).

To test this, we use data on export expectations, which provide a convenient testing ground for cross-domain extrapolation since they are orthogonal to domestic (public) procurement. A unique feature of the INVIND survey is that it elicits both expected and realized export revenues, allowing us to construct a measure of export sales forecast error, $\mathcal{E}_{it}^X$, following the same definition as in (1). Because the number of exporting firms is a subset of our main sample, for our baseline specification we rely on a static difference-in-

¹⁶Gennaioli et al. (2016) find comparable results for expectations of managers of publicly traded companies. In their sample a one-percentage-point increase of return on assets correlates with a 3-percentage-point reduction in the forecast error.

Table 2: Response of Export Sales Forecast Errors

	Narrow window	Broad window
$Post_t \times z_i$	-0.084** (0.042)	-0.098** (0.041)
Observations	196	844
Firm FEs	✓	✓
Sector-Year FEs	✓	✓

Note: estimates from the static difference-in-differences specification (9). The outcome variable is the export sales forecast error $\mathcal{E}_{it}^X$. The first column reports results for a narrow window, defined over 2014-2015, the second column reports results for a broader time window, defined over 2011-2018. Exposure robust standard errors in parentheses.

differences design:

$$\mathcal{E}_{it}^X = \alpha_i + \gamma_{s(i)t} + \beta z_i \times Post_t + \varepsilon_{it}, \quad (9)$$

where $\mathcal{E}_{it}^X$ is the export sales forecast error and all other variables are defined as before.

Table 2 reports the estimated $\hat{\beta}$ from this regression. The first column shows the results for a narrow window around the reform, 2014-2015, while the second column reports results for a broader window, i.e., 2011-2018. Following the reform, firms exposed to municipalities with more fiscal space become overly optimistic about export revenues. The effect is quantitatively large: a one-standard-deviation increase in z_i reduces the export forecast error by about 8 p.p. post-2015.

We interpret this as evidence of cross-domain extrapolation in firms' revenue expectations. Total sales represent a salient variable for managers, so when government spending raises revenues optimism propagates to expectations of unrelated sales streams. Under rational expectations, instead, government spending shocks should have no predictive power for export forecast errors.¹⁷ In the next section, we develop a theory of expectation formation that formalizes this mechanism.

2.7 Taking stock

Taken together, our results point to a sentiment channel of fiscal policy. Government spending raises firm revenues (Figure 3), and our key empirical finding is that this shock to one component of sales—public sales—induces managers to become overly optimistic

¹⁷While such results could be consistent with a story of imperfect information, we think that is unlikely since in practice it is easy for managers to observe sales by revenue source.

about future total sales (Figure 4). To investigate cross-domain extrapolation as a possible mechanism behind this finding, we additionally test whether optimism extends to export sales (Table 2). The evidence of such spillovers supports this interpretation. These findings motivate the model we develop in the next section.

3 A Theory of Cross-Domain Extrapolation

Our findings in Figure 4 and Table 2, together with the evidence in Panel (b) of Figure 2, call for a framework that can account for three central facts: (i) firms overreact to procurement shocks, raising expectations about total sales by more than realizations warrant, (ii) they extrapolate these shocks to unrelated domains, (iii) and finally firms' forecast errors are positively auto-correlated. We develop a model of Cross-Domain Extrapolation (XDE), in which recent news about total sales shapes forecasts about the components of sales. We discuss alternative candidate explanations using imperfect-information or Bayesian-learning models in Sections B.4 and B.5. Our theory of XDE combines two central ideas from the expectation formation literature: *salience* and *representativeness*.

A variable is *salient* when it stands out in the agent's memory relative to its underlying components, in the spirit of Bordalo et al. (2023b). The *representativeness* heuristic instead captures the tendency of agents to overweight the probability of events that have become more likely given recent news (Tversky and Kahneman, 1983).

Our framework also connects to recent work on memory and expectation formation. Bordalo et al. (2025) show how selective recall and mental simulation allow experiences in one domain to shape beliefs in another. Bordalo et al. (2026b) develop this perspective for macroeconomic expectations, showing how contextual cues shape beliefs and how spending can generate a confidence multiplier. We complement this work with a tractable, recursive continuous-time formulation of cross-domain extrapolation that can be embedded in macroeconomic models.¹⁸

3.1 Continuous-time setup

Time is continuous, $t \geq 0$. The firm observes private demand z_t and public demand g_t , together with total demand, which is defined as the sum of the private and public components, $x_t \equiv z_t + g_t$. We assume z_t and g_t each follow Ornstein-Uhlenbeck processes,

¹⁸Section B.1 describes the discrete-time case for illustration.

the continuous-time analogue of AR(1):

$$dz_t = -\mu_z z_t dt + dZ_t \quad (10)$$

$$dg_t = -\mu_g g_t dt + dG_t, \quad (11)$$

where dZ_t and dG_t are independent Brownian motions scaled by the volatility of the shocks σ_z^2 and σ_g^2 .

Expectations under cross-domain extrapolation depart from rationality through two biases: representativeness and salience. Representativeness implies that agents overweight the probability of events that become more likely in light of recent news, *relative* to a reference case in which no new information arrives. In this context, salience determines which news will actually trigger this overweighting. In particular, we assume that total demand x_t is salient: although x_t is redundant information for a forecaster observing z_t and g_t , it sways expectations through representativeness.

To formalize this, we need to introduce a notion of background context, i.e., the reference level of x_t used to characterize representativeness. Intuitively, the background is what agents would expect in the absence of innovations (Bordalo et al., 2018). In continuous time, agents are hit with shocks at every instant; we therefore need to define a notion of “recent news” that captures the realization of recent shocks to the salient variable. To do so, we follow Maxted (2023) and introduce a variable $\mathcal{S}_t$ as a measure of recent information, which we define as the exponentially-weighted average of past shocks to the salient variable:

$$\mathcal{S}_t \equiv \int_{-\infty}^t e^{-\kappa(t-s)} (dZ_s + dG_s),$$

where $\kappa > 0$ is a memory parameter governing the rate of decay of past shocks in the agent’s memory. We call this variable “sentiment”. In this setting, the evolution of sentiment may conveniently be expressed recursively as

$$d\mathcal{S}_t = -\kappa \mathcal{S}_t dt + dZ_t + dG_t. \quad (12)$$

We then define the background context as the level of x_t in the absence of recent news: $\mathcal{B}_t = x_t - \mathcal{S}_t$. With this structure in place, we can characterize beliefs under XDE. The forecast of future private demand is based on the distorted density:

$$\tilde{h}_t(z_{t+dt}) \propto h(z_{t+dt} \mid z_t = \hat{z}_t) \cdot \underbrace{\left(\frac{h(z_{t+dt} \mid x_t = \hat{x}_t)}{h(z_{t+dt} \mid x_t = \mathcal{B}_t)} \right)^\chi}_{\text{representativeness ratio}}, \quad (13)$$

where $\tilde{h}(\cdot|\cdot)$ is the distorted density, $h(\cdot|\cdot)$ represents the correct probability density function, $\hat{z}_t$ and $\hat{x}_t$ represent respectively the current realizations of z_t and x_t , and $\mathcal{B}_t$ is the background context described above.¹⁹ The proportionality constant is chosen to ensure that $\tilde{h}_t(\cdot)$ integrates to one.

Equation (13) shows that XDE forecasters have in mind a “kernel of truth”, that is, the true conditional distribution of future states $h(z_{t+dt} \mid z_t = \hat{z}_t)$, but distort it through the representativeness ratio. After a sales surprise $\hat{x}_t - \mathcal{B}_t \equiv \mathcal{S}_t$, realizations of z_{t+dt} that become more likely are overweighted, while those that become less likely are underweighted. The magnitude of this bias is governed by the behavioral parameter χ . When $\chi = 0$, (13) nests rational expectations. When $\chi > 0$, shocks to public demand g_t spill over into beliefs about private demand—even though g_t is uninformative for z_{t+dt} to a rational and fully informed agent.

This setup allows us to state how cross-domain extrapolation shapes the perceived evolution of variables.

Proposition 1 (Continuous-time XDE with OU processes). *When the fundamentals follow (10) and (11), the perceived laws of motion for z_t and g_t under cross-domain extrapolation are*

$$d\tilde{z}_t = \left(-\mu_z z_t + \underbrace{\chi_z \mathcal{S}_t}_{\text{XDE bias}} \right) dt + dZ_t, \quad \chi_z \equiv \chi \beta_z \frac{\sigma_z^2}{\sigma_{z|x}^2} \quad (14)$$

$$d\tilde{g}_t = \left(-\mu_g g_t + \underbrace{\chi_g \mathcal{S}_t}_{\text{XDE bias}} \right) dt + dG_t, \quad \chi_g \equiv \chi \beta_g \frac{\sigma_g^2}{\sigma_{g|x}^2}, \quad (15)$$

where β_z is the coefficient from an OLS regression of z_t on x_t , $\sigma_{z|x}^2$ is the variance of z_t conditional on x_t , β_g is the coefficient from an OLS regression of g_t on x_t , and $\sigma_{g|x}^2$ is the variance of g_t conditional on x_t .²⁰

Proof. See Section B. □

Proposition 1 provides a decomposition of cross-domain beliefs into a rational expectations component and a systematic bias. This bias depends on four components. First, the behavioral parameter χ governs the strength of the representativeness heuristic (i.e., how much the agent lets resemblance override the true conditional distribution). Second,

¹⁹We can equivalently define the distorted beliefs for public demand g_t , given by $\tilde{h}_t(\hat{g}_{t+dt}) \propto h(\hat{g}_{t+dt} \mid g_t = \hat{g}_t) \cdot \left(\frac{h(\hat{g}_{t+dt} \mid x_t = \hat{x}_t)}{h(\hat{g}_{t+dt} \mid x_t = \mathcal{B}_t)} \right)^\chi$

²⁰ $\beta_z = \frac{\sigma_z^2 / \mu_z}{\sigma_z^2 / \mu_z + \sigma_g^2 / \mu_g}$, $\sigma_{z|x}^2 = \frac{\left(\frac{\sigma_z^2}{2\mu_z} \right) \left(\frac{\sigma_g^2}{2\mu_g} \right)}{\frac{\sigma_z^2}{2\mu_z} + \frac{\sigma_g^2}{2\mu_g}}$, $\beta_g = \frac{\sigma_g^2 / \mu_g}{\sigma_z^2 / \mu_z + \sigma_g^2 / \mu_g}$, $\sigma_{g|x}^2 = \sigma_{z|x}^2$.

the OLS coefficients, β_z and β_g , capture the co-movement between the forecast variable (z_t or g_t) and the salient outcome (x_t). They summarize the extent to which total sales are representative of public and private sales in the agent's memory. Third, the variance ratio $\sigma_z^2/\sigma_{z|x}^2$ scales the bias by the precision of x as a predictor for z or g . Finally, through its recursive law of motion (12), sentiment then becomes the relevant state variable summarizing agents' optimism or pessimism in expectations. Together, these components map behavioral heuristics into forecast errors.

Finally, cross-domain extrapolation for the two fundamentals generates diagnostic expectations for the salient variable x_t :

$$d\tilde{x}_t = (-\mu_z z_t - \mu_g g_t + \chi_x \mathcal{S}_t) dt + dZ_t + dG_t, \quad \chi_x \equiv \chi_z + \chi_g = \chi \left(\beta_z \frac{\sigma_z^2}{\sigma_{z|x}^2} + \beta_g \frac{\sigma_g^2}{\sigma_{g|x}^2} \right).$$

Taken together, (12), (14) and (15) define the stochastic process governing the perceived evolution of private and public demand under XDE. This process depends on structural parameters governing the private and public demand process as well as on the two psychological parameters, κ and χ . In Section 5.2 we leverage our empirical estimates and micro-data to discipline these parameters. The state vector consists of private demand z_t , public demand g_t , and sentiment $\mathcal{S}_t$. To provide some additional intuition for how XDE operates, we illustrate how our theory of expectations works by simulating our model.

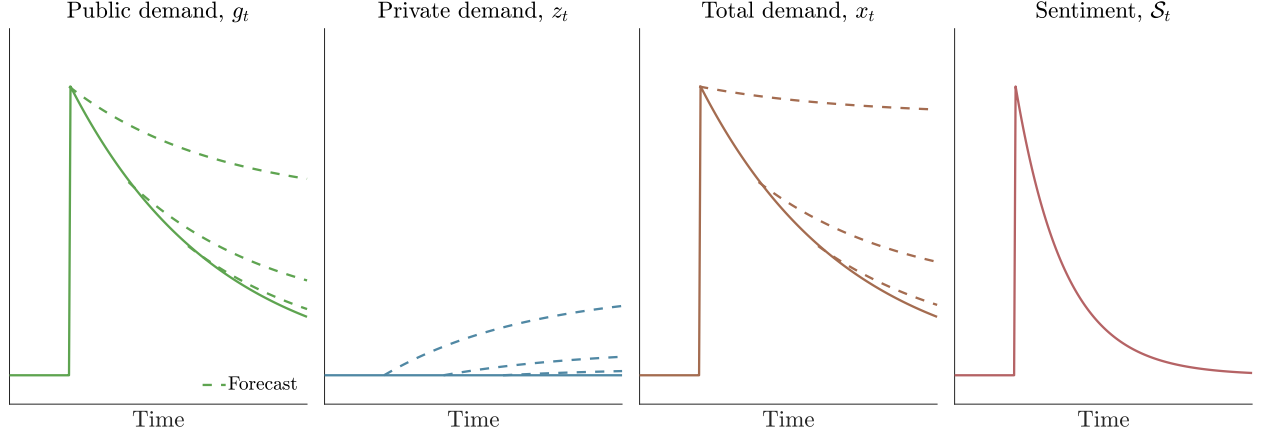
3.2 Simulation of cross-domain extrapolation

To illustrate how an XDE agent forms forecasts, the four panels in Figure 5 plot simulated paths for public demand g_t , private demand z_t , total sales x_t and the latent sentiment variable $\mathcal{S}_t$, respectively. To simplify the exposition, we only plot the behavior of the model given a single shock to public sales. In each panel, the solid lines show the realized path of a variable, and the dashed lines show the forecast path of the same variable at different forecasting times.

The first panel depicts a positive shock to public demand that raises g_t . Private sales receive no shocks. As total sales are the sum of public and private sales, total sales also jump. Through (12), this shock induces a jump in sentiment $\mathcal{S}_t$ (fourth panel). Elevated sentiment then tilts beliefs upward, implying that the XDE forecast for public demand (dashed green) exceeds the rational benchmark (solid green), which quickly mean-reverts toward the unconditional mean at the correct rate.

The second panel shows how this optimism spills over to beliefs about private demand. Even though z_t experienced no positive shocks, the agent extrapolates the positive

Figure 5: Simulation of cross-domain extrapolation



Note: simulated paths for public, private and total demand, as well as sentiment, given a single shock to public sales. Solid lines denote the realized path of each variable, while dashed lines denote the forecast path of each variable at different forecasting times. Calibration is illustrative.

sales news across domains, so that the XDE forecast (dashed blue line) lies above the rational path (solid blue).

The third panel combines the two components: since expectations for both g_t and z_t are biased upward, the bias in expectations cumulates, so the forecast for total sales (dashed red) lies above the rational benchmark by more than either component on its own.

Finally, we show that sentiment gradually wanes as the behavioral memory of the shock fades—a process reflected in the mean reversion of sentiment. Consequently, forecast paths converge to the true realization as sentiment reverts back to zero.

4 A q -theory of Investment with Sentiment

Having developed our model of expectation formation under cross-domain extrapolation, we now examine its implications for firm investment. We embed sentiment into an otherwise standard partial equilibrium q -theory of investment with financial frictions. This allows us to characterize analytically not only the impact of sentiment on investment decisions but also its interaction with financial frictions.

4.1 Partial equilibrium model of the firm

Time $t \geq 0$ is continuous and runs forever. There is a continuum of monopolistically competitive firms $i \in [0, 1]$, each producing a differentiated good. Firms face stochastic fluctuations in public and private demand for their good, denoted by g and z , and use

capital k as an input.²¹ This gives rise to a profit function of the form $\pi(k, g, z)$, which is increasing in g and z and concave and single-peaked in capital.²²

Firms own their capital stock and accumulate it through investment. To install ι units of capital, a firm must purchase $\iota + \Phi(\iota, k)$ units of the final good, where $\Phi(\iota, k)$ denotes investment adjustment costs. These costs make the investment decision forward-looking. Capital depreciates at rate δ .

Investment is subject to financial frictions in the form of an external-finance premium. We capture this with a cost function $\Gamma(d) \leq 0$, where $d \equiv \pi(k, g, z) - \iota - \Phi(\iota, k)$ denotes dividends paid to shareholders. The function $\Gamma(d)$ satisfies $\Gamma(d) = 0$ when $d \geq 0$ and $\Gamma(d) < 0$ when $d < 0$ (when the firm resorts to external financing). The net dividend paid to shareholders is therefore $d + \Gamma(d)$.

We depart from the rational expectations benchmark by allowing firms to hold biased expectations about future demand (g, z) . To do so, we embed our theory of expectations formation into this environment and let sentiment $\mathcal{S}$ bias firms' expectations, as described in (14) and (15).²³ Throughout this section, we treat sentiment as exogenous, which allows us to cleanly isolate the way in which biased expectations affect firms' investment decisions.

Firms maximize the net present value of dividends and discount the future at the interest rate r .²⁴ In a stationary economy, they solve the following problem:

$$\begin{aligned} V(k_0, \mathcal{S}_0, g_0, z_0) = \max_{\{\iota_t\}_{t \geq 0}} \quad & \tilde{\mathbb{E}}_0 \int_0^\infty e^{-rt} (d_t + \Gamma(d_t)) dt \\ \text{s.t.} \quad & \dot{k}_t = \iota_t - \delta k_t, \\ & d_t = \pi(k_t, g_t, z_t) - \iota_t - \Phi(\iota_t, k_t), \end{aligned} \tag{16}$$

where $\tilde{\mathbb{E}}_0$ denotes the time-0 expectation over idiosyncratic demand g_t and z_t under firms' subjective beliefs shaped by sentiment $\mathcal{S}$, as described in Proposition 1.

4.2 Sentiment and investment

Armed with our theoretical framework, we now analytically characterize the way in which sentiment affects firms' investment behavior.

²¹The inclusion of other frictionlessly adjustable inputs does not affect our analysis.

²²We microfound this profit function in Section 5.1.

²³Hence, firms are "internally rational" as in Adam and Marcet (2011).

²⁴We define this problem in steady-state, generalizing it to the non-stationary case with aggregate shocks in the next section.

Proposition 2 (Euler equation). *Let $\tilde{q}$ denote the marginal value of capital perceived by the firm, i.e., $\tilde{q} \equiv V_k$. The evolution of $\tilde{q}_t$ can be decomposed as*

$$(r + \delta)\tilde{q}_t = (\pi_k(k_t, g_t, z_t) - \Phi_k(l_t, k_t)) (1 + \Gamma'(d_t)) + \mathbb{E}_t d\tilde{q}_t/dt + (\tilde{\mathbb{E}}_t - \mathbb{E}_t) d\tilde{q}_t/dt, \quad (17)$$

where $\mathbb{E}_t$ and $\tilde{\mathbb{E}}_t$ are the rational and biased expectation operators respectively. Under our expectations formation process (12), (14) and (15), we can rewrite the last term as

$$(\tilde{\mathbb{E}}_t - \mathbb{E}_t) d\tilde{q}_t/dt = \left(\chi_g \frac{\partial \tilde{q}_t}{\partial g} + \chi_z \frac{\partial \tilde{q}_t}{\partial z} \right) \mathcal{S}_t.$$

Proof. See Section C.2. □

Proposition 2 shows that in the presence of sentiment, the marginal- q condition takes the familiar form of an Euler equation, augmented by a novel expectation term. Absent financial frictions and under rational expectations, this expression nests the classic Euler equation of neoclassical investment theory (Abel and Eberly, 1994).

The left-hand side represents the required gross return on a marginal unit of capital: the interest rate plus depreciation. The right-hand side is the expected return from that unit, and consists of two terms. The first is the instantaneous cash yield (the marginal increase in profits from an additional unit of capital, net of the adjustment cost and further reduced by the fact that higher investment lowers dividends, increasing the external-finance premium). The second is the expected capital gain (the anticipated appreciation in the value of installed capital, i.e., the change in perceived marginal q).

Under rational expectations, the capital gain component coincides with the true expected path of future returns. The presence of sentiment, however, introduces an additional term: an expectation wedge capturing an “expected forecast error”, a predictable bias embedded in firms’ subjective beliefs. This wedge is the channel through which sentiment affects investment dynamics. For optimistic firms featuring positive $\mathcal{S}$ the wedge is positive, raising the perceived marginal value of capital and thereby stimulating investment, as the next proposition shows.

Proposition 3 (Sentiment and investment). *The optimal investment decision when firms have sentiment is given by*

$$\begin{aligned} \Phi_l(l_t, k_t) &= \frac{q_t + \mathcal{F}_t}{1 + \Gamma'(d_t)} - 1 \\ \mathcal{F}_t &\equiv \int_0^\infty e^{-(r+\delta)s} (\tilde{\mathbb{E}}_t - \mathbb{E}_t) (1 + \Gamma'(d_{t+s})) (\pi_k(k_{t+s}, g_{t+s}, z_{t+s}) - \Phi_k(l_{t+s}, k_{t+s})) ds, \end{aligned} \quad (18)$$

where $q_t = V_k(k_t, 0, g_t, z_t)$ is the rationally perceived marginal value of capital and $\mathcal{F}_t$ is the forecast error in perceived marginal q and can be expressed in terms of sentiment as

$$\mathcal{F}_t = V_k(k_t, \mathcal{S}_t, g_t, z_t) - V_k(k_t, 0, g_t, z_t). \quad (19)$$

Proof. See Section C.2. □

Proposition 3 characterizes how sentiment distorts firms' investment behavior through an additive term $\mathcal{F}_t$, which represents the misperception in the marginal value of capital (the marginal q). As shown in (19), this misperception is increasing in sentiment. Therefore, higher sentiment raises investment, driven by optimistic expectations of future returns.

A key implication of (18) is that financial constraints dampen the relationship between investment and sentiment. When firms are constrained, they must rely on costly external finance to invest more, as reflected in the $\Gamma'(d_t)$ term in the denominator. This raises the effective marginal cost of investment, thus muting the sensitivity of investment to expectations. We refer to the sensitivity of investment to sentiment as the *marginal propensity to invest out of sentiment* (MPIS). Section C.3 formalizes this measure, illustrates how financial constraints dampen it, and presents supporting evidence from our firm-level survey data.

4.3 Taking stock

In this section we have analytically characterized how sentiment shapes investment by embedding our model of expectation formation in a q -theory of investment with financial frictions. The analysis shows that investment is increasing in sentiment and that this sensitivity is muted for firms facing tighter financial constraints, consistent with the firm-level evidence reported in Section C.3. We now proceed to embed this model of investment with sentiment in a full general equilibrium environment, which will enable us to study the role of the sentiment channel for the aggregate transmission of fiscal policy.

5 The Sentiment Channel in General Equilibrium

Having characterized the effect of sentiment on investment, we now turn to the quantification of our framework. To do so, we first embed both the firm investment problem of Section 4 and the XDE model of Section 3 within a general equilibrium heterogeneous-firm New Keynesian model. We calibrate the model to our data, quantify the sentiment channel and run policy experiments.

5.1 General equilibrium model

We now proceed to embed the problem of the firm (detailed in Section 4.1) in a general equilibrium model and describe the different blocks of the model.

Public and private demand There are two final goods: a private good (used for household consumption and investment), and public goods (consumed by the government). Let Y_t^z and G_t denote, respectively, the aggregators for these two goods. Both are CES composites of the differentiated intermediate varieties y_{it} :

$$Y_t^z = \left(\int_0^1 e^{\frac{z_{it}}{\varepsilon}} y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad \text{and} \quad G_t = \left(\int_0^1 e^{\frac{g_{it}}{\varepsilon}} y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad (20)$$

where z_{it} and g_{it} denote, respectively, idiosyncratic taste shifters of private and public demand for variety i . These aggregators deliver standard downward-sloping firm-level demands:

$$y_{it}^z = e^{z_{it}} \left(\frac{p_{it}}{P_t^z} \right)^{-\varepsilon} Y_t^z \quad \text{and} \quad y_{it}^g = e^{g_{it}} \left(\frac{p_{it}}{P_t^g} \right)^{-\varepsilon} G_t, \quad (21)$$

with corresponding price indices

$$P_t^z = \left(\int_0^1 e^{z_{it}} p_{it}^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}, \quad P_t^g = \left(\int_0^1 e^{g_{it}} p_{it}^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}.$$

Total demand for firm i is therefore $y_{it} = y_{it}^z + y_{it}^g$. To map this structure to our theory of expectation formation, we let the taste shifters z_{it} and g_{it} evolve stochastically according to the OU processes described in (10) and (11).

Production Each firm is a monopolistically competitive producer of a differentiated variety i and uses a constant-returns-to-scale technology that combines k_{it} units of capital and ℓ_{it} units of labor:

$$y_{it} = k_{it}^\alpha \ell_{it}^{1-\alpha},$$

where $\alpha \in (0, 1)$. Labor is hired on a competitive market at the real wage w_t and can be adjusted frictionlessly within a period. Capital is owned by the firm and evolves according to the law of motion described in (16). Normalizing the price index of the private good to one, the firm's static optimization over labor and prices yields a profit function of the

form

$$\pi_t(g, z, k) = \Omega_t(k) \cdot (e^z + \alpha_{gt} e^g)^{\frac{1}{1-\alpha+\epsilon\alpha}}, \quad (22)$$

where $\Omega_t(k)$ is a function of capital, aggregate variables (captured by the time index) and structural parameters, and $\alpha_{gt} \equiv G_t(P_t^g)^\epsilon / Y_t^z$.²⁵ This expression incorporates the firm's optimal static labor and pricing decisions.

The problem of the firm and expectations As the time-dependence of the profit function (22) makes clear, solving the dynamic capital accumulation problem requires firms to form expectations over sequences of aggregate variables $\{\Upsilon_t\}_{t \geq 0} = \{w_t, r_t, Y_t^z, G_t, P_t^g\}_{t \geq 0}$. Regarding these aggregate variables, we maintain the assumption of rational expectations.²⁶

We introduce departures from rational expectations in the way firms forecast their own idiosyncratic private and public demand shifters z_{it} and g_{it} . We model expectations over these variables using the recursive formulation of cross-domain extrapolation defined in (12), (14) and (15).²⁷ This formulation introduces sentiment $\mathcal{S}$ as an additional state, which means that the firm's idiosyncratic state vector becomes $\mathbf{s} = (k, \mathcal{S}, g, z)$.

A key challenge when departing from rational expectations is that the *perceived* and *true* law of motion of the states no longer coincide. Our recursive formulation with sentiment provides a tractable way to capture this discrepancy in the Hamilton-Jacobi-Bellman (HJB) equation through an additive term, which we label the rationality wedge:

$$\begin{aligned} r_t V_t(\mathbf{s}) - \partial_t V_t(\mathbf{s}) = & \max_t \underbrace{\pi_t(k, g, z) - \iota - \Phi(\iota, k) + \Gamma(d_t)}_{\text{flow to households}} \\ & + \underbrace{(\iota - \delta k) \partial_k V_t(\mathbf{s}) - \mu_z z \partial_z V_t(\mathbf{s}) + \frac{\sigma_z^2}{2} \partial_{zz}^2 V_t(\mathbf{s}) - \mu_g g \partial_g V_t(\mathbf{s}) + \frac{\sigma_g^2}{2} \partial_{gg}^2 V_t(\mathbf{s})}_{\text{continuation}} \\ & + \underbrace{\chi_z \mathcal{S} \partial_z V_t(\mathbf{s}) + \chi_g \mathcal{S} \partial_g V_t(\mathbf{s})}_{\text{rationality wedge}}, \end{aligned} \quad (23)$$

where $d_t = \pi_t(k, g, z) - \iota - \Phi(\iota, k)$ denotes dividends. The rationality wedge in (23) reflects the behavioral distortions introduced by sentiment, leading firms to misperceive the future evolution of z and g . Note that the law of motion for $\mathcal{S}$ itself is absent from the HJB, since we assume that firms are naïve: they are not aware about their sentiment bias, and

²⁵See Section C.1 for derivations.

²⁶In what follows we focus on a so-called MIT shock, which implies that after date 0 firms have perfect foresight with respect to sequences of aggregate variables.

²⁷To cast the firm's problem in the log-additive setting of Section 3, we take a first-order approximation of the profit function (22) around steady-state, which shows that the salient variable shifting total profits is $z + \alpha_g g$.

therefore do not track its dynamics.

The solution to the HJB (23) yields a sequence of policy functions for firm investment, labor demand and dividends: $\{\iota_t(s), \ell_t(s), d_t(s)\}_{t \geq 0}$. Together with the sequence of aggregate variables $\{\Upsilon\}_{t \geq 0}$, these fully characterize the evolution of the distribution of firms over the state space, $\mu_t(s)$, as described by the Kolmogorov Forward equation:

$$\begin{aligned} \partial_t \mu_t(s) = & \underbrace{-\partial_k[(\iota_t(s) - \delta k)\mu_t(s)] + \mu_z \partial_z[z\mu_t(s)] + \mu_g \partial_g[g\mu_t(s)] + \kappa \partial_S[\mathcal{S}\mu_t(s)]}_{\text{true drifts}} \\ & \underbrace{+ \frac{1}{2} \sigma_z^2 \partial_{zz}^2 \mu_t(s) + \frac{1}{2} \sigma_g^2 \partial_{gg}^2 \mu_t(s) + \frac{\sigma_z^2 + \sigma_g^2}{2} \partial_{SS}^2 \mu_t(s) + \sigma_z^2 \partial_{zS}^2 \mu_t(s) + \sigma_g^2 \partial_{gS}^2 \mu_t(s)}_{\text{true diffusion}} \end{aligned} \quad (24)$$

Because the Kolmogorov Forward equation describes the *true* evolution of the distribution, it omits the rationality wedge present in the HJB, while instead featuring the law of motion for sentiment. Misperceptions of z and g enter only indirectly through the biased policy functions $\iota_t(s)$, while the processes for z , g and $\mathcal{S}$ remain themselves correct.

This concludes the description of the firms' problem. We now turn to the remaining blocks of the model to characterize general equilibrium.

Household There is a representative Ricardian household that discounts the future at rate ρ and has preferences over hours worked and consumption. The household owns the firms and can save in a riskless real bond issued by the government A_t . The household solves the following problem:

$$\begin{aligned} \max_{\{C_t, \dot{A}_t\}_{t \geq 0}} \quad & \mathbb{E}_0 \int_0^\infty e^{-\rho t} (u(C_t) - v(L_t)) dt \\ \text{s.t.} \quad & \dot{A}_t = w_t L_t + r_t A_t + D_t - T_t - C_t, \end{aligned} \quad (25)$$

where C_t is consumption, L_t is hours worked, D_t are aggregate dividends rebated by firms, and T_t are government lump-sum taxes. Labor supply is determined by unions, which we describe next.

Unions We introduce nominal rigidities in the form of sticky wages. As is standard in the New Keynesian sticky-wage literature, we assume that households' hours worked L_t are determined by labor unions (Erceg et al., 2000, Schmitt-Grohé and Uribe, 2005, Auclert et al., 2024). Specifically, the household supplies a measure of labor ℓ_{kt} to a continuum of monopolistically competitive unions indexed by $k \in [0, 1]$, with $\int_0^1 \ell_{kt} dk = L_t$. Each union

provides a differentiated labor service, which is combined into aggregate labor by a competitive labor packer using a CES aggregator with elasticity of substitution ϵ . Unions set wages for their variety to maximize lifetime utility of the household, subject to Rotemberg adjustment costs:

$$\max_{\{\pi_{kt}^w\}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} \left(u(C_t) - v(\ell_{kt}) - \frac{\psi}{2} (\pi_{kt}^w)^2 L_t \right) dt,$$

where π_{kt}^w denotes nominal wage inflation. This problem gives rise to the following non-linear wage New Keynesian Phillips Curve:²⁸

$$\epsilon v'(L_t) L_t - (\epsilon - 1) u'(C_t) w_t L_t + \psi (\dot{\pi}_t^w - \rho \pi_t^w) = 0.$$

Government The government finances G_t via lump-sum taxes on households T_t and debt issuance. The budget constraint of the government is

$$\dot{B}_t = r_t B_t + P_t^g G_t - T_t \quad (26)$$

We assume the government runs a balanced budget in every period.²⁹

Government spending shocks We model government spending shocks as a collection of correlated shocks to individual firms' public procurement. At the time of the shock, firm i 's public-demand shifter increases from g_{i0} to $g_{i0} + \hat{g}_i$, with the additional demand subsequently reverting at rate μ_g . These shocks enter firms' demand functions in (21) without directly changing the aggregator governing government preferences in (20). In the baseline exercise, we consider a uniform procurement shock, $\hat{g}_i = \hat{g} = 0.01$ for all firms. Aggregate government consumption therefore rises by approximately 1% on impact, as shown in Figure 6a. The shock decays at the calibrated annual rate $\mu_g = 1.12$, corresponding to a standard quarterly persistence of around 75%.

Procurement shocks also affect firms' expectations through sentiment. Under the calibrated sentiment process in (12), the innovation $\hat{g}_i$ raises firm i 's sentiment by $\alpha_g \hat{g}_i$, inducing optimism about both future private and public demand. Together, the shifts in public demand and sentiment determine the initial post-shock distribution of firms, $\mu_0(s)$, from which we compute the general equilibrium transition. To isolate the sentiment channel, we compare this response with a counterfactual in which the same procurement shock

²⁸See Section D.1 for details on the derivation.

²⁹Because our model features a Ricardian household, the financing mix between taxes and debt is irrelevant for equilibrium outcomes.

leaves sentiment unchanged on impact.

Our aggregate exercise assumes that the mapping from a firm's procurement news into sentiment remains unchanged when these innovations are correlated across firms. Conditional on its own state and procurement innovation, a firm applies the same expectation-formation rule as in the calibration, without adjusting this rule to the cross-sectional correlation of shocks. This is a maintained assumption connecting the firm-level evidence to the aggregate counterfactual which allows us to quantify the aggregate effects of the sentiment response. It parallels the use of microeconomic evidence on marginal propensities to consume to discipline household responses to aggregate shocks in the extensive HANK literature. In both cases, the microeconomic empirical response supplies an input to the aggregate analysis, while equilibrium changes in prices, incomes, and quantities determine the resulting multiplier.

Monetary authority Monetary policy follows a standard Taylor rule:

$$i_t = r + \phi^\pi \pi_t,$$

where r is the steady-state interest rate and we assume $\phi^\pi > 1$ to ensure determinacy. Given inflation and the nominal interest rate, the real interest rate is then given by the continuous-time Fisher equation $r_t = i_t - \pi_t$.

Equilibrium We are now ready to define the equilibrium of our model.

Definition 1 (Equilibrium). *Given a government spending shock $\{\widehat{g}_i\}_i$, an equilibrium is a sequence of firm policy functions $\{\iota_t(\mathbf{s}), \ell_t(\mathbf{s}), \mathbf{d}_t(\mathbf{s})\}_{t \geq 0}$, firm distributions $\{\mu_t(\mathbf{s})\}_{t \geq 0}$, prices $\{w_t, r_t, P_t^g\}_{t \geq 0}$, and aggregate quantities $\{G_t, C_t, L_t, A_t, I_t, D_t, T_t, B_t, Y_t\}_{t \geq 0}$ such that the post-shock distribution of firms $\mu_0(\mathbf{s})$ is consistent with the endogenous response of sentiment to the shocks $\{\widehat{g}_i\}_i$, the path of Government expenditures G_t is consistent with the initial government demand shock, firms optimize given prices and sentiment, the Fisher and Taylor equations hold, the government runs a balanced budget, the firm distribution evolves according to the Kolmogorov forward equation with initial condition $\mu_0(\mathbf{s})$, and the goods, labor, and asset markets clear:*

$$Y_t = C_t + I_t + P_t^g G_t, \tag{27}$$

$$L_t = \int \ell_t(\mathbf{s}) d\mu_t(\mathbf{s}), \tag{28}$$

$$A_t = B_t. \tag{29}$$

Aggregate investment is defined as $I_t \equiv \int [\iota_t(\mathbf{s}) + \Phi(\iota_t(\mathbf{s}), k(\mathbf{s})) - \Gamma(\mathbf{d}_t(\mathbf{s}))] d\mu_t(\mathbf{s})$.

Since $\Gamma(d) \leq 0$, the resource cost of external finance is $-\Gamma(d) \geq 0$. Here $k(s)$ is the capital stock component of the firm state, and nominal output is $Y_t \equiv \int p_t(s) y_t(s) d\mu_t(s)$.

5.2 Calibration

We now turn to the calibration of the model. Parameters are expressed at an annual frequency and reported in Table 3. We distinguish parameters calibrated externally, parameters calibrated to individual aggregate moments, and seven parameters governing demand and expectation formation that are calibrated jointly. Table D.1 in the appendix reports the empirical targets and their model counterparts.

Table 3: Model calibration

Description		Value	Comment
<i>Externally calibrated</i>			
ρ	Household discount rate	0.02	Target 2% steady-state real rate
γ	EIS	1	Standard
φ	Frisch elasticity	1	Chetty et al. (2011)
θ	Household labor disutility	1	Normalization
α	Capital share	0.33	Standard
ε	Demand elasticity	6	Product-market elasticity
ϵ	Union market power	21	Schmitt-Grohé and Uribe (2005)
ϕ^π	Taylor coefficient	1.5	Standard
ϕ	Investment adjustment costs	1.5	Chodorow-Reich (2025)
<i>Internally calibrated</i>			
δ	Depreciation rate	0.1	Target aggregate investment rate
δ^{exit}	Firm exit rate	0.07	Target annual exit rate
ψ	Wage adjustment cost	24.57	Wage Phillips slope 1.04; Fitzgerald et al. (2024)
ζ	External finance cost	0.35	Target excess bond premium; Gilchrist and Zakrajšek (2012)
<i>Jointly calibrated</i>			
μ_z	Private-demand mean reversion	0.04	Jointly match private-sales moments
σ_z	Private-demand volatility	0.53	Jointly match private-sales moments
μ_g	Public-demand mean reversion	1.12	Jointly match public-sales moments
σ_g	Public-demand volatility	2.32	Jointly match public-sales moments
χ	Cross-domain extrapolation	0.35	Target IV slope in Table D.1
κ	Sentiment mean reversion	0.19	Target forecast-error autocorrelation
$\bar{G}$	Government demand	0.33	Target government expenditure/GDP of 0.18

Note: Parameters are expressed at an annual frequency. The parameters μ_z , μ_g , and κ are continuous-time mean-reversion rates; σ_z and σ_g are diffusion volatilities. The seven jointly calibrated parameters target the moments in Table D.1.

Household preferences We externally calibrate a number of parameters governing households' preferences. Following standard values in the literature, we set the household discount rate $\rho = 0.02$ to target a 2% yearly real interest rate. We specify a standard separable CRRA functional form for households' utility, $u(C_t, L_t) = \frac{C_t^{1-\gamma}-1}{1-\gamma} - \vartheta \frac{L_t^{1+\phi}}{1+\phi}$, with log utility over consumption when $\gamma = 1$. We set the elasticity of intertemporal substitution and the Frisch elasticity of labor supply to 1, the latter in line with [Chetty et al. \(2011\)](#), and normalize the labor-disutility parameter ϑ to 1. The demand elasticity ε is set to 6.

Technology parameters We calibrate the capital share in production α to 0.33, as standard. We posit a quadratic function for investment adjustment costs, $\Phi(\iota, k) = \frac{\phi}{2} (\frac{\iota}{k} - \delta)^2 k$, and set $\phi = 1.5$ based on the evidence reviewed by [Chodorow-Reich \(2025\)](#). We also assume a quadratic cost of external finance, $\Gamma(d) = -\frac{\zeta}{2} d^2 \cdot \mathbb{1}(d < 0)$, with $\zeta = 0.35$ calibrated to the excess bond premium in [Gilchrist and Zakrajšek \(2012\)](#). The depreciation rate δ is set to 0.1 to match the aggregate investment rate. Firms exit at a Poisson rate δ^{exit} , calibrated to the annual exit rate and equal to 0.07. Each exit is offset by the entry of a new firm, preserving the total mass of firms. We set union market power ϵ to 21 following [Schmitt-Grohé and Uribe \(2005\)](#) and the Taylor coefficient ϕ^π to 1.5 as is standard. The wage-adjustment parameter $\psi = 24.57$ implements a wage Phillips curve slope of 1.04, obtained by mapping the estimates of [Fitzgerald et al. \(2024\)](#) into the model.

Joint calibration of demand and XDE We jointly calibrate the four parameters governing idiosyncratic private and public demand, $(\mu_z, \sigma_z, \mu_g, \sigma_g)$, the two parameters governing sentiment, (χ, κ) , and the level of government demand, $\bar{G}$. These seven parameters target seven moments: the persistence and innovation variance of annual private and public sales, the autocorrelation of forecast errors, the response of forecast errors to sales variation induced by public-demand shocks, and the government expenditure share of GDP. Joint calibration allows the demand processes and expectation formation to affect these moments simultaneously through firms' sales and investment decisions.

From the contract-level procurement data, we construct an annual measure of firm-level public sales, $Sales_{it}^g$, and define private sales as total sales from the income statement less public sales, $Sales_{it}^z = Sales_{it} - Sales_{it}^g$. Section [D.2](#) details the reconstruction of annual public sales from contract values and execution dates, the estimation sample, and the empirical procedure. We estimate a pooled AR(1) process with an intercept for each sales component:

$$\log Sales_{it}^z = a_z + \rho^z \log Sales_{it-1}^z + u_{it}^z, \quad (30)$$

$$\log Sales_{it}^g = a_g + \rho^g \log Sales_{it-1}^g + u_{it}^g, \quad (31)$$

We target the estimated persistence coefficients and the variances of the innovations u_{it}^z and u_{it}^g . The demand parameters are therefore disciplined by the dynamics of simulated sales rather than equated directly to the estimated annual AR(1) coefficients.

The two forecast-error moments discipline expectation formation. We target an autocorrelation of 0.17 and an IV semi-elasticity of forecast errors with respect to lagged sales of -0.65 , as reported in Table 1. The IV estimate uses sales variation induced by public-demand shocks: a 1% increase in sales lowers the following year’s forecast error by 0.65 percentage points, indicating greater overoptimism. These moments are particularly informative about sentiment persistence and the strength of cross-domain extrapolation, respectively. Finally, $\bar{G}$ is chosen within the joint calibration so that government expenditure represents 18% of steady-state GDP, as observed in the data.

For each candidate parameter vector, we solve the stationary equilibrium, simulate firms’ annual sales and forecast errors, and calculate the model moments. Section D.3 explains why these parameters are calibrated jointly, defines the simulated moments, and describes the simulated method of moments procedure. Table D.1 in that section reports the fit.

Relation to the literature A closely related paper is [Bordalo et al. \(2026a\)](#), who estimate firms’ overreaction to news and study its implications for investment and credit cycles. Their estimated diagnosticity parameter implies that one-year-ahead log-productivity forecasts respond to news approximately 91% more strongly than under rational expectations. Our calibration likewise implies substantial overreaction: pooling private- and public-demand news, subjective annual sales forecasts respond approximately 23% more strongly than their objective counterparts. Likewise, [Maxted \(2023\)](#) features an exponentially decaying “sentiment” state, whose half life is about 6 years compared to 3.57 years in our setting. Our calibration is therefore broadly consistent with the magnitude of belief distortions documented in this literature. Its distinctive feature is that news about one component of demand spills over into expectations about another through persistent sentiment. Section D.3.1 details the calculation and its interpretation.

Stationary equilibrium Section D.6 presents the stationary distributions of capital, private demand, public demand, and sentiment, together with firms’ policy functions and simulated dynamics. The policy functions illustrate how sentiment distorts investment decisions: positive sentiment stimulates additional investment, while negative sentiment

dampens it. We also show that our model is successful at matching untargetted moments of the stationary distribution of firms.

5.3 The sentiment channel and the government spending multiplier

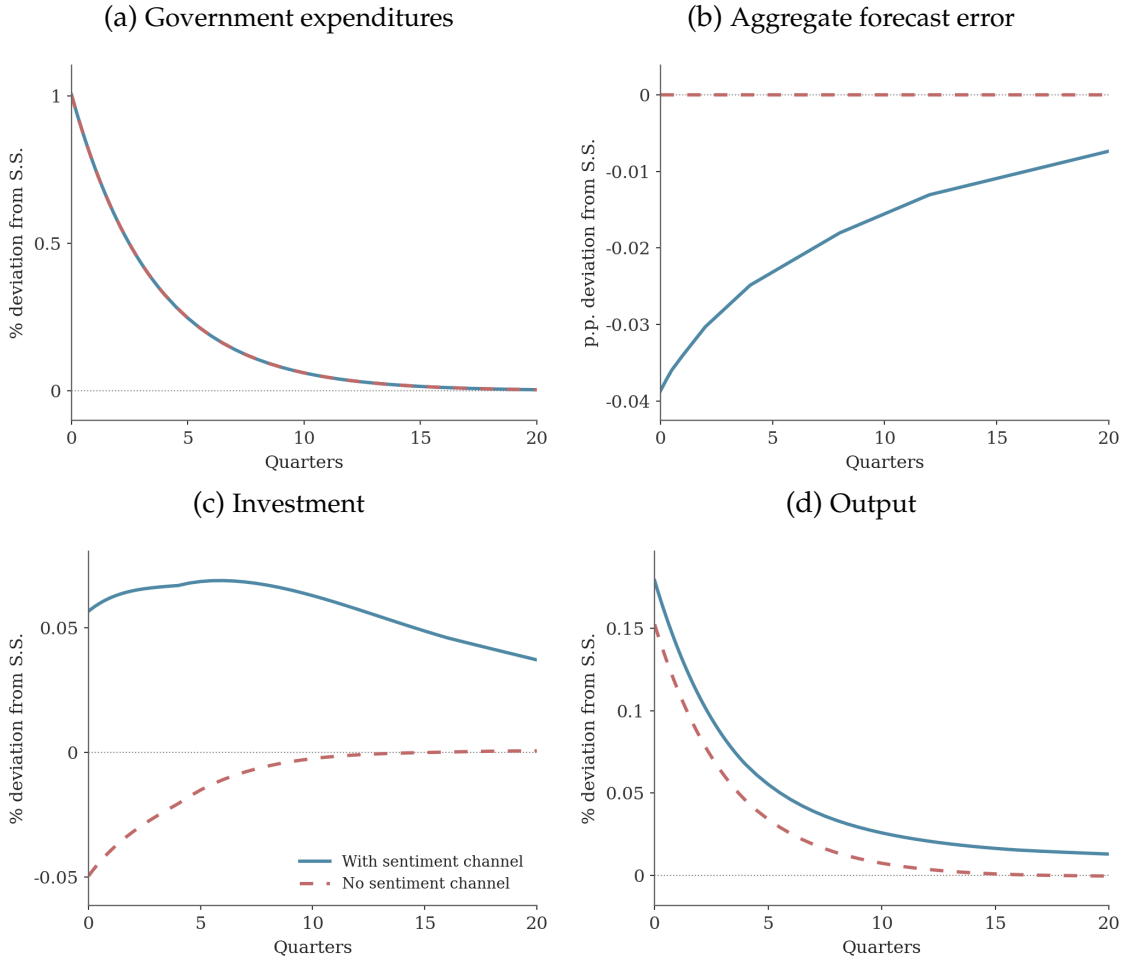
We now study the role of the sentiment channel in the transmission of fiscal policy. We consider the uniform procurement shock described in Section 5.1, which raises aggregate government consumption by approximately 1% on impact, as shown in Figure 6a.

The transmission of fiscal policy with sentiment Figure 6 plots the impulse responses to this shock. The red dashed line shows the benchmark case where sentiment does not respond to government spending in the aggregate. Because of this, the average forecast error that firms make in forecasting their own future sales does not respond to the government spending shock, as is clear from Figure 6b. Hence, the usual transmission mechanism is at play: a rise in Government expenditures induces an income effect on households' labor supply, thus increasing output. At the same time, as households try to smooth their consumption path, the interest rate rises, leading to a fall in investment. Overall, this crowding-out of investment makes the fiscal multiplier small: 0.77 on impact. This reflects the standard crowding-out mechanism in a New Keynesian model with capital and a Taylor rule; see for example [Christiano et al. \(2011\)](#).

The blue solid line depicts the responses in our baseline model, where sentiment endogenously responds to government spending. The rise in Government procurements induces firms to become overoptimistic about future private and public idiosyncratic demand, as depicted in Figure 6b. This boost in optimism is sufficiently strong to overturn the crowding-out effect of government spending on investment, yielding a positive investment response in general equilibrium: Sentiment has a “*crowding-in*” effect on investment, as is clear from Figure 6c. In turn, the positive investment response boosts aggregate demand and increases the five-year cumulative fiscal multiplier by 75%, as depicted in Figure 6d. Overall, the sentiment channel substantially amplifies the five-year cumulative government spending multiplier predicted by the model, from 0.72 in the baseline without sentiment to 1.26 once we introduce the sentiment channel.³⁰ This magnitude is broadly in line with the empirical estimates in the literature ([Ramey, 2016](#)), and matches recent estimates in [McClure and Yding \(2025b\)](#) who find a cumulative government spending multiplier of 1.2 over a 5 year window. Therefore, our sentiment channel offers an alternative to Heterogeneous Agents New Keynesian (HANK) models—which

³⁰Our baseline multiplier is the ratio of the present discounted value of the output response to that of government purchases over the first five years, using the annual household discount rate of 2%.

Figure 6: The sentiment channel of fiscal policy



Note: IRFs to a homogeneous procurement shock of 0.01 log points, raising real government purchases by approximately 1% on impact. The shock decays at an annual rate of 1.12, corresponding to quarterly persistence of 0.76. Blue solid lines allow sentiment to respond to the shock; red dashed lines suppress the aggregate sentiment response in the same calibrated economy. Aggregate forecast errors are set to zero in the no-sentiment case. The x-axis measures quarters since the shock. Panels (a), (c), and (d) show percentage deviations from steady state; panel (b) shows the response of the one-year-ahead forecast error in percentage points.

rely on incomplete markets and household borrowing constraints—to generate large multipliers (Auclert et al., 2024, Hagedorn et al., 2019). Our mechanism is instead rooted in firms’ belief formation, and has different implications for the behavior of investment in response to government spending shocks.

Figure D.4 in Section D.6 shows additional impulse responses for the other variables in the model. A noteworthy implication of the sentiment channel concerns consumption. In the benchmark New Keynesian economy, a rise in government spending raises the natural real interest rate; under a Taylor rule, policy tightens, and private consumption is

crowded out. With sentiment, firms' optimism raises investment and output, but also increases the initial response of interest rates. Consumption initially falls by 0.09%, compared with 0.06% without the aggregate sentiment channel. The stronger initial crowding-out is subsequently reversed as consumption recovers more rapidly in the sentiment economy. This is because the long-lasting boost in accumulated capital depresses the interest rate in the medium-run, thus crowding-in consumption. When nominal frictions are sufficiently strong in our model, the additional aggregate demand induced by sentiment is in fact able to crowd-in consumption on impact, as the interest rate does not respond as strongly after the shock.

To isolate the role of nominal frictions, we solve the model with flexible wages and plot the corresponding impulse responses in Figure D.5, with and without the sentiment channel. When wages are flexible, higher investment induced by optimism mainly raises the real interest rate without boosting demand, so consumption is crowded out more strongly. In other words, when nominal frictions are absent, the sentiment channel has no multiplier effects on private consumption. That being said, the overall effects on output and investment remain substantial.

5.4 State dependency

The amplification of fiscal policy through sentiment depends on firms' ability to translate optimism into real investment decisions, in other words firms need to have access to sufficiently cheap funds. When external finance becomes sufficiently costly, the investment response to improved expectations may therefore weaken. We assess this mechanism by comparing the contribution of sentiment to the fiscal multiplier under normal financing conditions and during a severe tightening of financial constraints.³¹

Starting from the calibrated stationary equilibrium, we introduce a temporary increase in the external-financing cost parameter, γ . The increase subsequently decays at the same rate as the government-demand shock. We retain the baseline nominal frictions and monetary policy rule. The financing disturbance alone reduces investment by 12.1% and output by 4.62% on impact, which roughly corresponds to the dynamics of the economy during the global financial crisis.

We then introduce the baseline government-demand shock simultaneously with the financing disturbance and compute the non-linear equilibrium paths with and without the aggregate sentiment response. In both cases, government demand follows the same baseline AR(1) process. To isolate the incremental effects of fiscal policy, we measure each

³¹See Appendix D.5 for details the shocks, counterfactual equilibrium paths, and multiplier calculations.

response relative to the equilibrium path with the financing disturbance alone. Government spending thus continues to support firms' sales and cash flows in both exercises; the comparison measures the additional equilibrium response generated by sentiment. As in the baseline, we calculate five-year cumulative multipliers as the ratio of discounted additional output to discounted additional government expenditure. Financial stress substantially attenuates the contribution of sentiment. Relative to normal financing conditions, the sentiment channel's contribution to the five-year cumulative fiscal multiplier is 25% smaller during the financial crisis compared to the baseline case. On impact, its contribution is 35% smaller.

These results suggest a potentially important source of variation in fiscal transmission over the business cycle: the sentiment channel is weaker when financing conditions limit firms' ability to act on improved expectations. Its aggregate strength may therefore depend on the financial conditions prevailing when spending occurs and on the distribution of firms. Our exercise illustrates this dependence through a temporary tightening of financing conditions, starting from a common stationary distribution. Studying how the distributions that arise over the business cycle further shape this channel is a natural next step.

These results also suggest that the appropriate mix of stabilization policies may vary with financial conditions over the business cycle. During downturns accompanied by tighter financing constraints, procurement expansions could be complemented by credit support, such as loan guarantees or targeted lending facilities. By easing these constraints, credit policy may help firms translate the optimism induced by public purchases into investment, reinforcing the sentiment channel precisely when financial stress weakens it. While our analysis does not characterize the optimal policy mix, this mechanism points to a potential complementarity between procurement and credit policy in stabilizing investment and output.

6 Conclusion

In this paper, we built a new dataset linking firms' sales forecasts, realized sales, and public procurement contracts to identify a sentiment channel of fiscal policy. Positive government spending shocks generate excessive optimism that extends beyond the directly affected market to other domains of demand, including export sales. We develop a tractable theory of cross-domain extrapolation to interpret this evidence. Embedded in a general equilibrium model, the resulting sentiment channel allows government spending to crowd in investment and substantially amplifies its effects on output. However the

magnitude of the sentiment channel might be state-dependent.

These findings raise several questions about expectation formation and policy design. The first concerns the distinction between responses to policy shocks and responses to policy rules. How much of the sentiment response to an unexpected procurement expansion would survive if that expansion were anticipated, routinely implemented, or systematically tied to economic conditions? Within our framework, predictable spending may generate no sentiment response if firms internalize the policy rule and correctly attribute the resulting demand to government intervention. This raises a behavioral version of the Lucas critique: the mapping from public spending into beliefs may itself depend on whether the variation in spending is a shock (discretionary policy) or a systematic response to economic conditions. This raises the question of the extent to which estimated economic responses to policy shocks can inform the design of systematic stabilization rules, as in (McKay and Wolf, 2023).

A second question is whether the sentiment channel can be influenced through the targeting of policy. Which firms should receive additional demand if the objective is to support investment during a recession? The relevant heterogeneity lies not only in how strongly firms revise their expectations, but also in their capacity to act on those revisions. A large confidence response may generate little investment when financing constraints bind, suggesting potential complementarities between procurement and access to finance. Our model provides a conceptual framework for studying how policy should be targeted when firms differ along both of these dimensions.

A third question concerns the communication of policy. Can transparent communication about the source, duration, and scope of a fiscal expansion shape how firms extrapolate from it? Building on information experiments that identify the effects of expectations on firms' decisions (Coibion et al., 2020), treatments that vary these features could help distinguish cross-domain extrapolation from learning about fundamentals. They could also reveal whether communication can support activity without inducing systematically mistaken beliefs.

Finally, this paper raises questions about the possibility of a confidence multiplier. Can the initial optimism generated by public spending propagate beyond its direct recipients, as their investment creates sales and favorable news for other firms? Such feedback would connect cross-domain extrapolation to the interaction between confidence and economic activity studied by Angeletos and Lian (2022). Optimism might also spread through communication and the adoption of shared narratives, as in Flynn and Sastry (2024). Finally, it could spread through overreaction to endogenous variables (Bianchi et al., 2024). These mechanisms could turn a localized demand intervention into a broader shift in expecta-

tions. Their strength would depend on production and customer relationships, the information firms observe, and the domains across which they extrapolate. Understanding these interactions would clarify when microeconomic belief responses are amplified or attenuated in the aggregate.

These questions open avenues for studying how policy shapes beliefs and how those beliefs, in turn, shape policy transmission. As a step in this direction, we show that a theory of sentiment grounded in firm-level expectations data can be integrated into a tractable macroeconomic framework and substantially reshape its predictions for the effects of fiscal policy.

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The Sentiment Channel of Fiscal Policy

Online Appendix

Marco Bellifemine

Adrien Couturier

Jacopo Tozzo

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A Appendix to Section 2

In this Section we present additional figures and tables, as well as robustness exercises, related to our empirical exercises. We also report the details of the reform to the Italian internal stability pact that we use as our natural experiment.

A.1 Summary statistics

Table A.1 presents summary statistics for the two merged datasets used in our analysis: the CADS-ANAC dataset (Panel A) and the INVIND-CADS-ANAC dataset (Panel B).

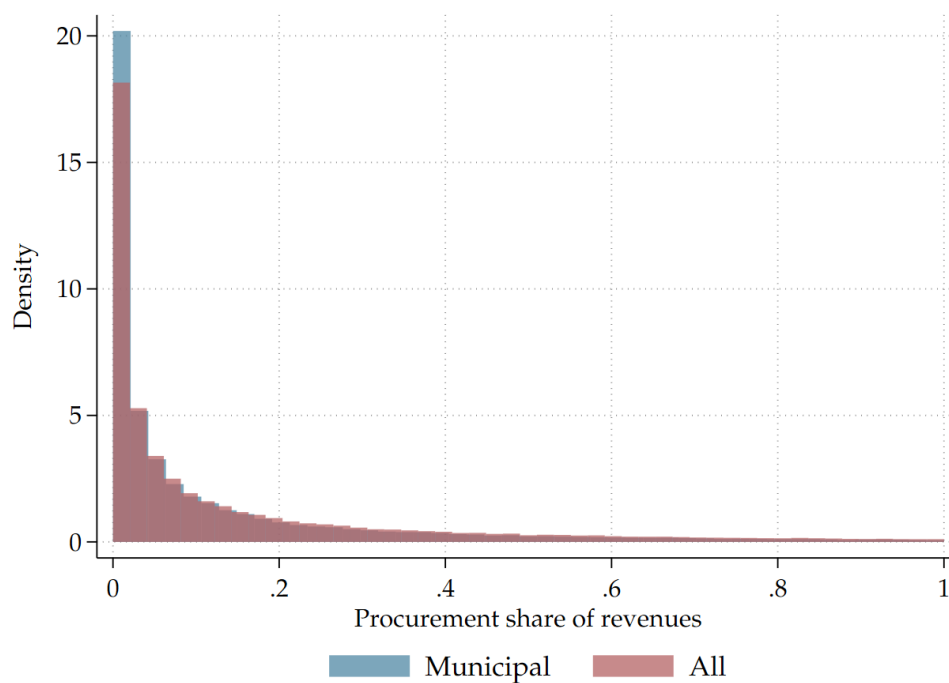
Figure A.1 plots the histogram of the share of procurement revenues over total revenues for all firms in our sample, respectively for procurements awarded by municipalities versus other public bodies. Figure A.2 plots the distribution (interquartile range and mean) of firms engaging in municipal and non-municipal procurement, versus that of firms that do not obtain procurement contracts from the public sector.

Table A.1: Summary statistics

	Obs.	Mean	p25	p50	p75
Panel A: ANAC-CERVED sample					
Employment	173,261	71	4	10	24
Revenues	175,541	17231	434	1,150	3202
Assets	175,541	26,415	519	1,352	3846
Leverage	175,149	12.57	2.57	5.08	11.33
Age	175,541	19	9	16	26
Panel B: ANAC-CERVED-INVIND sample					
Employment	5,344	134	16	27	49
Revenues	5,345	33,046	2,386	4,637	10,390
Assets	5,345	51,300	3,282	7,372	16,478
Leverage	5,345	10	3	5	8
Age	5,345	31	16	29	37
Sales forecast error	4,086	-2.47	-19.68	-1.60	15.58
Export sales forecast error	844	-14.39	-36.07	-6.16	12.68

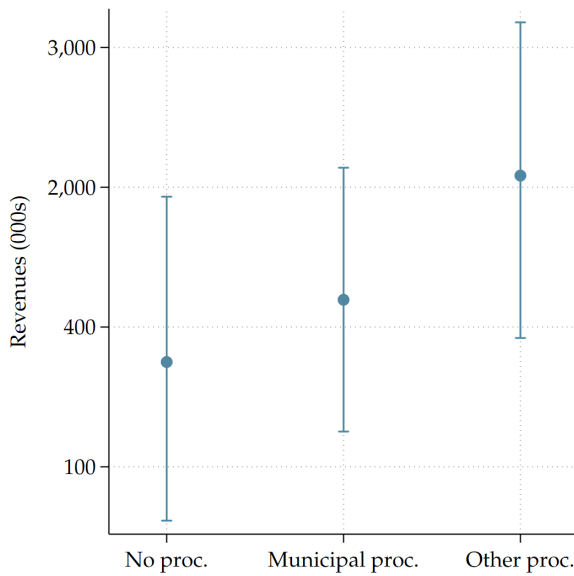
Note: Panel A reports summary statistics of the merged CADS and ANAC datasets, while panel B for the merged CADS, INVIND and ANAC datasets. Revenues and assets are expressed in thousands of euros, employment in units, age in years, and leverage is defined as assets over equity.

Figure A.1: Share of Procurement in Firms' Revenues

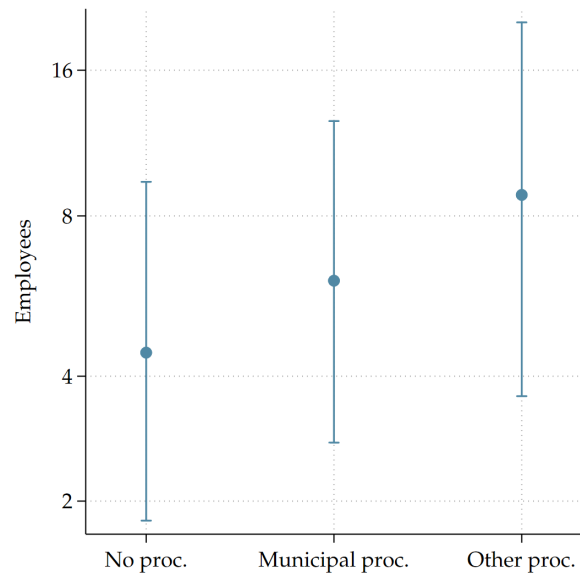


Note: Share of procurement contracts in firms' revenues for municipal contracts and all contracts. The sample of firms is conditional on having a public procurement contract for the red bars and conditional on having a municipal public procurement contract for the blue bars.

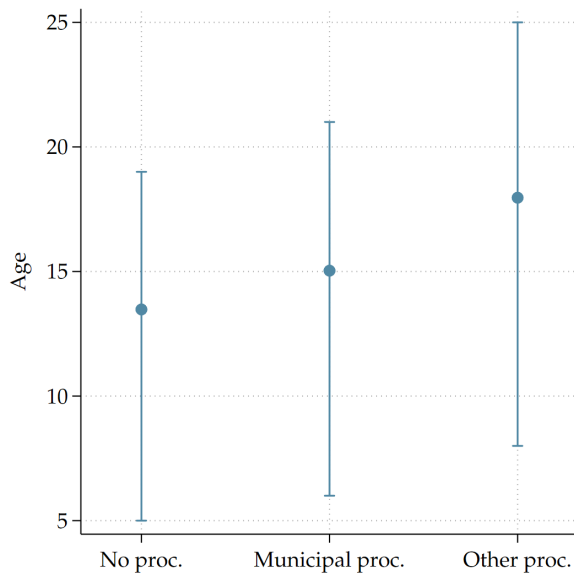
Figure A.2: Characteristics of Municipal Procurement, Non-Procurement and Non-Municipal procurement Firms



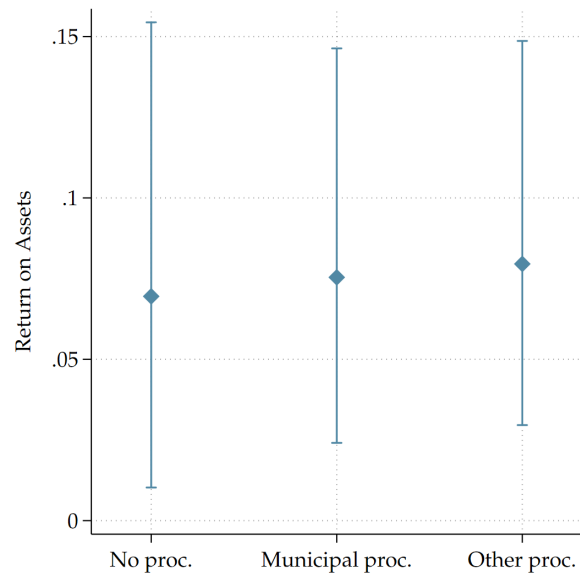
(a) Revenues



(b) Employment



(c) Age



(d) Return on assets

Note: characteristics of firms without procurement contracts, with municipal procurement contracts and with procurement contracts with non municipal bodies. The bands show the interquartile range, while squares show the mean and diamonds the median of the distribution. For readability, we plot the distribution of revenues and employment in logarithms. We define return on assets as Earnings Before Interest, Taxation, Depreciation and Amortization over total assets.

A.2 Stability Pact

Art. 31 of Legge 12 Novembre 2011, n.183

For the purposes of determining the specific financial balance objective, provinces and municipalities with a population greater than 1,000 inhabitants shall apply, to the average current expenditure recorded in the years 2006–2008 (for the year 2012), in the years 2007–2009 (for the year 2013), in the years 2009–2011 (for the year 2014), and in the years 2010–2012 (for the years 2015 to 2018), as derived from the certified financial statements, the following percentages:

a) for provinces, the percentages are 16.5 percent for the year 2012, 18.8 percent for the year 2013, 19.25 percent for the year 2014, 17.20 percent for the year 2015, and 18.03 percent for the years 2016, 2017, and 2018;

b) for municipalities with a population greater than 5,000 inhabitants, the percentages are 15.6 percent for the year 2012, 14.8 percent for the year 2013, 14.07 percent for the year 2014, 8.60 percent for the year 2015, and 9.15 percent for the years 2016, 2017, and 2018;

c) for municipalities with a population between 1,001 and 5,000 inhabitants, the percentages are 12.0 percent for the year 2013, 14.07 percent for the year 2014, 8.60 percent for the year 2015, and 9.15 percent for the years 2016, 2017, and 2018.

Art .1, comma 532 of Legge 27 Dicembre 2013, n. 147

In paragraph 2 of Article 31 of Law No. 183 of 12 November 2011, the following amendments are made:

a) the words: “and recorded in the years 2007–2009, for the years from 2013 to 2016,” are replaced by the following: “, recorded in the years 2007–2009, for the year 2013, and recorded in the years 2009–2011 for the years from 2014 to 2017,”;

b) the words: “and 18.8 percent for the year 2013 and thereafter” are replaced by the following: “, 18.8 percent for the year 2013, 19.25 percent for the years 2014 and 2015, and 20.05 percent for the years 2016 and 2017”;

c) the words: “and 14.8 percent for the year 2013 and thereafter” are replaced by the following: “, 14.8 percent for the year 2013, 14.07 percent for the years 2014 and 2015, and 14.62 percent for the years 2016 and 2017”;

d) the words: “and 14.8 percent for the years from 2014 to 2016” are replaced by the following: “, 14.07 percent for the years 2014 and 2015, and 14.62 percent for the years 2016 and 2017.”

Art. 1, comma 489 of Legge 23 Dicembre 2014, n.190

In paragraph 2 of Article 31 of Law No. 183 of 12 November 2011, as subsequently amended, the following modifications are made:

a) the words: “and recorded in the years 2009–2011 for the years from 2014 to 2017” are replaced by the following: “recorded in the years 2009–2011, for the year 2014, and recorded in the years 2010–2012, for the years from 2015 to 2018”;

b) in letter a), the words: “, 19.25 percent for the years 2014 and 2015, and 20.05 percent for the years 2016 and 2017” are replaced by the following: “, 19.25 percent for the year 2014, 17.20 percent for the year 2015, and 18.03 percent for the years 2016, 2017, and 2018”;

c) in letter b), the words: “, 14.07 percent for the years 2014 and 2015, and 14.62 percent for the years 2016 and 2017” are replaced by the following: “, 14.07 percent for the year 2014, 8.60 percent for the year 2015, and 9.15 percent for the years 2016, 2017, and 2018”;

d) in letter c), the words: “, 14.07 percent for the years 2014 and 2015, and 14.62 percent for the years 2016 and 2017” are replaced by the following: “, 14.07 percent for the year 2014, 8.60 percent for the year 2015, and 9.15 percent for the years 2016, 2017, and 2018”;

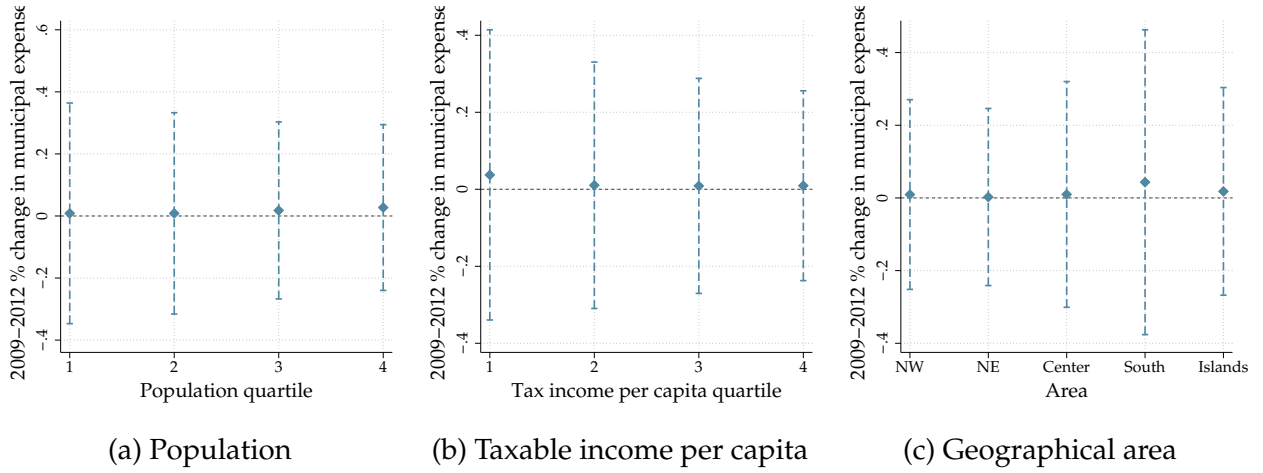
e) the following sentences are added at the end: “By decree of the Minister of Economy and Finance, subject to agreement within the State-City and Local Autonomies Conference, the objectives of each entity referred to in this paragraph may be redefined—on the proposal of ANCI and UPI—by 31 January 2015, without prejudice to the overall objective for the sector, also taking into account the additional functions assigned to metropolitan cities and the increased costs related to natural disasters, safety interventions for school buildings and the territory, the exercise of the role of lead entity, as well as the costs arising from final judgments resulting from expropriation procedures or disputes related to structural failures. After this deadline, the objectives of each entity shall be those determined by applying the percentages referred to in letters a), b), and c) of this paragraph.”

A.3 Balance tests

We run balance tests for our shift-share instrument. We check whether the municipal-level shift that we construct in Section 2 shows any systematic difference across observable characteristics of municipalities. To do so, we split municipalities into groups based on three observables: population, taxable income per capita, and geographical area. We then compute the average shift for municipalities in each group, as well as 95% confidence intervals. The results are reported in Figure A.3. There is no evidence of any systematic difference in the distribution of the shift along municipal observables—both in terms of averages and confidence bands.

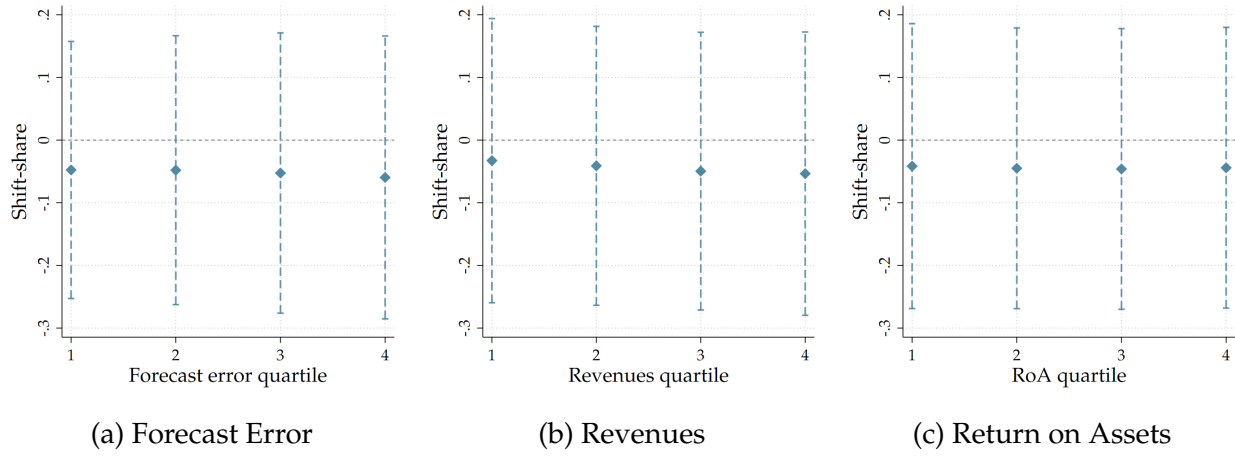
Figure A.4 reports results for balance tests at the firm level. We split firms into quartiles based on their pre-2015 forecast error, revenues and return on assets. For each group we then compute the average shift-share z_i , as well as the 95% confidence bands. Overall, there is no evidence of any systematic difference in the distribution of our shift-share measure across firms' observables.

Figure A.3: Balance test—municipal level



Note: balance test for the shift g_m defined in (2). We split municipalities in quartiles based on population, taxable income per capita and geographical area. For each quartile we plot the average shift, together with 95% confidence bands.

Figure A.4: Balance test—firm level



Note: balance test for the shift-share z_i defined in Section 2.4. We split firms in quartiles based on pre-2015 forecast error, revenues and return on assets. For each quartile we plot the average shift-share, together with 95% confidence bands.

A.4 Descriptive statistics for the shift-share

In this Section we report some descriptive statistics for the shifts discussed in Section 2, as well as the associated weights. Figure A.5 plots the distribution of municipal shifts g_m as defined in (2). Figure A.6 plots the distribution of average municipal shares s_m defined in (5) for both our CADS and CADS-INVIND merged datasets. Figure A.7 does the same for the distribution of the shift-share instrument z_i , as defined in (4). Finally, Figure A.8 plots the geographical distribution of our shifts, as well as the average exposure shares, over space.

Figure A.5: Distribution of Shifts

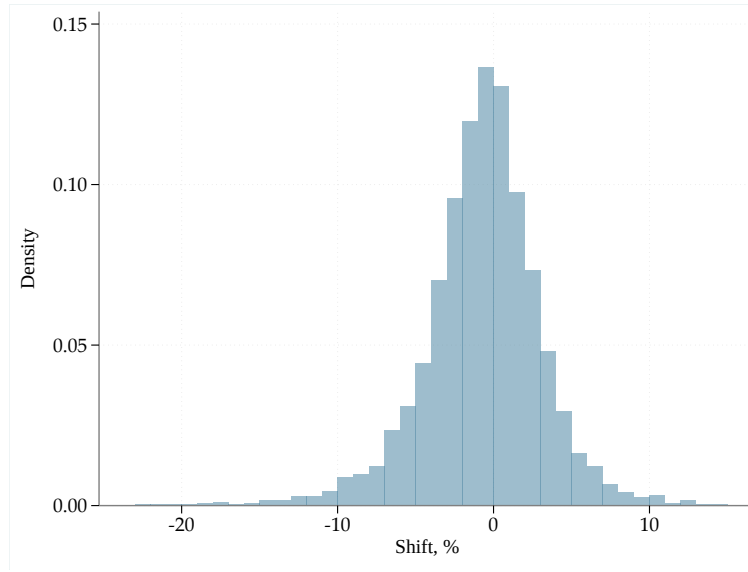
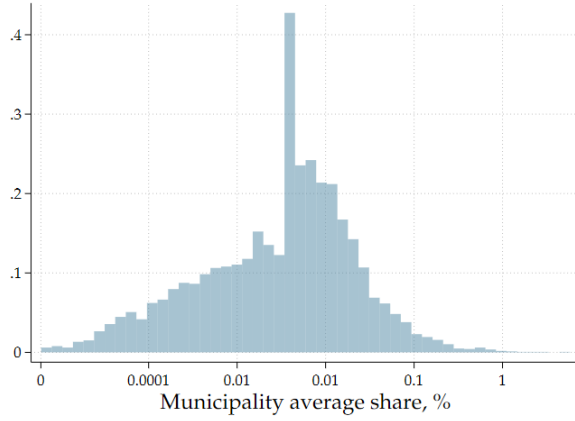
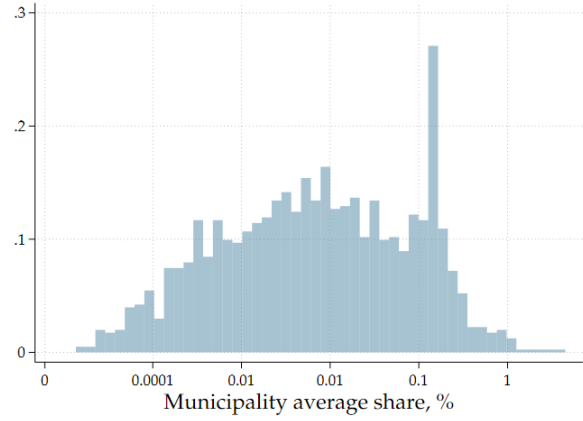


Figure A.6: Distribution of Average Municipal Shares



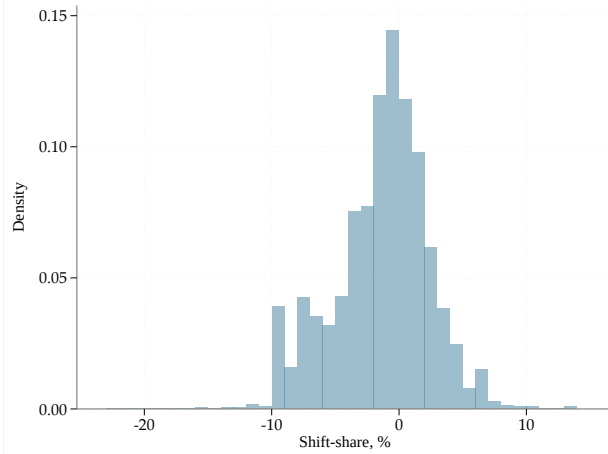
(a) CADS dataset



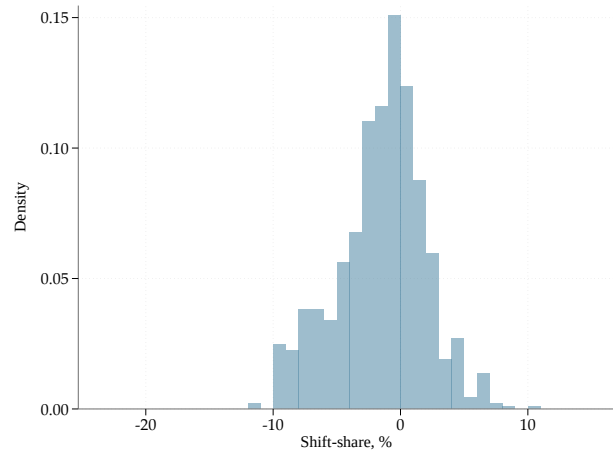
(b) CADS-INVIND dataset

Note: distribution of average municipal shares s_m defined in (5) for the merged CADS-ANAC sample and the merged CADS-INVIND-ANAC sample.

Figure A.7: Distribution of Shift-Share



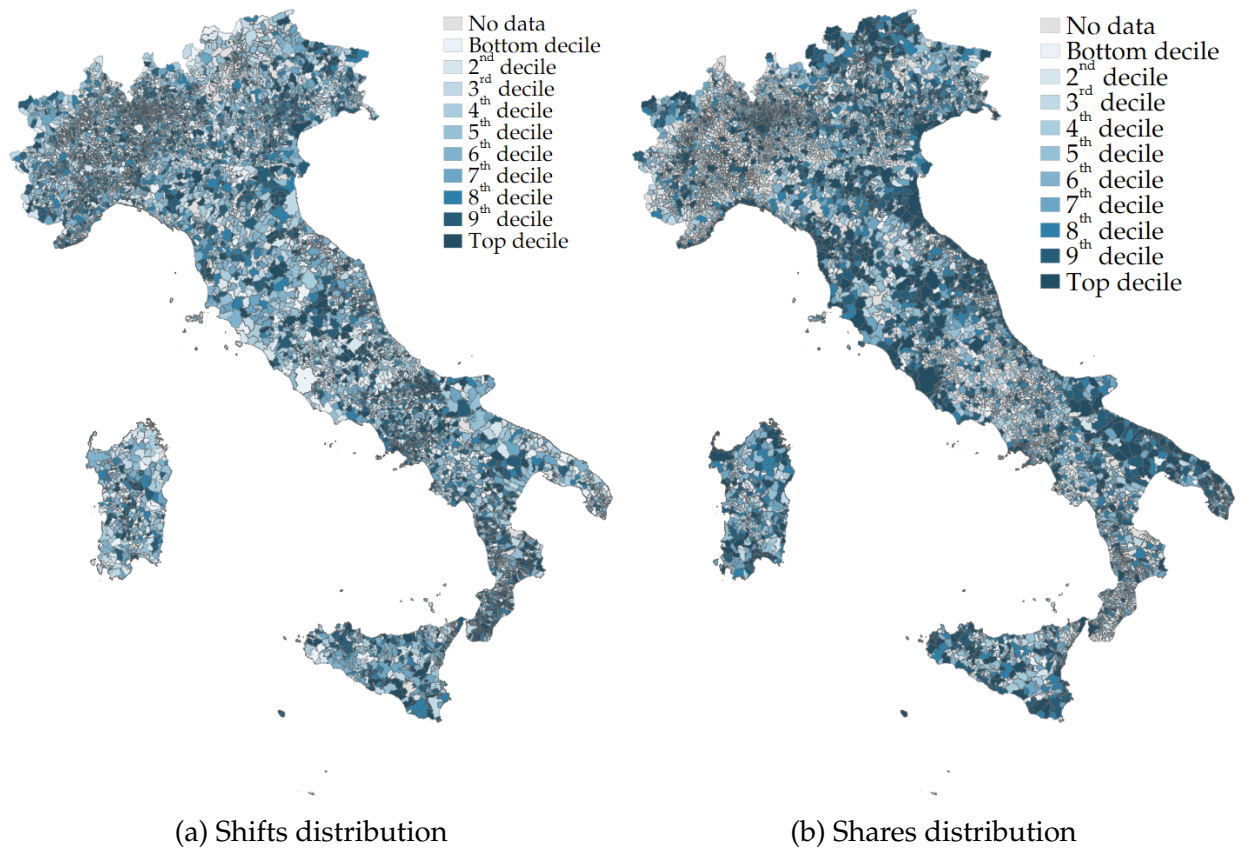
(a) CADS dataset



(b) CADS-INVIND dataset

Note: distribution of shift-share instrument z_i defined in (4) across firms for the merged CADS-ANAC sample and the merged CADS-INVIND-ANAC sample.

Figure A.8: Geographical Distribution of our Shift-Share Instrument



Note: panel (a) plots the geographical distribution of the shifts g_m across municipalities. Panel (b) plots the geographical distribution of the share s_{im} across municipalities.

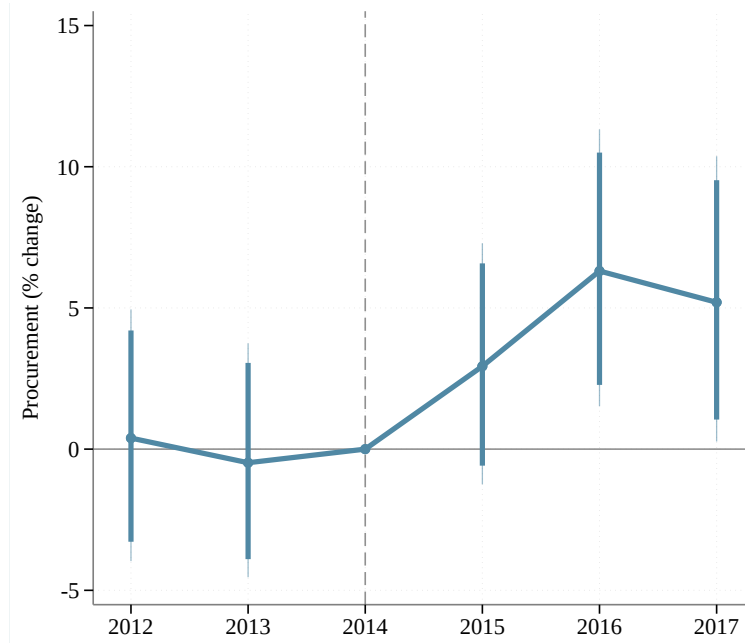
A.5 Robustness

Response of municipal procurement expenses Figure A.9 plots the dynamic response of municipal procurement expenses to our reform. We estimate the following regression:

$$\log(\text{ProcExp}_{mt}) = \alpha_m + \delta_t + \sum_{s=2012, \neq 2014}^{2017} \beta_s (\mathbb{1}_{s=t} \times g_m) + \varepsilon_{mt}, \quad (\text{A.1})$$

where α_m and δ_t denote respectively municipality and year fixed effects, and g_m is the shift measure defined in (2), normalized to have standard deviation of one. Consistent with our identifying assumption, municipalities that experienced a larger increase in fiscal space as a result of the reform increased their procurement expenses by more following 2015, with a one-standard-deviation increase in g_m leading to a 5% increase in procurement spending. The estimated coefficients prior to 2015 are both economically and statistically insignificant, showing no signs of differential behavior or anticipation.

Figure A.9: Response of Municipal Procurement Expenses

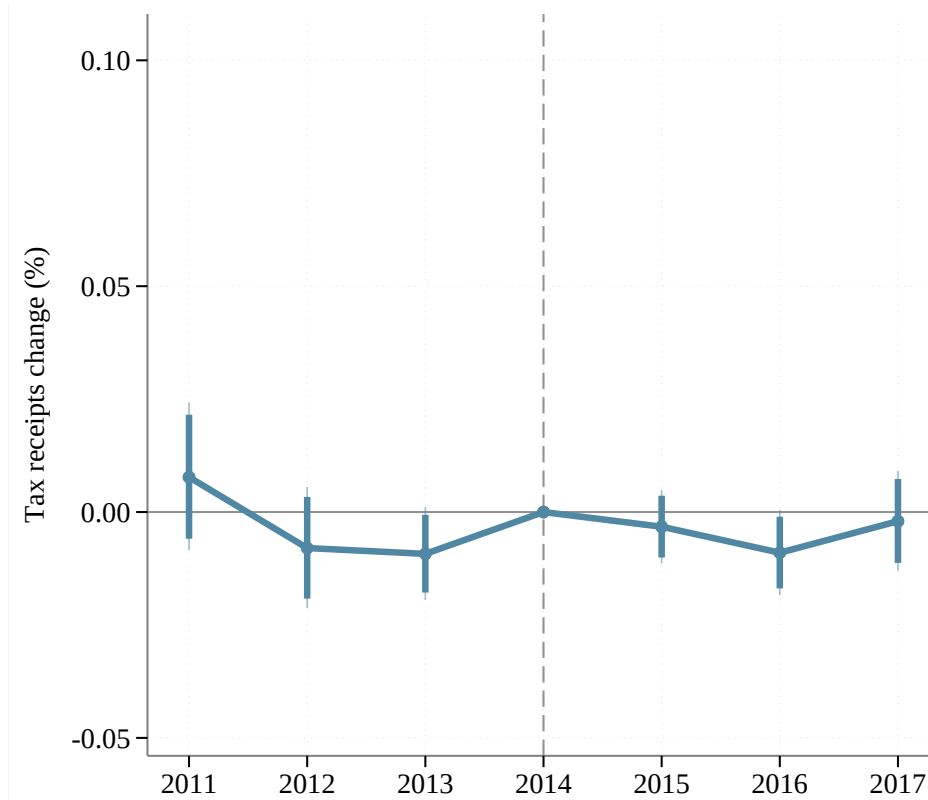


Note: estimated $\hat{\beta}_a$ from the event study design in (A.1). The y-axis denotes the percentage change (relative to the 2014 reference year) in municipal procurement expenditures. Shift measure g_m normalized to have standard deviation of one. Wide and thin lines represent 90% and 95% confidence bands, respectively. We cluster standard errors at the municipality level.

Response of municipal tax revenues In Figure A.10 we run the same specification as in (A.1), but use log municipal tax revenues as the dependent variable. The behavior

of tax revenues is both economically and statistically insignificant both before and after the reform. First, this lends further support to the parallel trend identifying assumption. Second, it also confirms that the main way via which municipalities responded to less stringent surplus targets is via increased expenses, as opposed to reduced taxes.

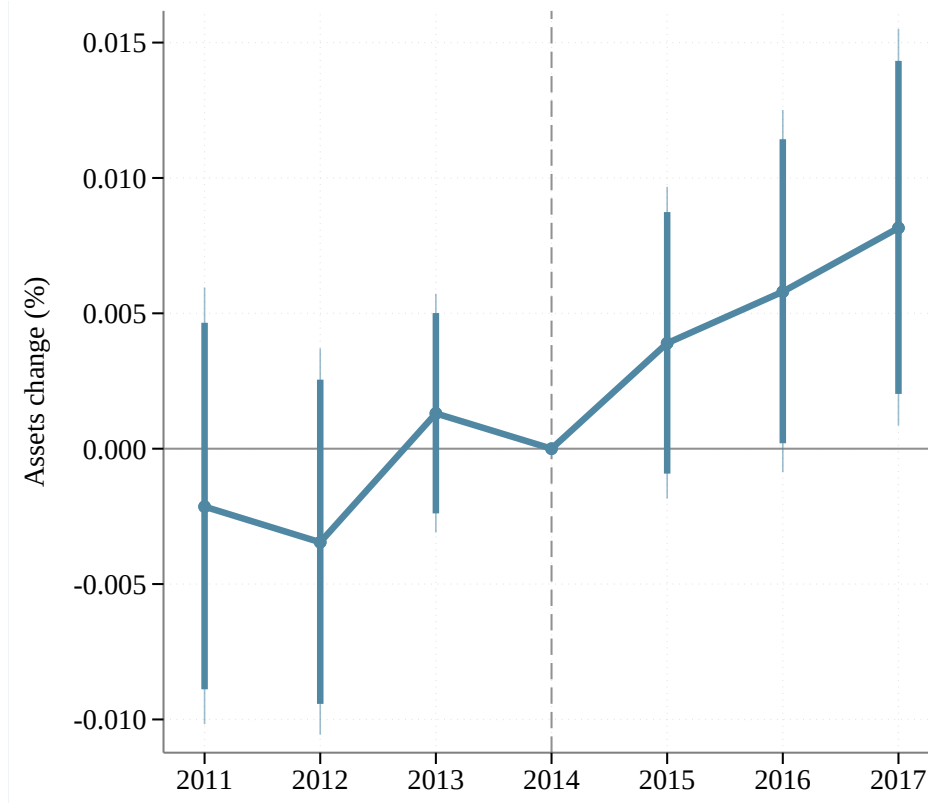
Figure A.10: Response of Municipal Tax Revenues



Note: response of tax revenues from the event study design in (A.1). The y-axis denotes the percentage change (relative to the 2014 reference year) in municipal tax revenues. Wide and thin lines represent 90% and 95% confidence bands, respectively. Shift measure g_m normalized to have standard deviation of one. We cluster standard errors at the municipality level.

Response of firm investment In Figure A.11 we run the same specification as in (6), but use log total firm assets as the dependent variable. As in the case of revenues, the estimated coefficients are insignificant prior to the reform, while following 2015 firms more exposed to municipalities that saw a larger increase in fiscal space increased their assets, with a one-standard-deviation increase in z_i leading to an increase in firm assets of roughly 1%.

Figure A.11: Response of Firms' Assets



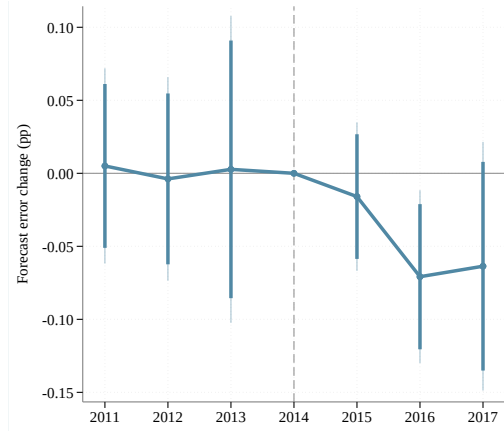
Note: estimated $\hat{\beta}_t$ from the event study design in (6) based on the full CADS sample, using total assets as the outcome variable. The y-axis denotes the percentage change (relative to the 2014 reference year) in total assets. Wide and thin lines represent 90% and 95% confidence bands, respectively, based on exposure robust standard errors.

Forecast error response We now perform a battery of robustness tests for the response of firms' forecast errors presented in Figure 4. In Figure A.12a, we run the same event study as in (7), but include province $\times$ year fixed effects, to control for potential local general equilibrium spillovers. Figure A.12b shows results for the case in which we do not winsorize our dependent variable (the forecast error). Similarly, in Figure A.12c we plot our estimates when we do not trim the shift-share variable. Figure A.12d reports the results of (7) when we remove the sector $\times$ year fixed effects from the controls. Finally, Figures A.12e and A.12f show results for the case in which we define our sector $\times$ time fixed effects at the 2 and 4 digits level, respectively. Across all specifications, the core pattern we identify in the main text remains robust.

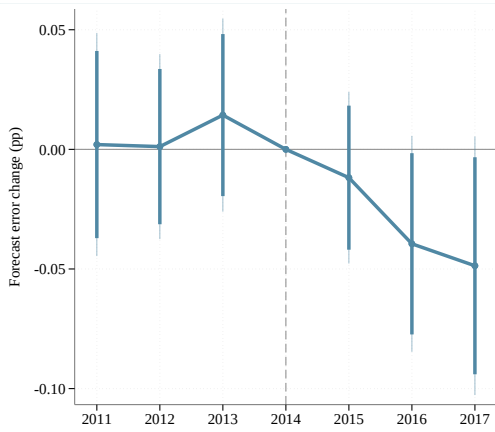
Figure A.12: Response of Forecast Error—Robustness



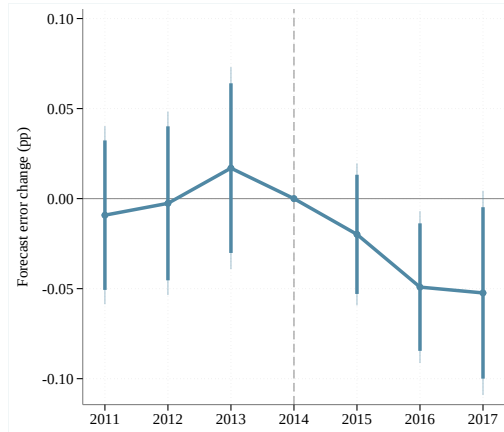
(a) Province $\times$ Time Fixed Effects



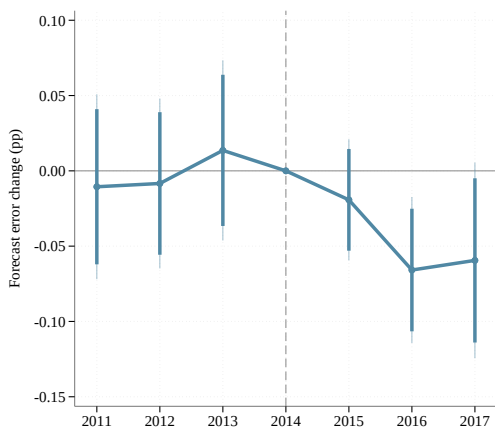
(b) No Winsorizing of Forecast Errors



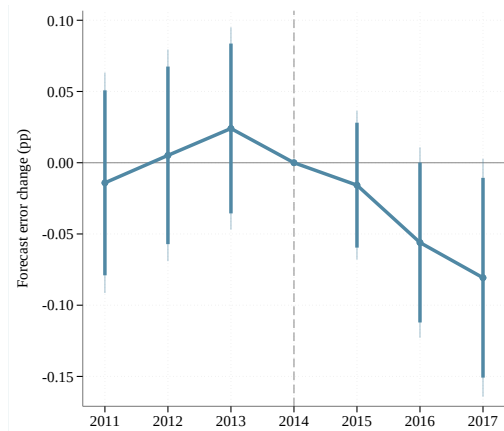
(c) No Trimming of Shift-Share



(d) No Sector $\times$ Time Fixed Effects



(e) 2-digit Sector $\times$ Time Fixed Effects



(f) 4-digit Sector $\times$ Time Fixed Effects

Note: robustness for the estimated $\hat{\beta}_h$ from the event study design in (7). The y-axis denotes the change in forecast error (relative to the base year 2014) in percentage points. Wide and thin lines represent 90% and 95% confidence bands, respectively, based on exposure robust standard errors.

Decomposition of price and quantity forecast error response Table A.2 reports the estimates of the price and quantity forecast error response to our government spending shock. It does so by reporting the $\hat{\beta}$ from the following regression:

$$\mathcal{E}_{it}^y = \alpha_i + \gamma_{s(i)t} + \beta z_i \times Post_t + \varepsilon_{it}, \quad y \in \{P, Q\} \quad (\text{A.2})$$

where $\mathcal{E}_{it}^P$ and $\mathcal{E}_{it}^Q$ are the forecast errors on the price and quantity changes, respectively, defined along the same lines of (1). To construct the forecast error on the price change we leverage a feature of our survey, that directly asks respondents to report their expectation of the change in the average price over the next 12 months as well as the realized average change in prices over the past 12 months. To construct a measure of forecast error on the quantity change we do a log-linearization of firms realized and expected sales and extract realized and expected quantity changes as a residual:

$$\hat{Q}_t = \hat{Y}_t - \hat{P}_t \quad (\text{A.3})$$

Where $\hat{X}_t$ denotes the log change in variable X and Y_t denotes revenues.

Table A.2: Response of Quantity and Price Forecast Errors

	Quantities	Prices
$Post_t \times z_i$	-0.407** (0.121)	-0.005 (0.012)
Observations	869	872
Firm FEs	✓	✓
Sector-Time FEs	✓	✓

Note: estimates from (A.2). Exposure robust standard errors in parenthesis. See the discussion of (A.3) for a description of how we construct the price and quantity forecast error.

B Appendix to Section 3

B.1 A simple AR(1) example

To build intuition, we consider a simple case in discrete time where both fundamentals follow AR(1) processes:

$$z_{t+1} = \rho_z z_t + \varepsilon_{t+1}^z, \quad \varepsilon_{t+1}^z \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_z^2) \quad (\text{B.1})$$

$$g_{t+1} = \rho_g g_t + \varepsilon_{t+1}^g, \quad \varepsilon_{t+1}^g \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_g^2) \quad (\text{B.2})$$

Where $x_t = \alpha_z z_t + \alpha_g g_t$ like in the main text. In a similar way to [Bordalo et al. \(2018\)](#), we define in the rest of our analysis the background context as the value x_t would take absent current shocks, so that $\hat{x} - \mathcal{B} = \alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g$, and $\hat{x}$ is the value of x in period t . We define the distortion in a similar fashion to the continuous time version

$$\tilde{h}_t(y_{t+1}) = h(y_{t+1} | y_t = \hat{y}_t) \cdot \underbrace{\left(\frac{h(y_{t+1} | x_t = \hat{x}_t)}{h(y_{t+1} | x_t = \mathcal{B}_t)} \right)^\chi}_{\text{representativeness ratio}} \frac{1}{Z_x(\chi)} \quad (\text{B.3})$$

Given this Gaussian setting, we can now derive the perceived distributions under XDE.

Proposition 4 (Cross-Domain Extrapolation with AR(1) Fundamentals). *When the fundamentals z_t and g_t follow the process in (B.1) and (B.2), the perceived distribution of z_{t+1} and g_{t+1} under cross-domain extrapolation are:*

$$z_{t+1} | z_t, g_t \stackrel{\chi}{\sim} \mathcal{N} \left(\rho_z z_t + \frac{\sigma_z^2}{\sigma_{z|x}^2} \chi \beta_z (\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g), \sigma_z^2 \right) \quad (\text{B.4})$$

$$g_{t+1} | z_t, g_t \stackrel{\chi}{\sim} \mathcal{N} \left(\rho_g g_t + \frac{\sigma_g^2}{\sigma_{g|x}^2} \chi \beta_g (\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g), \sigma_g^2 \right) \quad (\text{B.5})$$

where $\beta_z = \frac{\text{Cov}(z_{t+1}, x_t)}{\text{Var}(x_t)}$ is the OLS coefficient of a regression of z_{t+1} on x_t , $\sigma_{z|x}^2 = \frac{\sigma_z^2}{1 - \rho_z^2} \left(1 - \frac{\rho_z^2 \alpha_z^2 \sigma_z^2 / (1 - \rho_z^2)}{\alpha_z^2 \sigma_z^2 / (1 - \rho_z^2) + \alpha_g^2 \sigma_g^2 / (1 - \rho_g^2)} \right)$ is the variance of z_{t+1} conditional on x_t , $\beta_g = \frac{\text{Cov}(g_{t+1}, x_t)}{\text{Var}(x_t)}$, and $\sigma_{g|x}^2 = \frac{\sigma_g^2}{1 - \rho_g^2} \left(1 - \frac{\alpha_g^2 \rho_g^2 \sigma_g^2 / (1 - \rho_g^2)}{\alpha_z^2 \sigma_z^2 / (1 - \rho_z^2) + \alpha_g^2 \sigma_g^2 / (1 - \rho_g^2)} \right)$ is the variance of g_{t+1} conditional on y_t .

Proof. See Section B. □

Proposition 4 delivers three key results. First, cross-domain extrapolation preserves normality of perceived distributions. Second, it does not distort perceived variances.

Third, the only bias introduced by XDE is in the perception of expected values. The following corollary shows that this first-moment distortion admits a useful decomposition.

Corollary 1 (Perceived First-Moment under XDE with AR(1) Fundamentals). *When the fundamentals z_t and g_t follow the process in (B.1) and (B.2), the perceived expected value of z_{t+1} and g_{t+1} under cross-domain extrapolation can be decomposed as:*

$$\mathbb{E}_t^\chi(z_{t+1}) = \mathbb{E}_t(z_{t+1}) + \chi\beta_z \frac{\sigma_z^2}{\sigma_{z|x}^2} (\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g) \quad (\text{B.6})$$

$$\mathbb{E}_t^\chi(g_{t+1}) = \mathbb{E}_t(g_{t+1}) + \chi\beta_g \frac{\sigma_g^2}{\sigma_{g|x}^2} (\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g) \quad (\text{B.7})$$

Where $\mathbb{E}_t(z_{t+1}) \equiv \rho_z z_t$ and $\mathbb{E}_t(g_{t+1}) \equiv \rho_g g_t$ denote expectations under rational expectations.

Corollary 1 provides a decomposition of cross-domain beliefs into the rational expectations benchmark plus an additive bias. This bias is the product of four components: (i) the behavioral parameter χ , governing the degree of cross-domain extrapolation, (ii) the OLS coefficient β capturing the strength of the co-movement between the target variable and the salient outcome, (iii) a variance ratio adjusting for relative precisions, and (iv) the total surprise in the salient variable relative to the background, $\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g$. Each component has a distinct psychological or statistical meaning. χ is the behavioural parameter: it measures the strength of the representativeness heuristic, i.e. how much the agent lets resemblance override the objective conditional distribution. β is purely mnemonic: it is the slope the agent has learned from past co-movements between x and the target variable z ; it captures how diagnostic the agent's memory of x is for z . Finally, the variance ratio $\frac{\sigma_z^2}{\sigma_{z|x}^2}$ is a purely statistical scaling that tells the agent how precisely x pins down z : the more precise the link, the larger the leverage of any surprise in x .

Finally, note that cross-domain extrapolation for the fundamentals z_t, g_t generates standard diagnostic expectations for the salient outcome x_t . (B.4) and (B.5) together imply that the perceived expected value of x_{t+1} can be expressed as:

$$\mathbb{E}_t^\chi(x_{t+1}) = \mathbb{E}_t(x_{t+1}) + \tilde{\chi}(\alpha_z \varepsilon_t^z + \alpha_g \varepsilon_t^g)$$

Where $\tilde{\chi} \equiv \chi \left[\alpha_z \frac{\sigma_z^2}{\sigma_{z|x}^2} \beta_z + \alpha_g \frac{\sigma_g^2}{\sigma_{g|x}^2} \beta_g \right]$.

B.2 Proof of Proposition 4

In equation (B.3) we define Z as

$$Z_x(\chi) \equiv \int h_x(s|x) \left(\frac{h_x(s|y = \hat{y})}{h_x(s|y = y_0)} \right)^\chi ds$$

Is a normalizing constant that guarantees the density integrates to 1. For convenience we can rewrite the expression above as

$$h_x^\chi(x'|\mathcal{I}) = \frac{1}{Z_x(\chi)} h_x(x'|x) e^{\chi T_x(x')}$$

Where

$$\begin{aligned} T_x(x') &\equiv \log \left(\frac{h_x(x'|y = \hat{y})}{h_x(x'|y = y_0)} \right) \\ &= \log(h_x(x'|y = \hat{y})) - \log(h_x(x'|y = y_0)) \end{aligned}$$

Now consider the discrete-time AR(1) gaussian setting in the main text:

$$\begin{aligned} z_{t+1} &= \rho_z z_t + \varepsilon_{t+1}^z \\ g_{t+1} &= \rho_g g_t + \varepsilon_{t+1}^g \\ x_t &= \alpha_z z_t + \alpha_g g_t \end{aligned}$$

Where $\varepsilon_{t+1}^z \sim \mathcal{N}(0, \sigma_z^2)$ iid and $\varepsilon_{t+1}^g \sim \mathcal{N}(0, \sigma_g^2)$. From this it follows that $\mathbb{V}_z = \frac{\sigma_z^2}{1-\rho_z^2}$ and $\mathbb{V}_g = \frac{\sigma_g^2}{1-\rho_g^2}$. Moreover $\text{Cov}(z_t, x_t) = \alpha_z \mathbb{V}_z$. Then it follows that

$$\begin{bmatrix} z_t \\ x_t \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbb{V}_z & \alpha_z \mathbb{V}_z \\ \alpha_z \mathbb{V}_z & \alpha_z^2 \mathbb{V}_z + \alpha_g^2 \mathbb{V}_g \end{bmatrix} \right)$$

Remember that if $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$. Then

$$X_1 \mid X_2 = a \sim \mathcal{N}(\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (a - \mu_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}).$$

Thus, it follows that

$$z_t \mid x_t \sim \mathcal{N} \left(\frac{\alpha_z \mathbb{V}_z}{\alpha_z^2 \mathbb{V}_z + \alpha_g^2 \mathbb{V}_g} x_t, \mathbb{V}_z - \frac{\alpha_z^2 \mathbb{V}_z^2}{\alpha_z^2 \mathbb{V}_z + \alpha_g^2 \mathbb{V}_g} \right)$$

Now let $\beta = \frac{\alpha_z \mathbb{V}_z}{\alpha_z^2 \mathbb{V}_z + \alpha_g^2 \mathbb{V}_g}$. Then

$$z_t \mid x_t \sim \mathcal{N}(\beta x_t, \mathbb{V}_z(1 - \alpha_z \beta))$$

Moreover

$$z_{t+1} \mid x_t \sim \mathcal{N}(\rho_z \beta x_t, \mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta))$$

So we can see that the variance does not depend on y_t . Let $\mathcal{F}_t \equiv \sigma\{y_s : s \leq t\}$. The smallest σ -field in which the realized y_t is measurable is $\mathcal{F}_t$, while the smallest σ -field in which the no-news counterfactual y_t^0 is measurable is $\mathcal{F}_{t-1} \subset \mathcal{F}_t$. Crucially, both y_t and y_t^0 become deterministic elements of $\mathcal{F}_t$ once t is reached so they can be plugged into any $\mathcal{F}_t$ -measurable function without ambiguity. The likelihood tilt is computed within a single filtration, $\mathcal{F}_t$, the numerator and denominator densities condition on the same σ -field at the moment they are compared, even though the benchmark argument was chosen using the smaller σ -field $\mathcal{F}_{t-1}$. See footnote 8 in the original Journal of Finance diagnostic expectation paper to see that this is indeed what Bordalo et al. assume. So that means that our numerator and denominator share equal variances by design.

Given this we can define

$$\begin{aligned} T_y(y') &\equiv \log \left(\frac{h_y(y' \mid x = x_t)}{h_y(y' \mid x = \mathcal{B})} \right) \\ &= \log(h_y(y' \mid x = x_t)) - \log(h_y(y' \mid x = \mathcal{B})) \end{aligned}$$

In our specific AR(1) Gaussian case. The tilting is:

$$\begin{aligned} T_y(y') &= -\frac{(y' - \rho_z \beta x_t)^2}{2\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} + \frac{(y' - \rho_z \beta \mathcal{B})^2}{2\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} \\ &= \frac{1}{2\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} \left[-y'^2 + 2y' \rho_z \beta x_t - \rho_z^2 \beta^2 x_t^2 + y'^2 - 2y' \rho_z \beta \mathcal{B} + \rho_z^2 \beta^2 \mathcal{B}^2 \right] \\ &= \frac{1}{2\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} \left[2y'(\rho_z \beta x_t - \rho_z \beta \mathcal{B}) + (\rho_z^2 \beta^2 \mathcal{B}_0^2 - \rho_z^2 \beta^2 x_t^2) \right] \\ &= \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} y' + \frac{\rho_z^2 \beta^2}{2\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} (\mathcal{B}^2 - x_t^2) \end{aligned}$$

And the rational density is

$$h_y(y'|z_t) = \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(y' - \rho_z z_t)^2}{2\sigma_z^2}} \quad (\text{B.8})$$

From which it follows that $Z_y(\chi)$ is

$$\begin{aligned} Z_y(\chi) &\equiv \int h_y(s|z_t) \left(\frac{h_y(s|x = x_t)}{h_y(s|x = \mathcal{B})} \right)^\chi ds \\ &= e^{\chi \frac{\rho_z^2 \beta^2}{2\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} (\mathcal{B}^2 - x_t^2)} \int \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(s - \rho_z z_t)^2}{2\sigma_z^2}} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} s} ds \end{aligned}$$

Note that

$$\frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(s - \rho_z z_t)^2}{2\sigma_z^2}} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} s}$$

Is a moment generating function, and it's integral is equal to

$$\int \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(s - \rho_z z_t)^2}{2\sigma_z^2}} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} s} ds = e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B}) \rho_z z_t}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} + \frac{1}{2} \left(\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} \right)^2 \sigma_z^2}$$

From which it follows that the normalizing term is

$$Z_y(\chi) = e^{\chi \frac{\rho_z^2 \beta^2}{2\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} (\mathcal{B}^2 - x_t^2)} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B}) \rho_z z_t}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} + \frac{1}{2} \left(\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} \right)^2 \sigma_z^2}$$

Then

$$\begin{aligned} h_y^\chi(y'|\mathcal{I}) &= \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(y' - \rho_z z_t)^2}{2\sigma_z^2}} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} y' + \chi \frac{\rho_z^2 \beta^2}{2\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} (\mathcal{B}^2 - x_t^2)} e^{-\chi \frac{\rho_z^2 \beta^2}{2\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} (\mathcal{B}^2 - x_t^2)} \\ &\quad e^{-\chi \frac{\rho_z \beta (x_t - \mathcal{B}) \rho_z z_t}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} - \frac{1}{2} \left(\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} \right)^2 \sigma_z^2} \end{aligned}$$

After simplifying:

$$h_y^\chi(y'|\mathcal{I}) = \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{(y' - \rho_z z_t)^2}{2\sigma_z^2}} e^{\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} y'} e^{-\chi \frac{\rho_z \beta (x_t - \mathcal{B}) \rho_z z_t}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} - \frac{1}{2} \left(\chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1-\rho_z^2 \alpha_z \beta)} \right)^2 \sigma_z^2}$$

Let $A = \chi \frac{\rho_z \beta (x_t - \mathcal{B})}{\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)}$. Rewriting we get:

$$h_y^\chi(y'|\mathcal{I}) = \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{1}{2\sigma_z^2}[(y' - \rho_z z_t)^2 - 2A\sigma_z^2(y' - \rho_z z_t)] - \frac{1}{2}A^2\sigma_z^2}$$

Add and subtract $A^2\sigma_z^4$ inside the square brackets

$$\begin{aligned} h_y^\chi(y'|\mathcal{I}) &= \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{1}{2\sigma_z^2}[y' - \rho_z z_t - A\sigma_z^2]^2 + \frac{1}{2}A^2\sigma_z^2 - \frac{1}{2}A^2\sigma_z^2} \\ &= \frac{1}{\sqrt{2\pi\sigma_z^2}} e^{-\frac{1}{2\sigma_z^2}[y' - \rho_z z_t - A\sigma_z^2]^2} \end{aligned}$$

This is just the pdf of a normal distribution with mean $\rho_z z_t + A\sigma_z^2 = \rho_z \left(z_t + \frac{\sigma_z^2}{\mathbb{V}_z(1 - \rho_z^2 \alpha_z \beta)} \chi \beta (x_t - \mathcal{B}) \right)$ and variance σ_z^2 . We can rewrite the mean as $\rho_z \left(z_t + \frac{(1 - \rho_z^2)}{(1 - \rho_z^2 \alpha_z \beta)} \chi \beta (x_t - \mathcal{B}) \right)$. Remember that the reason why we have some objects premultiplying $\beta \chi$ is that the variance of the tilted likelihood is larger than the variance of the rational markovian forecast, so you take this into account when updating your mean.

B.3 Proof of Proposition 1

Consider the following true environment:

$$dz_t = -\mu_z z_t dt + \sigma_z dW_t^z, \quad dg_t = -\mu_g g_t dt + \sigma_g dW_t^g, \quad W^z \perp W^g, \quad \mu_z, \mu_g > 0.$$

We'll prove the result under the more general setup where $x = \alpha_g g + \alpha_z z$, this will be convenient later on in the paper. Note that this implies that the long-run variances are $\text{Var}(z_t) = \sigma_z^2 / (2\mu_z)$ and $\text{Var}(g_t) = \sigma_g^2 / (2\mu_g)$. Hence

$$\beta_z \equiv \frac{\text{Cov}(z_t, x_t)}{\text{Var}(x_t)} = \frac{\alpha_z \sigma_z^2 / \mu_z}{\alpha_z^2 \sigma_z^2 / \mu_z + \alpha_g^2 \sigma_g^2 / \mu_g}, \quad 0 < \beta_z < 1,$$

with companion weight β_g for g .

Define the background context as

$$\mathcal{B}_t \equiv x_t - \mathcal{S}_t$$

And

$$\mathcal{S}_t \equiv \int_0^t e^{-\kappa(t-s)} d\Delta_s, \quad \kappa \geq 0.$$

Where

$$d\Delta_t = \alpha_z \sigma_z dW_t^z + \alpha_g \sigma_g dW_t^g.$$

Note that the limit $\kappa \rightarrow 0$ we get

$$\mathcal{S}_t = d\Delta_t$$

which brings us back to our discrete-time benchmark where there is no memory. In fact, in this limit case the background just boils down to the rationally predicted x_t , i.e., x_t in the absence of shocks. It's just the realization minus the shock which by definition is the rationally predicted FIRE realization of x_t .

Because (z_t, g_t) is Gaussian, the joint process for (z_t, y_t) is also Gaussian with

$$\begin{bmatrix} z_t \\ x_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_z^2 / (2\mu_z) & \alpha_z \sigma_z^2 / (2\mu_z) \\ \alpha_z \sigma_z^2 / (2\mu_z) & \alpha_z^2 \sigma_z^2 / (2\mu_z) + \alpha_g^2 \sigma_g^2 / (2\mu_g) \end{bmatrix}\right)$$

Remember that if $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right)$. Then

$$X_1 \mid X_2 = a \sim \mathcal{N}(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(a - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}).$$

Thus, it follows that

$$z_t \mid x_t \sim \mathcal{N}\left(\beta_z x_t, \sigma_z^2 / (2\mu_z) - \frac{(\alpha_z \sigma_z^2 / (2\mu_z))^2}{\alpha_z^2 \sigma_z^2 / (2\mu_z) + \alpha_g^2 \sigma_g^2 / (2\mu_g)}\right)$$

That is

$$z_t \mid x_t \sim \mathcal{N}(\beta y_t, \frac{\sigma_z^2}{2\mu_z}(1 - \alpha_z \beta))$$

For simplicity, call

$$\Sigma_{z,0} \equiv \frac{\sigma_z^2}{2\mu_z}(1 - \alpha_z \beta).$$

Propagating one step of length dt forward, we can write

$$z_{t+dt} = z_t + dz_t = (1 - \mu_z dt)z_t + \sigma_z(W_{t+dt}^z - W_t^z)$$

Notice that $W_{t+dt}^z - W_t^z \sim \mathcal{N}(0, dt)$ and is independent of $\mathcal{F}_t$. Thus it follows that

$$z_{t+dt} | z_t \sim \mathcal{N}(m_t^z, \text{Var}(z_{t+dt} | z_t)), \quad m_t^z \equiv (1 - \mu_z dt)z_t, \quad \text{Var}(z_{t+dt} | z_t) = \frac{\sigma_z^2}{2\mu_z} (1 - e^{-2\mu_z dt}).$$

Which for small dt boils down to

$$z_{t+dt} | z_t \sim \mathcal{N}(m_t^z, \text{Var}(z_{t+dt} | z_t)), \quad m_t^z \equiv (1 - \mu_z dt)z_t, \quad \text{Var}(z_{t+dt} | z_t) = \sigma_z^2 dt.$$

Now however we want to change the conditioning set from z_t to either x_t or $\mathcal{B}_t$. To do so, we'll use the law of total expectations and the law of total variance. By the law of total expectations we have that

$$\begin{aligned} m_t^x &= E(z_{t+dt} | x_t) \\ &= E(E(z_{t+dt} | z_t, x_t) | x_t) \\ &= E((1 - \mu_z dt)z_t | x_t) \\ &= (1 - \mu_z dt)E(z_t | x_t) \\ &= (1 - \mu_z dt)\beta_z x_t \end{aligned}$$

Now law of total variance

$$\begin{aligned} \text{Var}(z_{t+dt} | x_t) &= E(\text{Var}(z_{t+dt} | z_t, x_t) | x_t) + \text{Var}(E(z_{t+dt} | z_t, x_t) | x_t) \\ &= E\left(\frac{\sigma_z^2}{2\mu_z} (1 - e^{-2\mu_z dt}) | x_t\right) + \text{Var}(e^{-\mu_z dt} z_t | x_t) \\ &= \frac{\sigma_z^2}{2\mu_z} (1 - e^{-2\mu_z dt}) + e^{-2\mu_z dt} \text{Var}(z_t | x_t) \\ &= \frac{\sigma_z^2}{2\mu_z} (1 - e^{-2\mu_z dt}) + e^{-2\mu_z dt} \frac{\sigma_z^2}{2\mu_z} (1 - \alpha_z \beta_z) \end{aligned}$$

Which for small dt becomes

$$\begin{aligned} \text{Var}(z_{t+dt} | x_t) &= \frac{\sigma_z^2}{2\mu_z} (2\mu_z dt) + (1 - 2\mu_z dt) \frac{\sigma_z^2}{2\mu_z} (1 - \alpha_z \beta_z) \\ &= \sigma_z^2 dt - \sigma_z^2 (1 - \alpha_z \beta_z) dt + \underbrace{\frac{\sigma_z^2}{2\mu_z} (1 - \alpha_z \beta_z)}_{=\Sigma_{z,0}} \\ &= \Sigma_{z,0} + \beta_z \alpha_z \sigma_z^2 dt \end{aligned}$$

So now it immediately follows that

$$z_{t+dt} \mid x_t \sim \mathcal{N}(m_t^x, \Sigma_z(dt)), \quad m_t^x \equiv (1 - \mu_z dt) \beta_z x_t, \quad \Sigma_z(dt) = \Sigma_{z,0} + \beta_z \alpha_z \sigma_z^2 dt + o(dt).$$

Similarly,

$$z_{t+dt} \mid \mathcal{B}_t \sim \mathcal{N}(m_t^{\mathcal{B}}, \Sigma_z(dt)), \quad m_t^{\mathcal{B}} \equiv (1 - \mu_z dt) \beta_z \mathcal{B}_t, \quad \Sigma_z(dt) = \Sigma_{z,0} + \beta_z \alpha_z \sigma_z^2 dt + o(dt).$$

Now we have all the results needed to proceed with our calculation of the XDE density. First, note that

$$h_z(z_{t+dt} \mid z_t) = \frac{1}{\sqrt{2\pi \sigma_z^2 dt}} \exp\left(-\frac{(z_{t+dt} - (1 - \mu_z dt) z_t)^2}{2 \sigma_z^2 dt}\right).$$

Moreover

$$h_z(z_{t+dt} \mid x_t) = \frac{1}{\sqrt{2\pi \Sigma_z(dt)}} \exp\left(-\frac{(z_{t+dt} - (1 - \mu_z dt) \beta_z x_t)^2}{2 \Sigma_z(dt)}\right).$$

And similarly

$$h_z(z_{t+dt} \mid \mathcal{B}_t) = \frac{1}{\sqrt{2\pi \Sigma_z(dt)}} \exp\left(-\frac{(z_{t+dt} - (1 - \mu_z dt) \beta_z \mathcal{B}_t)^2}{2 \Sigma_z(dt)}\right).$$

Now let's compute the tilting

$$\begin{aligned} T_{z,t}(z') &\equiv \log \frac{h_z(z' \mid x_t)}{h_z(z' \mid \mathcal{B}_t)} \\ &= -\log(\sqrt{2\pi \Sigma_z(dt)}) - \frac{(z_{t+dt} - (1 - \mu_z dt) \beta_z x_t)^2}{2 \Sigma_z(dt)} + \log(\sqrt{2\pi \Sigma_z(dt)}) \\ &\quad + \frac{(z_{t+dt} - (1 - \mu_z dt) \beta_z \mathcal{B}_t)^2}{2 \Sigma_z(dt)} \\ &= \frac{(z_{t+dt} - (1 - \mu_z dt) \beta_z \mathcal{B}_t)^2 - (z_{t+dt} - (1 - \mu_z dt) \beta_z x_t)^2}{2 \Sigma_z(dt)} \\ &= \frac{(1 - \mu_z dt)^2 \beta_z^2 \mathcal{B}_t^2 - (1 - \mu_z dt)^2 \beta_z^2 x_t^2 + 2z_{t+dt}(1 - \mu_z dt) \beta_z (x_t - \mathcal{B}_t)}{2 \Sigma_z(dt)} \\ &= \frac{z_{t+dt}(1 - \mu_z dt) \beta_z (x_t - \mathcal{B}_t)}{\Sigma_z(dt)} + \frac{(1 - \mu_z dt)^2 \beta_z^2 (\mathcal{B}_t^2 - x_t^2)}{2 \Sigma_z(dt)} \end{aligned}$$

Then, the normalizing constant is

$$\begin{aligned}
Z_z(\chi) &= \int \frac{1}{\sqrt{2\pi\sigma_z^2 dt}} \exp\left(-\frac{(s - (1 - \mu_z dt) z_t)^2}{2\sigma_z^2 dt}\right) + \frac{\chi s(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)} \\
&\quad + \frac{\chi(1 - \mu_z dt)^2\beta_z^2(\mathcal{B}_t^2 - x_t^2)}{2\Sigma_z(dt)} ds \\
&= \exp\left(\frac{\chi(1 - \mu_z dt)^2\beta_z^2(\mathcal{B}_t^2 - x_t^2)}{2\Sigma_z(dt)}\right) \int \frac{1}{\sqrt{2\pi\sigma_z^2 dt}} \exp\left(-\frac{(s - (1 - \mu_z dt) z_t)^2}{2\sigma_z^2 dt}\right) \\
&\quad + \frac{\chi s(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)} ds
\end{aligned}$$

Notice that

$$\frac{1}{\sqrt{2\pi\sigma_z^2 dt}} \exp\left(-\frac{(s - (1 - \mu_z dt) z_t)^2}{2\sigma_z^2 dt}\right) \exp\left(\chi \frac{(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)} s\right)$$

Is a moment generating function for the moment $w = \chi \frac{(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)}$. Remember that if $x \sim \mathcal{N}(m, \sigma^2)$ then

$$E(e^{ax}) = \exp(am + \frac{1}{2}\sigma^2 a^2)$$

Then it follows that

$$\begin{aligned}
&\int \frac{1}{\sqrt{2\pi\sigma_z^2 dt}} \exp\left(-\frac{(s - (1 - \mu_z dt) z_t)^2}{2\sigma_z^2 dt}\right) \exp\left(\chi \frac{(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)} s\right) ds = \\
&\exp\left(\chi \frac{(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)(1 - \mu_z dt)z_t}{\Sigma_z(dt)} + \frac{1}{2}\left(\chi \frac{(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)}\right)^2 \sigma_z^2 dt\right)
\end{aligned}$$

Plugging all of this in we have that

$$\begin{aligned}
Z_z(\chi) &= \exp\left[\chi \frac{(1 - \mu_z dt)^2\beta_z^2}{2\Sigma_z(dt)}(\mathcal{B}_t^2 - x_t^2) + \chi \frac{(1 - \mu_z dt)^2\beta_z(x_t - \mathcal{B}_t)z_t}{\Sigma_z(dt)} \right. \\
&\quad \left. + \frac{1}{2}\chi^2 \frac{(1 - \mu_z dt)^2\beta_z^2(x_t - \mathcal{B}_t)^2}{\Sigma_z(dt)^2} \sigma_z^2 dt\right].
\end{aligned}$$

So now we can bring everything together

$$h_z^\chi(z_{t+dt} \mid \mathcal{F}_t) = \frac{1}{Z_z(\chi)} h_z(z_{t+dt} \mid z_t) \exp(\chi T_{z,t}(z_{t+dt}))$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi\sigma_z^2 dt}} \exp\left(-\frac{(z_{t+dt} - (1 - \mu_z dt) z_t)^2}{2\sigma_z^2 dt}\right) \\
&+ \frac{\chi z_{t+dt}(1 - \mu_z dt)\beta_z(x_t - \mathcal{B}_t)}{\Sigma_z(dt)} + \frac{\chi(1 - \mu_z dt)^2\beta_z^2(\mathcal{B}_t^2 - x_t^2)}{2\Sigma_z(dt)} \Big) \times \\
&\times \exp\left[-\chi \frac{(1 - \mu_z dt)^2\beta_z^2}{2\Sigma_z(dt)}(\mathcal{B}_t^2 - x_t^2) - \chi \frac{(1 - \mu_z dt)^2\beta_z(x_t - \mathcal{B}_t)z_t}{\Sigma_z(dt)}\right. \\
&\left. - \frac{1}{2}\chi^2 \frac{(1 - \mu_z dt)^2\beta_z^2(x_t - \mathcal{B}_t)^2}{\Sigma_z(dt)^2} \sigma_z^2 dt\right].
\end{aligned}$$

Now let's simplify terms to get the kernel of the normal distribution. First, define

$$m \equiv (1 - \mu_z dt) z_t, \quad \sigma^2 \equiv \sigma_z^2 dt, \quad a \equiv \chi \frac{(1 - \mu_z dt) \beta_z (x_t - \mathcal{B}_t)}{\Sigma_z(dt)}.$$

With these abbreviations the tilted density reads

$$h_z^\chi(z_{t+dt} | \mathcal{F}_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z_{t+dt}-m)^2}{2\sigma^2} + a z_{t+dt} - [m a + \frac{1}{2}a^2\sigma^2]\right).$$

Now complete the squares

$$\begin{aligned}
-\frac{(z_{t+dt} - m)^2}{2\sigma^2} + a(z_{t+dt} - m) - \frac{1}{2}a^2\sigma^2 &= -\frac{1}{2\sigma^2} \left[(z_{t+dt} - m)^2 - 2a\sigma^2(z_{t+dt} - m) + a^2\sigma^4 \right] \\
&= -\frac{1}{2\sigma^2} (z_{t+dt} - m - a\sigma^2)^2.
\end{aligned}$$

Insert the right-hand side above into the exponential:

$$h_z^\chi(z_{t+dt} | \mathcal{F}_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (z_{t+dt} - m - a\sigma^2)^2\right).$$

Write

$$\mu_{t+dt}^{(\chi)} \equiv m + a\sigma^2 = (1 - \mu_z dt) z_t + \chi (1 - \mu_z dt) \beta_z (x_t - \mathcal{B}_t) \frac{\sigma_z^2 dt}{\Sigma_z(dt)} = (1 - \mu_z dt) \left(z_t + \frac{\sigma_z^2 dt}{\Sigma_z(dt)} \chi \beta_z \mathcal{S}_t \right)$$

Then

$$h_z^\chi(z_{t+dt} | \mathcal{F}_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z_{t+dt} - \mu_{t+dt}^{(\chi)})^2}{2\sigma^2}\right).$$

Because the right-hand side is the canonical form of a normal pdf with variance σ^2 , we have

$$z_{t+dt} | \mathcal{F}_t, \chi \sim \mathcal{N}(\mu_{t+dt}^{(\chi)}, \sigma^2), \quad \sigma^2 = \sigma_z^2 dt.$$

Finally, it follows that

$$d\tilde{z}_t = \left[-\mu_z z_t + \chi \beta_z \mathcal{S}_t \frac{\sigma_z^2}{\Sigma_z(dt)} \right] dt + \sigma_z dW_t^z$$

Where, using the definition $\mathcal{S}_t = \int_0^t e^{-\kappa(t-s)} d\Delta_s$ and Ito's rule for deterministic kernels one obtains

$$d\mathcal{S}_t = -\kappa \mathcal{S}_t dt + d\Delta_t, \quad d\Delta_t = \alpha_z \sigma_z dW_t^z + \alpha_g \sigma_g dW_t^g.$$

B.4 Models of imperfect information

In this section, we present models of overreaction based on limited information, rather than on deviations from rational expectations.

In this section, we present models of overreaction based on limited information, rather than on deviations from rational expectations. Overreaction arises as a byproduct of a signal extraction problem rather than behavioral bias. We explain why, in the main analysis of this paper, we do not focus on these models.

Public and private demand shocks cannot be disentangled Let two latent demand components follow independent AR(1) laws,

$$z_t = \rho_z z_{t-1} + \varepsilon_t^z, \quad g_t = \rho_g g_{t-1} + \varepsilon_t^g,$$

with $|\rho_z|, |\rho_g| < 1$, innovations $\varepsilon_t^z \sim (0, \sigma_z^2)$ and $\varepsilon_t^g \sim (0, \sigma_g^2)$, mutually independent over time and across processes. The observed aggregate is $x_t = z_t + g_t$. The agent only observes $\{x_s\}_{s \leq t}$ and forms the optimal forecast of x_{t+1} . Define the period- t innovation in the observable as

$$u_t \equiv x_t - \mathbb{E}_{t-1}[x_t] = \varepsilon_t^z + \varepsilon_t^g,$$

Because everything is linear and Gaussian, the optimal one-step forecast revision is linear in u_t ,

$$\mathbb{E}_t[x_{t+1}] - \mathbb{E}_{t-1}[x_{t+1}] = \beta u_t,$$

where $\beta = \text{Cov}(x_{t+1}, u_t) / \text{Var}(u_t)$. Using $x_{t+1} = z_{t+1} + g_{t+1} = \rho_z z_t + \rho_g g_t + \varepsilon_{t+1}^z + \varepsilon_{t+1}^g$ and $u_t = \varepsilon_t^z + \varepsilon_t^g$, independence gives

$$\text{Cov}(x_{t+1}, u_t) = \rho_z \sigma_z^2 + \rho_g \sigma_g^2, \quad \text{Var}(u_t) = \sigma_z^2 + \sigma_g^2,$$

hence

$$\beta = \frac{\rho_z \sigma_z^2 + \rho_g \sigma_g^2}{\sigma_z^2 + \sigma_g^2}.$$

A simple reason for the linearity of the revision in u_t is that $\mathbb{E}_t[x_{t+1}]$ is the orthogonal projection of x_{t+1} onto the time- t information set. The only new information between $t-1$ and t is the scalar innovation u_t , which spans a one-dimensional subspace orthogonal to information at time $t-1$. By the projection theorem, the increment in the optimal forecast must lie in this span, hence it is proportional to u_t ; under joint normality, least-squares and conditional expectations coincide, which pins down the slope as β .

To assess reaction to a period- t shock in g_t , hold $\varepsilon_t^z = 0$ and perturb ε_t^g . The true effect on the next aggregate is $\partial x_{t+1} / \partial \varepsilon_t^g = \rho_g$, while the agent's revision in expected x_{t+1} per unit of u_t is β . The difference $\beta - \rho_g$ determines over- versus underreaction to g -news. Substituting the expression for β yields

$$\beta - \rho_g = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_g^2}(\rho_z - \rho_g).$$

Whenever $\sigma_z^2 > 0$, the sign is governed by the relative persistence of the latent components: if $\rho_z > \rho_g$ the agent overreacts to shocks in g_t because part of the innovation in x_t is (optimally) attributed to the more persistent z_t ; if $\rho_z < \rho_g$ the agent underreacts; if $\rho_z = \rho_g$ the reaction is exactly correct.

Now define $\text{FE}_{t+1} \equiv x_{t+1} - \mathbb{E}_t[x_{t+1}]$. Combine

$$x_{t+1} = \mathbb{E}_{t-1}[x_{t+1}] + \rho_z \varepsilon_t^z + \rho_g \varepsilon_t^g + \varepsilon_{t+1}^z + \varepsilon_{t+1}^g$$

with

$$\mathbb{E}_t[x_{t+1}] = \mathbb{E}_{t-1}[x_{t+1}] + \beta(\varepsilon_t^z + \varepsilon_t^g)$$

to obtain

$$\text{FE}_{t+1} = (\rho_z - \beta)\varepsilon_t^z + (\rho_g - \beta)\varepsilon_t^g + \varepsilon_{t+1}^z + \varepsilon_{t+1}^g.$$

This last equation makes it clear why we don't choose this model as our baseline model, in spite of it being able to generate overreaction in public spending shocks. First, it requires private demand to be more persistent than public demand ($\rho_z > \rho_g$), with the magnitude of overreaction increasing in the persistence gap and the variance share $\sigma_z^2 / (\sigma_z^2 + \sigma_g^2)$, yet empirical estimates typically do not show such a large persistence gap between aggregate (g) and idiosyncratic (z) components. Second, since β is a weighted average of the two persistence parameters, overreaction to public-demand news implies

$\rho_g < \beta < \rho_z$, and hence underreaction to private-demand news. This is at odds with much of the empirical literature, which tends to find overreaction to idiosyncratic shocks rather than underreaction. Third, this assumes that firms cannot disentangle public and private demand which is at odds with the data. Finally this framework does not allow us to think about the forecast of z separately and therefore cannot rationalize our results on export forecast.

Underlying quality Let $u_t = \rho_u u_{t-1} + \varepsilon_t^u$ with $|\rho_u| < 1$, and let

$$z_t = u_t + v_t^z, \quad v_t^z = \rho_z v_{t-1}^z + \varepsilon_t^z, \quad g_t = u_t + v_t^g, \quad v_t^g = \rho_g v_{t-1}^g + \varepsilon_t^g,$$

with mutually independent innovations $\varepsilon_t^u \sim (0, \sigma_u^2)$, $\varepsilon_t^z \sim (0, \sigma_z^2)$, $\varepsilon_t^g \sim (0, \sigma_g^2)$. The target is

$$x_{t+1} = z_{t+1} + g_{t+1} = 2u_{t+1} + v_{t+1}^z + v_{t+1}^g.$$

Define the time- t observation innovations

$$v_t^z \equiv z_t - \mathbb{E}_{t-1}[z_t] = \varepsilon_t^u + \varepsilon_t^z, \quad v_t^g \equiv g_t - \mathbb{E}_{t-1}[g_t] = \varepsilon_t^u + \varepsilon_t^g.$$

By linear-Gaussian projection,

$$\mathbb{E}_t[x_{t+1}] - \mathbb{E}_{t-1}[x_{t+1}] = b_z v_t^z + b_g v_t^g,$$

where $b_z = \text{Cov}(x_{t+1}, v_t^z)$ and $b_g = \text{Cov}(x_{t+1}, v_t^g)$ premultiplied by $\text{Var}(v_t)^{-1}$. Writing

$$\text{Var}(v_t) = \begin{pmatrix} \sigma_u^2 + \sigma_z^2 & \sigma_u^2 \\ \sigma_u^2 & \sigma_u^2 + \sigma_g^2 \end{pmatrix}, \quad \Delta \equiv (\sigma_u^2 + \sigma_z^2)(\sigma_u^2 + \sigma_g^2) - \sigma_u^4 = \sigma_u^2 \sigma_z^2 + \sigma_u^2 \sigma_g^2 + \sigma_z^2 \sigma_g^2,$$

and

$$\text{Cov}(x_{t+1}, v_t^z) = 2\rho_u \sigma_u^2 + \rho_z \sigma_z^2, \quad \text{Cov}(x_{t+1}, v_t^g) = 2\rho_u \sigma_u^2 + \rho_g \sigma_g^2,$$

a short calculation yields the closed-form coefficients

$$b_g = \rho_g + \frac{\sigma_u^2 \sigma_z^2}{\Delta} (2\rho_u - \rho_z - \rho_g), \quad b_z = \rho_z + \frac{\sigma_u^2 \sigma_g^2}{\Delta} (2\rho_u - \rho_z - \rho_g).$$

Hence the loadings on the primitive shocks in the time- t forecast revision are

$$\alpha_u = b_z + b_g, \quad \alpha_z = b_z, \quad \alpha_g = b_g,$$

so that the $t+1$ forecast error can be written as

$$\text{FE}_{t+1} \equiv x_{t+1} - \mathbb{E}_t[x_{t+1}] = (2\rho_u - \alpha_u)\varepsilon_t^u + (\rho_z - \alpha_z)\varepsilon_t^z + (\rho_g - \alpha_g)\varepsilon_t^g + 2\varepsilon_{t+1}^u + \varepsilon_{t+1}^z + \varepsilon_{t+1}^g.$$

Overreaction to g -news is the condition $b_g > \rho_g$. From the expression above,

$$b_g - \rho_g = \frac{\sigma_u^2 \sigma_z^2}{\Delta} (2\rho_u - \rho_z - \rho_g),$$

which is positive if and only if

$$2\rho_u > \rho_z + \rho_g.$$

Under the same condition $b_z > \rho_z$ and, correspondingly,

$$\alpha_u - 2\rho_u = \frac{\sigma_g^2 \sigma_z^2}{\Delta} (\rho_g + \rho_z - 2\rho_u) < 0,$$

so the forecast overreacts to the measurement-noise shocks $\varepsilon_t^z, \varepsilon_t^g$ while underreacting to the fundamental shock ε_t^u . Intuitively, a positive surprise in g_t raises the inferred u_t ; because x_{t+1} depends on u_t through $2\rho_u$, the optimal revision incorporates this channel in addition to the mechanical ρ_g propagation of v_t^g , making the loading on g -news exceed ρ_g .

Finally, shocks to the g -series also generate cross-reactions in expectations of z . From the forecast-revision equation

$$\mathbb{E}_t[x_{t+1}] - \mathbb{E}_{t-1}[x_{t+1}] = b_z v_t^z + b_g v_t^g,$$

the revision to expected z_{t+1} alone can be written as

$$\mathbb{E}_t[z_{t+1}] - \mathbb{E}_{t-1}[z_{t+1}] = c_z v_t^z + c_g v_t^g,$$

where c_g measures how a surprise in g_t affects the forecast of z_{t+1} .

Because both z_t and g_t load on the same latent factor u_t , a positive innovation in g_t raises the agent's inference about u_t and, therefore, their forecast of z_{t+1} . Formally, applying the same linear-Gaussian projection yields

$$c_g = \frac{\sigma_u^2 \sigma_z^2}{\Delta} (\rho_u - \rho_z),$$

which is positive whenever $\rho_u > \rho_z$. Thus, even though the shock originates in the idiosyncratic g -noise process, it induces a positive co-movement in expected z_{t+1} . This cross-channel response arises because g_t is an imperfect but correlated signal of the un-

derlying fundamental u_t ; a transitory improvement in g_t is partly interpreted as evidence of a stronger u_t , leading the agent to revise upward their expectation of z_{t+1} and, consequently, of the aggregate x_{t+1} .

In this limited information rational-expectations model, the one-step forecast error is an innovation, orthogonal to all information available at the time the forecast is made. This implies that its autocovariance must be zero, since the previous forecast error belongs to the information set. Formally, $\text{Cov}(\text{FE}_t, \text{FE}_{t-1}) = \mathbb{E}[\text{FE}_{t-1} \mathbb{E}(\text{FE}_t \mid \mathcal{F}_{t-1})] = 0$. Substituting the shock representation from the model, $\text{FE}_t = (2\rho_u - \alpha_u)\varepsilon_{t-1}^u + (\rho_z - \alpha_z)\varepsilon_{t-1}^z + (\rho_g - \alpha_g)\varepsilon_{t-1}^g + 2\varepsilon_t^u + \varepsilon_t^z + \varepsilon_t^g$, and an analogous expression for FE_{t-1} , only the $(t-1)$ shocks overlap across the two errors. This yields $\text{Cov}(\text{FE}_t, \text{FE}_{t-1}) = 2(2\rho_u - \alpha_u)\sigma_u^2 + (\rho_z - \alpha_z)\sigma_z^2 + (\rho_g - \alpha_g)\sigma_g^2 = 0$, because the projection coefficients b_z and b_g defining $\alpha_u = b_z + b_g$, $\alpha_z = b_z$, and $\alpha_g = b_g$ are chosen to ensure orthogonality. Hence, under rational expectations, forecast errors are serially uncorrelated. This theoretical result contrasts with the data, where forecast errors display positive autocorrelation.

B.5 Learning about persistent firm quality

The preceding imperfect-information models take the parameters of the demand processes as known. This subsection develops a Bayesian-learning mechanism through which an increase in public demand can raise expectations about private demand and generate subsequent forecast errors. Firms observe their sales but cannot perfectly distinguish changes in their underlying market appeal from market-specific demand shocks. They also learn about the persistence of their market appeal. Following the insight of [Farmer et al. \(2024\)](#), uncertainty about persistent, unobserved components can make learning sufficiently slow for forecast errors to remain predictable over extended periods.

The mechanism distinguishes two inference problems. Uncertainty about the *source* of demand generates cross-domain extrapolation following procurement shocks. Uncertainty about the *persistence* of underlying quality can generate positively autocorrelated forecast errors. Firm entry and exit sustain a population with heterogeneous learning histories, although the direction of forecast distortions also depends on entrants' priors.

B.5.1 Environment and information

Time is discrete. Firm i 's demand depends on an unobserved quality component q_{it} , which evolves according to

$$q_{i,t+1} = \rho_i q_{it} + \varepsilon_{i,t+1}, \quad \varepsilon_{i,t+1} \sim N(0, \sigma_q^2).$$

Quality represents customers' valuation of the firm's products, reputation, or reliability. It need not coincide with a technological characteristic directly observed by the firm. The persistence parameter ρ_i is fixed over the firm's lifetime but initially unknown to it.

The firm operates in two markets: public procurement and private demand. Log demand, net of market-specific constants, satisfies

$$g_{it} = q_{it} + \eta_{it}^g, \quad z_{it} = q_{it} + \eta_{it}^z,$$

where

$$\eta_{it}^g \sim N(0, \sigma_g^2), \quad \eta_{it}^z \sim N(0, \sigma_z^2).$$

In this subsection, g_{it} denotes observed public demand and η_{it}^g its market-specific component. Quality innovations and market-specific shocks are mutually independent and independent over time. The assumption that market-specific shocks are transitory simplifies the exposition. A richer specification could allow these shocks to be persistent while

remaining less persistent than quality.

Firms observe their public and private demand, but not the decomposition of either into quality and market-specific components. Their information set is

$$\mathcal{I}_{it} = \sigma(\{\dots, g_{it-1}, z_{it-1}, g_{it}, z_{it}\}).$$

Thus, a firm knows that it has received a public order but may not know whether the order reflects stronger underlying competitiveness or an expansion in the purchaser's budget. The latter distinction is essential: the mechanism requires that the purely fiscal component of public demand is not separately and perfectly observed.

Firms exit with probability δ each period and are replaced by entrants. Exit is independent of quality, demand shocks, and beliefs. Forecasts below are conditional on survival.

B.5.2 Priors and Bayesian updating

To obtain a tractable learning problem, suppose that persistence takes two values,

$$\rho_i \in \{\rho_L, \rho_H\}, \quad 0 < \rho_L < \rho_H < 1.$$

Entrants assign probability p_0 to the high-persistence type:

$$\Pr_i(\rho_i = \rho_H) = p_0, \quad 0 < p_0 < 1.$$

Conditional on each type, their initial belief about quality is its stationary distribution:

$$q_{i0} \mid \rho_i = \rho_j \sim N\left(0, \frac{\sigma_q^2}{1 - \rho_j^2}\right), \quad j \in \{L, H\}.$$

Both persistence types therefore remain in the support of beliefs. Forecast distortions arise from incomplete learning rather than from excluding the true parameter.

Let π_{it}^j denote the posterior probability of type j , and let m_{it}^j and P_{it}^j denote the posterior mean and variance of quality conditional on that type. Before observing current demand, the conditional predictive mean and variance are

$$\mu_{it}^j = \rho_j m_{i,t-1}^j, \quad R_{it}^j = \rho_j^2 P_{i,t-1}^j + \sigma_q^2.$$

At entry, these predictive moments are initialized at $\mu_{i0}^j = 0$ and $R_{i0}^j = \sigma_q^2 / (1 - \rho_j^2)$. After

observing both demand realizations, the quality estimates satisfy

$$P_{it}^j = \left(\frac{1}{R_{it}^j} + \frac{1}{\sigma_g^2} + \frac{1}{\sigma_z^2} \right)^{-1},$$

$$m_{it}^j = P_{it}^j \left(\frac{\mu_{it}^j}{R_{it}^j} + \frac{g_{it}}{\sigma_g^2} + \frac{z_{it}}{\sigma_z^2} \right).$$

Persistence beliefs are updated using the corresponding predictive likelihoods:

$$\pi_{it}^j = \frac{\pi_{i,t-1}^j f_j(g_{it}, z_{it} \mid \mathcal{I}_{i,t-1})}{\sum_{k \in \{L, H\}} \pi_{i,t-1}^k f_k(g_{it}, z_{it} \mid \mathcal{I}_{i,t-1})}.$$

At entry, the likelihoods are weighted by p_0 and $1 - p_0$. The forecasting problem is therefore a mixture of two Gaussian filtering problems. Expected private demand at horizon $h \geq 1$ is

$$F_{it} z_{i,t+h} = \sum_{j \in \{L, H\}} \pi_{it}^j \rho_j^h m_{it}^j.$$

An analogous expression holds for public demand because its market-specific component is transitory.

B.5.3 Cross-domain extrapolation following fiscal shocks

An increase in public demand raises the firm's estimate of quality, even when current private demand is unchanged. Conditional on persistence type j ,

$$\frac{\partial m_{it}^j}{\partial g_{it}} = K_{g,it}^j \equiv \frac{P_{it}^j}{\sigma_g^2} > 0.$$

The response of the full Bayesian private-demand forecast can be decomposed as

$$\frac{\partial F_{it} z_{i,t+h}}{\partial g_{it}} = \underbrace{\sum_j \pi_{it}^j \rho_j^h K_{g,it}^j}_{\text{updating estimated quality}} + \underbrace{\sum_j \rho_j^h m_{it}^j \frac{\partial \pi_{it}^j}{\partial g_{it}}}_{\text{updating persistence beliefs}}.$$

The first component is strictly positive. The second reflects changes in the relative likelihood of the two persistence types and depends on the firm's history. Cross-domain extrapolation occurs whenever this second component does not offset the positive quality-updating effect. In particular, at a symmetric history with zero filtered quality estimates

and zero current demand deviations, the second component vanishes and the response is strictly positive.

Consider now a purely fiscal innovation that increases η_{it}^g , holding quality and all other shocks fixed, without revealing its magnitude separately to the firm. This innovation has no effect on subsequent private demand. Nevertheless, when the forecast response above is positive, the firm raises its private-demand forecast.

Defining the forecast error as

$$e_{i,t+h|t}^z = z_{i,t+h} - F_{it}z_{i,t+h},$$

the fiscal innovation consequently produces a negative private-demand forecast-error response:

$$\frac{\partial e_{i,t+h|t}^z}{\partial \eta_{it}^g} = -\frac{\partial F_{it}z_{i,t+h}}{\partial g_{it}} < 0.$$

The forecast thus overreacts relative to the fiscal shock's actual effect on private demand. This need not represent a departure from Bayesian updating: the econometrician isolates a component of public demand that the firm cannot perfectly distinguish from news about quality.

B.5.4 Persistence learning and serially correlated forecast errors

Cross-domain extrapolation does not, by itself, imply positively autocorrelated forecast errors. A separate role is played by uncertainty about persistence.

Suppose that entrants place relatively little probability on high persistence, while the realized demand process is highly persistent. Following a genuine improvement in quality, firms may initially predict excessive mean reversion. Demand can then remain above forecasts over successive periods, generating positive serial correlation in forecast errors.

This mechanism is compatible with excessive optimism following a purely fiscal innovation. The two patterns concern different underlying shocks: firms can assign too much of a procurement innovation to quality while assigning too little persistence to a genuine quality innovation. Accordingly, overreaction to identified fiscal shocks need not imply overreaction to all demand news.

The distinction between subjective and objective probabilities is important. Bayesian forecasts satisfy

$$\mathbb{E}^i \left[e_{i,t+1|t}^z \mid \mathcal{I}_{it} \right] = 0$$

under the firm's subjective probability distribution. This does not imply the same con-

ditional restriction under a fixed true parameter when the firm has not yet learned that parameter.

However, let p^* denote the actual fraction of high-persistence entrants. If $p_0 = p^*$, and all remaining elements of the prior and model are correctly calibrated, forecasts are conditional expectations under the actual joint distribution. In that case,

$$\mathbb{E}^* \left[e_{i,t+1|t}^z e_{it|t-1}^z \right] = 0.$$

Thus, systematic forecast-error autocorrelation requires more than parameter uncertainty alone. In the construction considered here, it arises because entry beliefs differ from the actual distribution of persistence types and learning has not eliminated that discrepancy.

B.5.5 Interpretation, empirical discipline, and scope

The model illustrates how Bayesian learning can generate cross-domain extrapolation and predictable forecast errors. Its main limitation is the difficulty of independently disciplining the assumptions and parameters responsible for these patterns. Even the two-type specification requires two persistence parameters, three innovation variances, an entry prior, the actual distribution of persistence types, and an exit rate. These objects jointly determine the informativeness of public demand, the persistence attributed to inferred quality improvements, and the speed at which initial beliefs are corrected. Different combinations may therefore produce similar responses of forecasts and forecast errors.

Some parameters can, in principle, be disciplined using demand histories and observed entry and exit. Others are substantially harder to identify. In particular, the distribution of entrants' priors need not coincide with the actual distribution of firm types, and observed demand histories do not directly reveal that discrepancy. The information available to firms about the source of procurement is also imperfectly observed by the econometrician.

This distinguishes the learning account from the more parsimonious XDE framework developed in the main text. The learning model attributes observed responses to the interaction of latent fundamentals, signal precision, persistence uncertainty, and, most importantly, biased initial beliefs. The XDE specification instead places a more restrictive structure on how news propagates across demand domains. That parsimony facilitates empirical discipline and the interpretation of the mechanism within the general-equilibrium model.

The mechanism also depends on uncertainty about the source of public demand. If the purely fiscal component η_{it}^g is separately observed, firms can remove it from public

orders when estimating quality. A fiscal innovation then generates no spurious revision in private-demand expectations. More precise information about the origin of procurement should therefore attenuate the cross-domain response. Moreover, extrapolation is restricted to demand domains linked by a common latent quality component; the model does not predict an unrestricted increase in optimism across unrelated outcomes.

Finally, firm turnover preserves incomplete learning but does not explain the direction of prior errors. If entrants possess correctly calibrated priors, turnover alone does not generate systematic forecast-error autocorrelation. Nor does individual firm exit necessarily prevent learning when demand histories can be pooled across firms or knowledge transmitted to entrants. A quantitative application would therefore require an account of both the formation of entry beliefs and the limits to information sharing.

Taken together, these considerations make learning about persistent quality a relevant alternative interpretation of the evidence, but one whose quantitative assessment requires additional information on firms' priors and information sets. Its flexibility establishes that the documented patterns need not uniquely identify distorted updating; the more parsimonious XDE framework provides the disciplined representation of cross-domain extrapolation used in the main analysis.

C Appendix to Section 4

C.1 Derivation of profit function

Demand aggregation. From the CES aggregators for Y_t^z and G_t , the associated price indices are

$$P_t^z = \left(\int_0^1 e^{z_{it}} p_{it}^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}, \quad P_t^g = \left(\int_0^1 e^{g_{it}} p_{it}^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}.$$

Individual demands are therefore

$$y_{it}^z = e^{z_{it}} \left(\frac{p_{it}}{P_t^z} \right)^{-\varepsilon} Y_t^z, \quad y_{it}^g = e^{g_{it}} \left(\frac{p_{it}}{P_t^g} \right)^{-\varepsilon} G_t,$$

so total demand faced by firm i is isoelastic:

$$y_{it} = y_{it}^z + y_{it}^g = p_{it}^{-\varepsilon} [e^{z_{it}} Y_t^z (P_t^z)^\varepsilon + e^{g_{it}} G_t (P_t^g)^\varepsilon] \equiv \Lambda_{it} p_{it}^{-\varepsilon}. \quad (\text{C.1})$$

Technology and costs. Let production be Cobb–Douglas with predetermined capital k_{it} :

$$y_{it} = k_{it}^\alpha \ell_{it}^{1-\alpha},$$

and let w^r be the real wage. The conditional labor requirement for output y is

$$\ell(y, k) = \left(\frac{y}{k^\alpha} \right)^{\frac{1}{1-\alpha}},$$

so the variable cost is

$$VC(y, k) = w^r \ell(y, k) = w^r k^{-\frac{\alpha}{1-\alpha}} y^{\frac{1}{1-\alpha}}.$$

Hence marginal cost is

$$mc_{it}(y) = \frac{\partial VC}{\partial y} = \frac{w^r}{1-\alpha} k_{it}^{-\frac{\alpha}{1-\alpha}} y^{\frac{\alpha}{1-\alpha}}. \quad (\text{C.2})$$

Optimal pricing (markup condition). Given (C.1), profit as a function of price is

$$\pi_{it}(p) = p y_{it}(p) - VC(y_{it}(p), k_{it}), \quad y_{it}(p) = \Lambda_{it} p^{-\varepsilon}.$$

The first-order condition uses $dy/dp = -\varepsilon y/p$:

$$0 = \frac{d\pi}{dp} = y + (p - mc_{it}) \frac{dy}{dp} \implies \frac{p_{it} - mc_{it}}{p_{it}} = \frac{1}{\varepsilon} \implies p_{it} = \mu mc_{it}, \quad \mu \equiv \frac{\varepsilon}{\varepsilon - 1}.$$

Substituting (C.2) and $y_{it} = \Lambda_{it} p_{it}^{-\varepsilon}$ gives

$$p_{it} = \mu \cdot \frac{w^r}{1 - \alpha} k_{it}^{-\frac{\alpha}{1-\alpha}} (\Lambda_{it} p_{it}^{-\varepsilon})^{\frac{\alpha}{1-\alpha}} = \left[\mu \cdot \frac{w^r}{1 - \alpha} k_{it}^{-\frac{\alpha}{1-\alpha}} \right] \Lambda_{it}^{\frac{\alpha}{1-\alpha}} p_{it}^{-\frac{\varepsilon\alpha}{1-\alpha}}.$$

Collecting powers of p_{it} yields

$$p_{it}^{1 + \frac{\varepsilon\alpha}{1-\alpha}} = \left[\mu \cdot \frac{w^r}{1 - \alpha} k_{it}^{-\frac{\alpha}{1-\alpha}} \right] \Lambda_{it}^{\frac{\alpha}{1-\alpha}}.$$

Let $D \equiv 1 - \alpha + \varepsilon\alpha = 1 + (\varepsilon - 1)\alpha$. Solving for p_{it} ,

$$p_{it} = \left[\mu \cdot \frac{w^r}{1 - \alpha} \cdot k_{it}^{-\frac{\alpha}{1-\alpha}} \right]^{\frac{1-\alpha}{D}} \cdot \Lambda_{it}^{\frac{\alpha}{D}}, \quad \mu = \frac{\varepsilon}{\varepsilon - 1}. \quad (\text{C.3})$$

Profit in closed form. At the optimum, the Lerner condition implies $mc_{it} = (\varepsilon - 1)p_{it}/\varepsilon$. Since variable cost satisfies $VC(y_{it}, k_{it}) = (1 - \alpha)mc_{it}y_{it}$, operating profit is

$$\pi_{it}^* = \frac{D}{\varepsilon} p_{it}^* y_{it}^* = \frac{D}{\varepsilon} \Lambda_{it} (p_{it}^*)^{1-\varepsilon}.$$

Using (C.3) and $D = 1 - \alpha + \varepsilon\alpha$,

$$(p_{it}^*)^{1-\varepsilon} = \left[\mu \cdot \frac{w^r}{1 - \alpha} \cdot k_{it}^{-\frac{\alpha}{1-\alpha}} \right]^{(1-\varepsilon)\frac{1-\alpha}{D}} \cdot \Lambda_{it}^{(1-\varepsilon)\frac{\alpha}{D}}.$$

Multiplying by the leading Λ_{it} gives the exponent on Λ_{it} as

$$1 + (1 - \varepsilon)\frac{\alpha}{D} = \frac{D}{D} + (1 - \varepsilon)\frac{\alpha}{D} = \frac{1}{D}.$$

Therefore,

$$\pi_{it}^* = \frac{D}{\varepsilon} \left[\frac{\varepsilon}{\varepsilon - 1} \cdot \frac{w^r}{1 - \alpha} \cdot k_{it}^{-\frac{\alpha}{1-\alpha}} \right]^{(1-\varepsilon)\frac{1-\alpha}{1-\alpha+\varepsilon\alpha}} \cdot \Lambda_{it}^{\frac{1}{1-\alpha+\varepsilon\alpha}} \quad (\text{C.4})$$

with

$$\Lambda_{it} = e^{z_{it}} Y_t^z (P_t^z)^\varepsilon + e^{g_{it}} G_t (P_t^g)^\varepsilon.$$

Now using the normalization $P_t^z = 1$ we can rewrite Λ_{it} as

$$\Lambda_{it} = [e^{z_{it}} + \alpha_{gt} e^{g_{it}}] Y_t^z, \quad \alpha_{gt} \equiv \frac{G_t (P_t^g)^\varepsilon}{Y_t^z}$$

which yields the desired expression.

C.2 Proof of Propositions 2 and 3

Consistent with (23), firms treat their current sentiment as fixed when evaluating continuation values. Define the perceived generator of the demand states by

$$\begin{aligned} \tilde{\mathcal{L}}_S f = & (-\mu_z z + \chi_z \mathcal{S}) f_z + (-\mu_g g + \chi_g \mathcal{S}) f_g \\ & + \frac{\sigma_z^2}{2} f_{zz} + \frac{\sigma_g^2}{2} f_{gg}. \end{aligned}$$

The HJB equation is

$$\begin{aligned} rV(k, \mathcal{S}, g, z) = & \max_{\iota} \left\{ \pi(k, g, z) - \iota - \Phi(\iota, k) + \Gamma(d) \right. \\ & \left. + (\iota - \delta k) V_k + \tilde{\mathcal{L}}_S V \right\}, \end{aligned}$$

where $d = \pi(k, g, z) - \iota - \Phi(\iota, k)$. The first-order condition with respect to investment is

$$V_k = (1 + \Phi_\iota(\iota, k)) (1 + \Gamma'(d)). \quad (\text{C.5})$$

The envelope condition for capital is

$$\begin{aligned} (r + \delta) V_k = & (\pi_k(k, g, z) - \Phi_k(\iota, k)) (1 + \Gamma'(d)) \\ & + \dot{k} V_{kk} + \tilde{\mathcal{L}}_S V_k. \end{aligned}$$

As standard, we define marginal q as the perceived marginal value of capital, i.e., $\tilde{q} \equiv V_k$. Then we can rewrite the equation above as

$$(r + \delta) \tilde{q}_t = (\pi_k(k_t, g_t, z_t) - \Phi_k(\iota_t, k_t)) (1 + \Gamma'(d_t)) + \mathbb{E}_t^\chi (d\tilde{q}_t/dt) \quad (\text{C.6})$$

Where

$$\mathbb{E}^\chi (d\tilde{q}_t/dt) \equiv \dot{k} V_{kk} + \tilde{\mathcal{L}}_S V_k$$

Then (17) simply follows from (C.6) by decomposing $\mathbb{E}_t^\chi(d\tilde{q}_t/dt) = \mathbb{E}_t(d\tilde{q}_t/dt) + (\mathbb{E}_t^\chi - \mathbb{E}_t)(d\tilde{q}_t/dt)$.

To obtain (18) we apply the Feynman-Kac formula to (C.6) above to get

$$\tilde{q}_t = \int_0^\infty e^{-(r+\delta)s} \mathbb{E}_t^\chi (\pi_k(k_{t+s}, g_{t+s}, z_{t+s}) - \Phi_k(l_{t+s}, k_{t+s})) (1 + \Gamma'(d_{t+s})) ds$$

Finally, defining $q_t \equiv \int_0^\infty e^{-(r+\delta)s} \mathbb{E}_t (\pi_k(k_{t+s}, g_{t+s}, z_{t+s}) - \Phi_k(l_{t+s}, k_{t+s})) (1 + \Gamma'(d_{t+s})) ds$ and $\mathcal{F}_t \equiv \tilde{q}_t - q_t$ and substituting into (C.5) gives (18).

C.3 Financial frictions and the transmission of sentiment to investment

This subsection develops the implications of Proposition 3 for the sensitivity of investment to sentiment and presents supporting firm-level evidence.

The marginal propensity to invest out of sentiment We define the *marginal propensity to invest out of sentiment* (MPIS) as the marginal change in investment induced by an additional unit of sentiment:

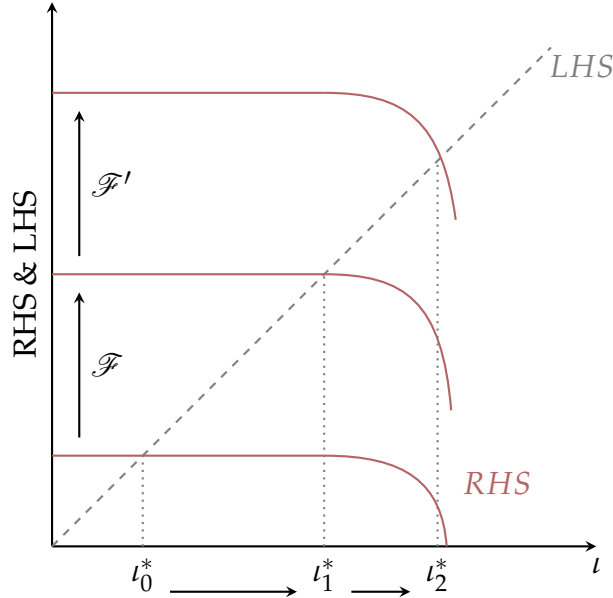
$$MPIS(k, \mathcal{S}, g, z) \equiv \frac{\partial \iota}{\partial \mathcal{S}} = \frac{\partial \iota}{\partial \mathcal{F}} \cdot \frac{\partial \mathcal{F}}{\partial \mathcal{S}}. \quad (\text{C.7})$$

By construction, the MPIS is always positive, since the perceived marginal value of capital is increasing in sentiment, i.e., $V_{k\mathcal{S}} > 0$. Moreover, the MPIS is lower for financially constrained firms, as tighter constraints raise the effective marginal cost of investment through $\Gamma'(d)$.

Figure C.1 illustrates the sentiment–investment relationship implied by (18), with investment on the horizontal axis. The grey dashed line plots the marginal cost of investment (left-hand side of (18)), which is increasing in investment. The red solid lines plot the perceived marginal benefit of capital (right-hand side of (18)) at different sentiment levels. The marginal benefit is flat in investment when firms are unconstrained, but declines with investment once external finance is needed, as lower dividends raise funding costs. The intersection of these two curves determines the optimal investment level.

To see how financial frictions affect the MPIS, start from some baseline level of investment ι_0^* and consider an increase in sentiment which shifts up the marginal benefit schedule (red line). When the firm is unconstrained, this leads to a large increase in investment to ι_1^* . When investment exceeds internal cash flows, the firm must turn to outside funding. From that point, the marginal benefit schedule slopes downward, so that further increases in sentiment yield smaller increases in investment, to ι_2^* . The MPIS is therefore

Figure C.1: Effect of sentiment on investment



Note: diagrammatic representation of the effect of an increase in sentiment on investment using Proposition 3. The right hand side (RHS) and left hand side of (18) are depicted in solid red and dashed gray, respectively.

lower for constrained firms, whose limited scope for intertemporal decisions makes beliefs about the future less relevant for current behavior. The interaction between financial frictions and the MPIS plays a central role in the analysis of aggregate fiscal transmission in Section 5.

Testing the theory in the data A central prediction of Proposition 3 is that financially constrained firms exhibit a flatter investment-sentiment relationship. Under a standard power form for the investment function, $\Phi(l, k) = \frac{\phi}{a} \left(\frac{l}{k} - \delta \right)^a k$, (18) maps directly into a regression specification of the log investment-capital ratio on marginal q_t and $\mathcal{F}_t$. Since $\mathcal{F}_t$ is the present value of forecast errors about future profitability, our survey data allow us to construct an empirical proxy for it and test this prediction.

Since most firms in our sample are privately owned, market-based measures of q are not readily available. Following Asker et al. (2014), we therefore proxy q_t with sales growth, for which both realizations and expectations are available in our data. Because $\mathcal{F}_t$ captures misperceptions in q_t , we construct its empirical counterpart as the forecast error in sales growth between t and $t + 1$, which we denote by $\widehat{\mathcal{F}}_t$.

To measure financial frictions, we use the CADS credit score, which is constructed based on Altman's Z-score. Within each sector-year, we split firms into two groups: those in the bottom half of the credit score distribution (corresponding to high default risk and

Table C.1: Financial frictions and sentiment in the data

	Dependent variable: $\log \frac{Inv_{it}}{K_{it}}$				
	(1)	(2)	(3)	(4)	(5)
$\Delta \log Sales_{it+1}$	1.207*** (0.339)	1.415*** (0.171)	0.690*** (0.115)	1.099*** (0.136)	1.189*** (0.152)
$\widehat{\mathcal{F}}_{it+1}$	-1.119*** (0.168)	-1.130*** (0.136)	-0.712*** (0.103)	-0.847*** (0.115)	-0.850*** (0.148)
Low score _{it}	-0.151*** (0.052)	-0.163*** (0.049)	-0.074*** (0.024)	-0.194*** (0.044)	-0.162*** (0.045)
$\widehat{\mathcal{F}}_{it+1} \times \text{Low score}_{it}$	0.386*** (0.135)	0.379*** (0.131)	0.413*** (0.104)	0.283** (0.123)	0.199* (0.106)
$\log \frac{Inv_{it-1}}{K_{it-1}}$			0.690*** (0.017)		
Observations	45,590	45,590	43,952	45,401	42,635
R-squared	0.01	0.04	0.49	0.20	0.21
Within R-squared	0.01	0.02	0.48	0.01	0.01
Year FEs		✓	✓		
Sector $\times$ Year FEs				✓	✓
Year of birth FEs					✓

Note: estimates from (C.8). $\widehat{\mathcal{F}}_t$ denotes the forecast error in sales growth between t and $t + 1$. $\Delta \log Sales_{it+1}$ is realized sales growth between t and $t + 1$. Low score is a dummy for firms in the bottom 50% of the credit score distribution within a 2-digit NACE sector and year. Standard errors are clustered at the firm and year levels. All estimates are weighted using survey weights.

strong frictions), and those in the top half (corresponding to low default risk and weak frictions).³² We then define an indicator variable D_{it}^{score} equal to 1 if firm i in year t is in the high-frictions bin.

Having constructed empirical counterparts to the objects defined in Proposition 3, we estimate the following regression:

$$\log \frac{Inv_{it}}{K_{it}} = \gamma \Delta \log Sales_{it+1} + \delta \widehat{\mathcal{F}}_{it+1} + \beta \widehat{\mathcal{F}}_{it+1} \times D_{it}^{\text{score}} + \psi D_{it}^{\text{score}} + \Xi \mathbf{X}'_{it} + u_{it}, \quad (\text{C.8})$$

where $\frac{Inv_{it}}{K_{it}}$ is total investment over total material assets in period t , $\Delta \log Sales_{it+1}$ denotes the growth rate in sales of firm i between years t and $t + 1$, $\widehat{\mathcal{F}}_{it+1}$ and D_{it}^{score} were defined above, and $\mathbf{X}_{it}$ denotes a vector of controls, specified below.

³²We construct groups within 2-digit NACE sectors and year to prevent industry composition or aggregate fluctuations driving our results.

Table C.1 reports the results from this regression. The different columns report specifications with different sets of controls in X_{it} : no controls, year fixed effects, lagged log investment, sector-year fixed effects, and firm-cohort fixed effects. We find that a one-percentage-point increase in realized sales growth—our proxy for marginal q —is associated with an increase in the investment rate between 0.7 and 1.4%. Similarly, firms that turn out to be ex-post overoptimistic about sales growth also invest more, with a 1 percentage point optimistic forecast error associated with a 0.7 to 1.1% increase in the investment rate. The estimated coefficients on realized growth and forecast errors are remarkably similar in magnitude, consistent with the prediction of (18) that q_t and $\mathcal{F}_t$ enter the investment equation symmetrically.

Our main coefficient of interest, $\hat{\beta}$, captures the differential sensitivity of investment to sentiment for financially constrained firms relative to unconstrained ones. Across all specifications, we find consistently positive and statistically significant estimates. A one-percentage-point optimistic forecast error is associated with a 0.2 to .38 % smaller investment rate for firms in the bottom half of the credit score distribution, relative to a baseline of 0.7 to 1.1% for firms in the top 50% of the distribution. These findings confirm our theoretical prediction that financial constraints dampen the transmission of sentiment to investment.

D Appendix to Section 5

D.1 Unions

We show that the sticky-wage union problem delivers the (non-linear) wage New-Keynesian Phillips Curve

$$\epsilon v'(L_t)L_t - (\epsilon - 1) u'(C_t) w_t L_t + \psi(\dot{\pi}_t^w - \rho \pi_t^w) = 0.$$

Consider a unit continuum of unions $k \in [0, 1]$. Each union sets a nominal wage path W_{kt} (equivalently, nominal wage inflation $\pi_{kt}^w \equiv \dot{W}_{kt}/W_{kt}$) to maximise the lifetime utility of its members,

$$\max_{\{\pi_{kt}^w\}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} \left[u(C_t) - v(L_{kt}) - \frac{\psi}{2} (\pi_{kt}^w)^2 L_t \right] dt,$$

subject to the wage law of motion $\dot{W}_{kt} = \pi_{kt}^w W_{kt}$ and the labor-demand schedule that comes from CES aggregation,

$$L_{kt} = (W_{kt}/W_t)^{-\epsilon} L_t, \quad L_t = \left[\int_0^1 L_{kt}^{\frac{\epsilon-1}{\epsilon}} dk \right]^{\frac{\epsilon}{\epsilon-1}}.$$

Taking W_t and L_t as given, the current-value Hamiltonian is

$$\mathcal{H}_{kt} = u(C_t) - v(L_{kt}) - \frac{\psi}{2} (\pi_{kt}^w)^2 L_t + \lambda_{kt} (\pi_{kt}^w W_{kt} - \dot{W}_{kt}),$$

where λ_{kt} is the co-state variable on W_{kt} .

The first-order condition with respect to π_{kt}^w gives

$$0 = -\psi \pi_{kt}^w L_t + \lambda_{kt} W_{kt}. \quad (\text{D.1})$$

The co-state equation is

$$\dot{\lambda}_{kt} - \rho \lambda_{kt} = v'(L_{kt}) \frac{\partial L_{kt}}{\partial W_{kt}} = -\epsilon v'(L_{kt}) \frac{L_{kt}}{W_{kt}}. \quad (\text{D.2})$$

In a symmetric equilibrium $W_{kt} = W_t$, $L_{kt} = L_t$, $\pi_{kt}^w = \pi_t^w$, $\lambda_{kt} = \lambda_t$. Because each household owns the unions, λ_t equals the nominal marginal utility of income: $\lambda_t = W_t u'(C_t)$.

Using λ_t from (1) and substituting into (2) yields

$$-\psi \dot{\pi}_t^w L_t + \rho \psi \pi_t^w L_t = -\epsilon v'(L_t) L_t + (\epsilon - 1) u'(C_t) W_t L_t.$$

Finally, dividing by the price level P_t to express real wages $w_t \equiv W_t/P_t$ and rearranging terms gives the desired Phillips Curve:

$$\epsilon v'(L_t)L_t - (\epsilon - 1) u'(C_t) w_t L_t + \psi(\dot{\pi}_t^w - \rho \pi_t^w) = 0.$$

D.2 Calibration of private and public demand processes

We construct empirical counterparts of firms' annual public and private sales by linking the procurement records to the company accounts data. The procurement data record the value and recipient of each awarded contract, but contract awards do not generally coincide with revenue recognition: contracts can begin after their award date and can extend over several fiscal years. We therefore reconstruct the execution period of each contract and allocate its value across years before estimating the sales processes.

Contract sample and recipients. We begin with the universe of successful, positive-value procurement awards observed through 2017. We retain purchase contracts for works, services, and supplies. We exclude framework-agreement ceilings while retaining contracts awarded under those agreements, since counting both would duplicate public demand. We also exclude concessions and other arrangements for which the reported contract value does not represent a direct public purchase, as well as contracts recorded as terminated or otherwise unusable.

We assign a contract to a firm when the procurement records identify a unique official recipient, the recipient is classified as an individual enterprise, and the award does not involve a consortium or temporary grouping for which the allocation of revenues across firms is unknown. Firms linked to an award with an ambiguous recipient allocation are excluded from the estimation sample. In particular, we do not divide group contracts equally across their participants.

Timing and annual revenue recognition. For each contract c , we denote its awarded value by V_c . We use the actual execution start and completion dates whenever both are available and internally consistent. When an actual completion date is unavailable, we use the contractual completion date. Many contracts still have incomplete timing information. For these contracts, we jointly impute the delay between award and execution and the duration of execution by drawing from contracts with observed timing.

The donor cells are defined using the type of contract, the two-digit CPV product category, an order-of-magnitude bin for contract value, and the award cohort. We require a minimum number of observations within a donor cell and sequentially broaden the

matching criteria when the most detailed cell is too small. Drawing the starting delay and duration jointly preserves their empirical dependence. We construct five completed panels using five fixed random seeds. The imputation procedure and donor cells are specified without reference to the estimated sales coefficients or to the calibration targets.

Within each completed panel, we allocate the awarded value uniformly over the days between the estimated start and completion dates. Let w_{ct} denote the share of contract c assigned to calendar year t . The weights account for partial years and satisfy

$$\sum_t w_{ct} = 1$$

over the complete execution period of the contract. The annual public sales of firm i are then

$$G_{it}^{\text{data}} = \sum_{c \in \mathcal{C}_i} V_c w_{ct},$$

where $\mathcal{C}_i$ is the set of eligible contracts awarded to firm i . Thus, the procedure preserves the total value of every contract while smoothing the timing mismatch between contract awards and the corresponding production of goods and services.

Let R_{it} denote total sales reported in the firm's income statement. We construct private sales residually as

$$X_{it}^{\text{data}} = R_{it} - G_{it}^{\text{data}}.$$

We do not cap public sales at total sales or replace nonpositive private sales with an arbitrary positive value. Instead, the log-sales estimation sample contains firm-years for which both reconstructed public sales and residual private sales are strictly positive in every timing imputation considered. We use observations from 2011 to 2017 and retain each firm's longest consecutive spell containing at least three annual observations. Applying the same sample definition across imputations keeps changes in the estimates from reflecting changes in sample composition. The resulting common sample contains 10,232 firms, 44,377 firm-year observations, and 34,145 usable annual transitions.

Estimation. For each completed panel, we estimate the following pooled annual AR(1) specifications:

$$\log G_{it}^{\text{data}} = a_g + \rho_g \log G_{i,t-1}^{\text{data}} + u_{it}^g, \quad (\text{D.3})$$

$$\log X_{it}^{\text{data}} = a_z + \rho_z \log X_{i,t-1}^{\text{data}} + u_{it}^z. \quad (\text{D.4})$$

The benchmark regressions do not include firm fixed effects. Accordingly, the persistence coefficients describe the unconditional persistence of observed firm-level sales, including persistent differences in sales levels across firms. This is also the object computed from the simulated model panels in the simulated method of moments procedure below. We cluster standard errors at the firm level.

For each process, the persistence target is the estimated autoregressive coefficient. The innovation-variance target is the residual variance from the corresponding regression,

$$\hat{\sigma}_j^2 = \frac{1}{N-k} \sum_{i,t} \hat{u}_{it}^{j2}, \quad j \in \{g, z\}.$$

We estimate the regressions separately in each of the five completed timing panels and average the resulting coefficients and innovation variances. We do not first average the imputed sales histories, since doing so would mechanically remove part of the uncertainty associated with missing execution dates.

The resulting annual targets are

$$\hat{\rho}_z = 0.9751, \quad \hat{\sigma}_z^2 = 0.2202, \quad \hat{\rho}_g = 0.6425, \quad \hat{\sigma}_g^2 = 1.1483.$$

Across the five timing imputations, the private-sales persistence coefficient ranges from 0.9742 to 0.9769, while the public-sales coefficient ranges from 0.6389 to 0.6446. The corresponding R^2 values average 0.931 for private sales and 0.426 for public sales. Adding year fixed effects leaves the results nearly unchanged: the estimated persistence coefficients become 0.9759 and 0.6425 for private and public sales, respectively.

These four estimates are moments of the reconstructed annual sales processes rather than direct estimates of the continuous-time demand parameters. In particular, reconstructed public sales remain an imputed measure of contract revenue, and private sales inherit any measurement error in that reconstruction through the residual definition. We therefore apply the same construction and estimator to the simulated model data. Section D.3 describes this mapping and the joint simulated method of moments procedure, while Table D.1 reports the empirical targets and model fit.

D.3 Joint calibration of demand and expectations

We jointly calibrate seven parameters,

$$\Theta = (\mu_z, \sigma_z, \mu_g, \sigma_g, \chi, \kappa, \bar{G}),$$

using four moments of annual private and public sales, two moments of forecast errors, and the government expenditure share. The remaining parameters are held at the values reported in Table 3. This section explains the economic rationale for joint calibration, defines the model moments, and describes the simulated method of moments procedure.

Why joint calibration? Observed sales are endogenous outcomes of the model. Their persistence and volatility reflect the underlying demand processes, firms' pricing and investment decisions, and the stationary distribution of capital. These decisions depend on expectations, so changing the extrapolation parameter χ or sentiment persistence κ also changes the sales moments. Conversely, the demand processes determine the innovations that generate sentiment and forecast errors.

The relative importance of public demand introduces an additional connection between these moments. In units of the private consumption good, a firm's instantaneous sales revenue is

$$\begin{aligned} r_i &= p_i^{1-\varepsilon} (e^{z_i} \bar{Y}^p + e^{g_i} \bar{G} (\bar{P}^g)^\varepsilon) \\ &= p_i^{1-\varepsilon} \bar{Y}^p (e^{z_i} + \alpha_g e^{g_i}), \end{aligned}$$

where

$$\alpha_g \equiv \frac{\bar{G} (\bar{P}^g)^\varepsilon}{\bar{Y}^p}$$

measures the relative weight of public demand in the firm's demand schedule. Substituting the optimal pricing rule gives

$$r_i = A k_i^{\frac{\alpha(\varepsilon-1)}{D}} (e^{z_i} + \alpha_g e^{g_i})^{\frac{1}{D}}, \quad D = 1 + \alpha(\varepsilon - 1),$$

where A depends on steady-state aggregate quantities, the wage, and technology parameters.

Holding capital and aggregate conditions fixed, the elasticity of total sales to public demand is therefore

$$\frac{\partial \log r_i}{\partial g_i} = \frac{1}{D} \frac{\alpha_g e^{g_i}}{e^{z_i} + \alpha_g e^{g_i}}.$$

The same public-demand innovation generates different movements in total sales depending on α_g . Public-demand exposure also enters the formation of sentiment. Under the normalization used in the calibration,

$$dS_i = -\kappa S_i dt + \sigma_z dW_i^z + \alpha_g \sigma_g dW_i^g.$$

Thus, α_g affects both the sales response to public-demand shocks and their contribution to sentiment and subsequent forecast errors.

Government expenditure and public-demand exposure. The government expenditure share disciplines α_g . Since steady-state output satisfies

$$\bar{Y} = \bar{Y}^p + \bar{P}^g \bar{G},$$

defining $s_G = \bar{P}^g \bar{G} / \bar{Y}$ gives

$$\alpha_g = \frac{s_G}{1 - s_G} (\bar{P}^g)^{\varepsilon - 1}.$$

We fix the expenditure-share target at $s_G = 0.18$. However, the relative price $\bar{P}^g$ depends on the stationary equilibrium and hence on the demand and expectations parameters. Government demand $\bar{G}$ must therefore be determined jointly with these parameters and equilibrium prices. The coefficient α_g is an implied equilibrium quantity, rather than an additional independently calibrated parameter.

Calibration targets. The private- and public-sales moments primarily discipline the corresponding demand processes. The response of forecast errors to public-demand-induced sales innovations and the persistence of forecast errors primarily discipline extrapolation and sentiment persistence. These relationships provide intuition for the targets, while the calibration evaluates their joint implications in equilibrium. Table D.1 reports all seven targets and the resulting model fit. Sales moments refer to annual log revenues, and innovation moments are variances. Calibration and convergence checks use unrounded values. We next define the model moments, including the sample restriction used for forecast-error persistence, before describing the numerical calibration procedure.

Annual sales and forecasts. The calibration uses annual revenue flows rather than point-in-time demand states. Let $r_i^z(u)$ and $r_i^g(u)$ denote instantaneous private and public revenues implied by the firm's state and optimal policies. Annual revenues are

$$R_{it}^j = \int_{t-1}^t r_i^j(u) du, \quad j \in \{z, g\},$$

and total revenues are $R_{it} = R_{it}^z + R_{it}^g$.

Initial firm states are drawn from the stationary population with entry and exit. We simulate two consecutive years conditional on survival. Since exit is state independent, conditioning on survival does not select firms according to their subsequent shocks. Sim-

Table D.1: Calibration targets and model fit

Moment	Data	Model
Forecast-error autocorrelation	0.17	0.17
IV slope: forecast error on public-demand-induced sales	−0.65	−0.63
Private-sales persistence, ρ^z	0.98	0.95
Public-sales persistence, ρ^g	0.64	0.64
Private-sales innovation variance	0.22	0.22
Public-sales innovation variance	1.15	1.13
Government expenditure/GDP	0.18	0.18

Note: Sales moments refer to annual log sales. Innovation moments are variances, not standard deviations. The IV target is reported in Table 1. Government expenditure is measured relative to steady-state GDP. All entries are rounded to at most two decimal places; the fit criterion is evaluated using unrounded moments.

ulated paths follow the transition rates of the discretized continuous-time model, and annual revenues accumulate along each path.

At the beginning of each year, firms forecast total revenues using their subjective transition law and the model’s nonlinear revenue function. Denoting this forecast by

$$F_{i,t-1} = \tilde{\mathbb{E}}_{i,t-1}[R_{it}],$$

the realized forecast error is

$$FE_{it} = \log R_{it} - \log F_{i,t-1}.$$

Negative forecast errors therefore indicate overoptimism. Forecasts integrate over future uncertainty and endogenous capital adjustment; they are not obtained by evaluating revenues along a deterministic path of expected demand.

Sales persistence and innovation variances. For each revenue component, we estimate

$$\log R_{i,t+1}^j = a_j + \rho_j \log R_{it}^j + \varepsilon_{i,t+1}^j, \quad j \in \{z, g\}.$$

The calibration moments are the slope ρ_j and the variance of the regression residual. These regressions use annual revenues generated by the full model. Their coefficients therefore need not equal the persistence and innovation variance of the underlying demand state.

Forecast-error persistence. We estimate

$$FE_{i,t+1} = a_{FE} + \rho_{FE} FE_{it} + v_{i,t+1},$$

Forecast errors and public-demand innovations. To isolate variation originating in public demand, we construct

$$U_{it}^g = g_{it} - \mathbb{E}_{i,t-1}[g_{it}],$$

where the expectation is computed under the model's true transition law. We use this innovation as an instrument for unexpected annual total sales,

$$X_{it} = \log R_{it} - \log \mathbb{E}_{i,t-1}[R_{it}],$$

and compute

$$\beta_{FE}^{IV} = \frac{\text{Cov}(U_{it}^g, FE_{i,t+1})}{\text{Cov}(U_{it}^g, X_{it})}.$$

This coefficient measures how forecast errors for the following year respond to current sales variation induced by public-demand innovations. It is the model counterpart of the empirical IV coefficient in Table 1, whose unrounded target is -0.652 . The denominator uses total sales, rather than public sales or the demand state itself; hence the coefficient is not the response of forecast errors per unit of the primitive public-demand shock. A 1% public-demand-induced increase in total sales is associated with a 0.65-percentage-point decline in the following year's forecast error in the data, compared with 0.63 in the calibrated model. Both covariances use the full simulated sample. Subtracting the component of sales predictable from initial information reduces simulation noise without changing the population IV coefficient, because the instrument has conditional mean zero.

Simulated method of moments. Let m^{data} collect the six firm-level targets and $m^{\text{sim}}(\Theta)$ their model counterparts. For each candidate vector of demand and expectations parameters, we solve firms' optimization problem and the stationary distribution jointly with equilibrium prices and aggregate quantities. Within this calculation, government demand is adjusted to satisfy

$$\frac{\bar{P}^g \bar{G}}{\bar{Y}} = 0.18.$$

Although seven parameters are jointly determined, the numerical search therefore operates over six parameters, with $\bar{G}$ determined by the expenditure-share restriction at each evaluation.

After solving the equilibrium, we simulate firm histories and re-estimate the six moments. Each discrepancy is scaled by the absolute value of its target:

$$e_j(\Theta) = \frac{m_j^{\text{sim}}(\Theta) - m_j^{\text{data}}}{|m_j^{\text{data}}|}.$$

This normalization makes discrepancies comparable across moments measured in different units. Innovation discrepancies are evaluated in variances.

The numerical search seeks a parameter vector that matches every target within the prescribed tolerance. It reduces the criterion

$$Q(\Theta) = \sum_{j=1}^6 [\max \{|e_j(\Theta)| - b_j, 0\}]^2,$$

subject to parameter bounds and the government expenditure-share restriction. Additional numerical penalties discourage excessive probability mass at the boundaries of the state grids. The search uses bands b_j of 2%, with a 2.75% band for private-sales persistence, and a stopping tolerance of 2.8% for every firm-level moment. These bands provide a margin for simulation uncertainty around the final 3% acceptance threshold.

Parameters are updated in logarithms using local numerical sensitivities and bounded adjustments. Common random numbers reduce simulation noise across candidate evaluations, each based on five million firms. The selected calibration is then checked using an independent simulation of twenty million firms and accepted only if $\max_j |e_j(\Theta)| \leq 0.03$.

D.3.1 Comparison with diagnostic expectations

[Bordalo et al. \(2026a\)](#) provide a closely related quantitative analysis of firms' overreaction. In their model, log productivity follows an annual AR(1) process,

$$a_{t+1} = \rho a_t + \varepsilon_{t+1},$$

and, under their baseline specification, diagnostic expectations satisfy

$$\tilde{\mathbb{E}}_t[a_{t+1}] = \rho a_t + \theta \rho \varepsilon_t.$$

The response of the subjective forecast to the current innovation is therefore $(1 + \theta)\rho$, compared with ρ under rational expectations. Their estimate, $\theta = 0.91$, implies approximately 91% excess responsiveness of one-year-ahead log-productivity forecasts. This parameter

is estimated jointly by simulated method of moments, including moments of earnings forecast errors; the log-productivity response is a structural implication of their estimated model.

Our model differs in both the information that generates overreaction and its persistence. Private- and public-demand news jointly determine sentiment, which distorts the perceived drifts of both demand processes. The strength of these distortions depends on χ , the informativeness of total demand about each component, and the demand-process parameters:

$$\chi_z = \chi \beta_z \frac{\sigma_z^2}{\sigma_{z|x}^2}, \quad \chi_g = \chi \beta_g \frac{\sigma_g^2}{\sigma_{g|x}^2}.$$

Thus, $\chi = 0.35$ does not itself measure the proportional excess response of an annual sales forecast. Moreover, sentiment accumulates past news with an exponentially declining weight. Our estimate of $\kappa = 0.19$ implies that the contribution of a given shock to sentiment has a half-life of approximately 3.57 years, computed using the unrounded parameter. This differs from the baseline specification in [Bordalo et al. \(2026a\)](#), in which the diagnostic distortion to next-period log productivity depends on the current annual innovation.

To summarize the implications for sales expectations, we construct a measure of forecast responsiveness that pools both sources of demand news. Let R_u denote the firm's instantaneous sales and define its objective and subjective forecasts of sales over the coming year as

$$F_t^O = \mathbb{E}_t \left[\int_t^{t+1} R_u du \right], \quad F_t^S = \tilde{\mathbb{E}}_t \left[\int_t^{t+1} R_u du \right].$$

Both forecasts are evaluated using the calibrated equilibrium prices and policy functions. The objective forecast replaces the subjective transition law with the true law of motion; it does not come from a separately solved rational-expectations equilibrium.

For each forecast, we isolate the component that was unpredictable one year earlier:

$$N_t^j = \log F_t^j - \mathbb{E}_{t-1} [\log F_t^j], \quad j \in \{O, S\},$$

where the expectation used to remove the predictable component is taken under the true law of motion in both cases. These quantities capture innovations in the forecast functions, rather than differences between forecasts referring to different target years. We then define excess forecast responsiveness as

$$\Omega = \frac{\text{Cov}(N_t^O, N_t^S)}{\text{Var}(N_t^O)} - 1. \quad (\text{D.5})$$

This projection pools private- and public-demand news under their calibrated joint distribution. Consequently, each source of news contributes according to its implications for objective sales forecasts, rather than receiving an arbitrary equal weight.

We evaluate these moments on the numerical state space, starting from the calibrated stationary distribution and following surviving firms under the true transition law. The resulting value is $\Omega = 0.23$: news-driven movements in subjective log sales forecasts are approximately 23% larger than their objective counterparts in this projection. In the linear log-productivity benchmark above, applying the same construction to expected log productivity yields $\Omega = \theta$. The value 0.23 is an additional model-implied comparison statistic, not a calibration target. In particular, it differs from the empirical IV coefficient of -0.65 , which measures subsequent forecast errors relative to public-demand-induced variation in current sales; Ω instead compares subjective and objective forecast innovations and pools both sources of demand news.

The numerical comparison with [Bordalo et al. \(2026a\)](#) is illustrative. Our statistic concerns the logarithm of expected sales accumulated over the coming year, whereas their parameter governs expected log productivity one annual period ahead. Their empirical forecast moments also concern earnings rather than sales. Temporal aggregation, the forecast variable, and the distinction between expected logs and logs of expectations can all affect measured responsiveness. The comparison therefore documents the magnitude of overreaction implied by our calibration without interpreting the two numbers as estimates of an identical structural object.

D.4 Algorithm

1. Form an initial guess for the following sequences

$$\{w_t, r_t, Y_t^{p,\text{in}}, P_t^{g,\text{in}}\}_{t=0}^T$$

These vectors are the four blocks of the Newton search vector $\mathbf{x} \in \mathbb{R}^{4N}$ in the code.

2. Solve firm block

- Solve the Hamilton–Jacobi–Bellman equation (23) given the current guess for aggregate sequences to obtain policy functions $\iota_t(s), \ell_t(s), d_t(s)$.
- Update the sequence of distributions of firms $\{\mu_t(s)\}_t$ via the Kolmogorov-Forward equation (24).
- Aggregate to get $Y_t^{p,\text{out}}, I_t^{\text{gross}}, L_t^D, D_t^{\text{net}}, P_t^{g,\text{out}}$, and capital K_t .

3. **Solve government block** Impose the period-by-period balanced-budget condition

$$0 = r_t B_t + P_t^g G_t - T_t$$

i.e. equation (26). Together with the guessed path for r_t and the computed G_t , this pins down the tax path $\{T_t\}$.

4. **Solve household block** Solve the representative household's problem (25), using the dividend flow D_t^{net} , taxes T_t , labor L_t^D , the wage rate w_t and interest rate path r_t . This yields consumption C_t , bond holdings A_t , and the bond flow $\dot{A}_t$.
5. **Solve union block** Backward-integrate the nonlinear Phillips curve

$$\epsilon v'(L_t) L_t - (\epsilon - 1) u'(C_t) w_t L_t + \psi (\dot{\pi}_t^w - \rho \pi_t^w) = 0,$$

This provides wage inflation π_t^w , price inflation π_t , nominal interest i_t , labor supply L_t^S , and the real wage w_t .

6. **Market-clearing**

$$\text{Goods: } Y_t = C_t + I_t + P_t^g G_t,$$

$$\text{Labor: } L_t^S = L_t^D,$$

$$\text{Assets: } B_t = A_t,$$

where I_t is the integral of investment and adjustment costs defined in Section 4.1. Compute the resulting excess-demand vectors.

7. **Update the guess and iterate.** Feed the excess demands into a solver; iterate on initial guess the norm of all residuals is below a chosen tolerance.

D.5 Computing the state dependence of the sentiment channel

This appendix describes the experiment underlying Section 5.4. The objective is to measure how a temporary tightening of firms' financing conditions changes the additional effect of government spending generated by sentiment. We retain the baseline calibration, including nominal frictions and the monetary policy rule. All simulations start from the same calibrated stationary distribution of firms. Consequently, differences across experiments reflect the financing disturbance and the subsequent equilibrium adjustment, rather than differences in initial conditions or recalibrated parameters.

Financing and government-demand shocks. We represent the tightening of financing conditions by a temporary increase in the external-financing cost parameter:

$$\gamma_t = \bar{\gamma} [1 + 99e^{-\mu_g t}], \quad \bar{\gamma} = 0.35. \quad (\text{D.6})$$

The parameter therefore rises to 35 on impact and gradually returns to its stationary value. Time is measured in years, and μ_g is the mean-reversion rate of government demand in the baseline calibration, approximately 1.12 annually. The financing disturbance and the government-demand shock thus have the same decay rate. The size of the financing disturbance is chosen to provide a severe illustrative stress test; it is not estimated from a particular credit-crunch episode. We introduce no cost-push shock.

The fiscal intervention is the same homogeneous public-demand shock used in the baseline:

$$\log G_t - \log \bar{G} = \varepsilon_G e^{-\mu_g t}, \quad \varepsilon_G = 0.01. \quad (\text{D.7})$$

It occurs at the onset of the financing disturbance. The exogenous government-demand path is identical in the experiments with and without the aggregate sentiment response.

Equilibrium paths and counterfactuals. Under stressed financing conditions, we solve three nonlinear equilibrium transitions: the financing disturbance alone; the financing disturbance combined with government spending, suppressing the aggregate sentiment response; and the same two disturbances allowing sentiment to respond as in the baseline. In the last experiment, the fiscal innovation induces the initial sentiment shift implied by the calibrated expectation-formation process, $\alpha_g \varepsilon_G$. The experiment without aggregate sentiment suppresses this initial shift while retaining the calibrated idiosyncratic environment and all other structural parameters. Prices, investment decisions, and the distribution of firms subsequently evolve in equilibrium in each experiment.

Let $c \in \{N, C\}$ denote normal financing conditions and the credit-crunch experiment, respectively, and let $j \in \{0, S\}$ indicate whether the aggregate sentiment response is suppressed or included. For any aggregate variable X , define the incremental fiscal response as

$$\Delta X_{c,t}^j = X_{c,t}^{j,G} - X_{c,t}^{\emptyset}, \quad (\text{D.8})$$

where $X_{c,t}^{\emptyset}$ is the corresponding path without the fiscal intervention. Under normal financing conditions, this counterfactual is the stationary equilibrium. Under stressed financing conditions, it is the time-varying equilibrium path with the financing disturbance alone. This subtraction isolates the incremental effect of government spending from the contraction caused by the financing disturbance. Because the fiscal intervention is finite, these are

generalized impulse responses rather than infinitesimal derivatives.

Government spending is present in both sentiment specifications. Its effects on sales and cash flows therefore operate in both cases, although their equilibrium magnitudes can differ as sentiment changes investment, prices, and activity. The difference between the two fiscal responses captures the additional equilibrium effect generated by the aggregate sentiment response.

Cumulative multipliers. For horizon $H > 0$, we calculate

$$M_c^j(H) = \frac{\int_0^H e^{-rt} \Delta Y_{c,t}^j dt}{\int_0^H e^{-rt} \Delta (P_t^g G_t)^j_{c,t} dt}, \quad r = 0.02. \quad (\text{D.9})$$

Government expenditure is measured in units of the private consumption good. Each multiplier uses the incremental government-expenditure path from its own equilibrium simulation. The impact multiplier is the corresponding ratio of impact responses.

We define the contribution of sentiment to the multiplier as

$$A_c(H) = M_c^S(H) - M_c^0(H), \quad (\text{D.10})$$

and its attenuation under financial stress as

$$100 \left[1 - \frac{A_c(H)}{A_N(H)} \right]. \quad (\text{D.11})$$

At five years, the multipliers without and with sentiment are 0.72 and 1.26 under normal financing conditions, compared with 1.00 and 1.42 under financial stress. Sentiment therefore contributes 0.55 and 0.41 multiplier points, respectively. Using the unrounded values, this represents a 24.49% reduction. The corresponding reduction on impact is 35.22%.

Because government-purchase prices respond endogenously, the difference between multipliers is not identical to the additional output effect of sentiment divided by a common expenditure denominator. We therefore also compare the discounted additional output generated by sentiment directly:

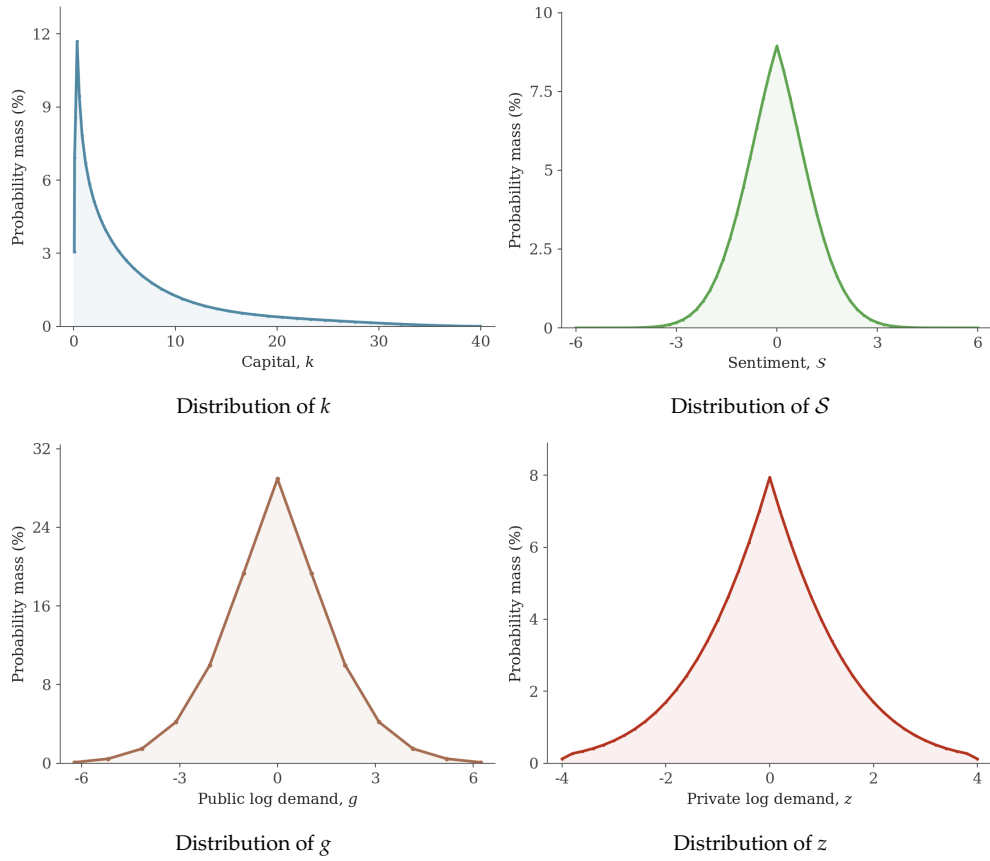
$$Q_c(H) = \int_0^H e^{-rt} [\Delta Y_{c,t}^S - \Delta Y_{c,t}^0] dt. \quad (\text{D.12})$$

At five years, this output contribution is 24.04% smaller under financial stress, confirming that the attenuation reflects a smaller output response to sentiment.

D.6 Additional model figures

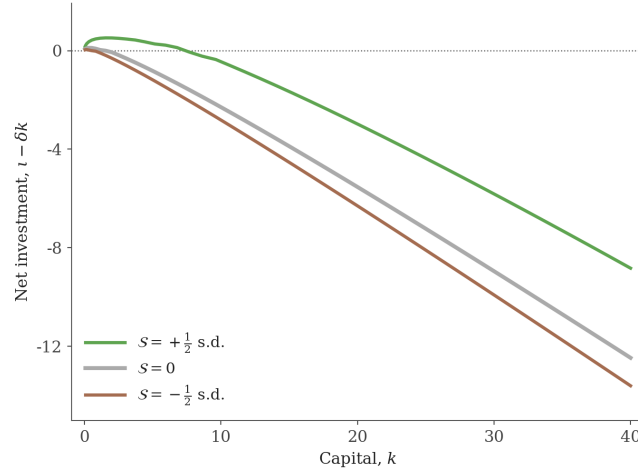
Figure D.1 plots the distribution of capital, sentiment, idiosyncratic public and private demand in the stationary equilibrium of our model. Figure D.2 plots the steady state investment policy functions for different levels of sentiment. Figure D.3 compares annual capital growth in the model, including firm exits, with that of continuing Cerved firms. Figure D.4 plots IRFs to the government spending shock considered in Figure 6 for additional variables. Figure D.5 plots the IRFs to the government spending shock considered in Figure 6 in a Real Business Cycle version of our model, i.e., a calibration with fully flexible wages.

Figure D.1: Stationary marginal distributions of idiosyncratic states



Note: Marginal probabilities in the stationary firm population. Lines connect probability masses at grid points; the capital grid is nonuniform. Public and private demand are expressed in log units.

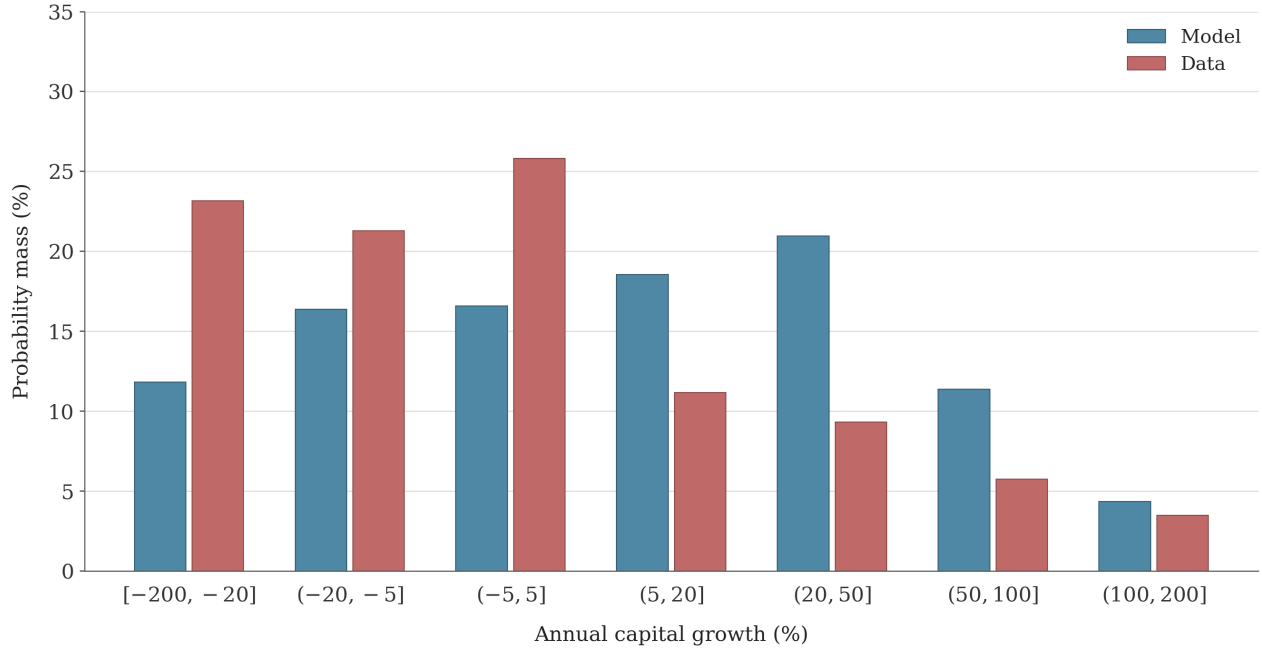
Figure D.2: Stationary Policy Functions for Investment



(a) Net investment policy function

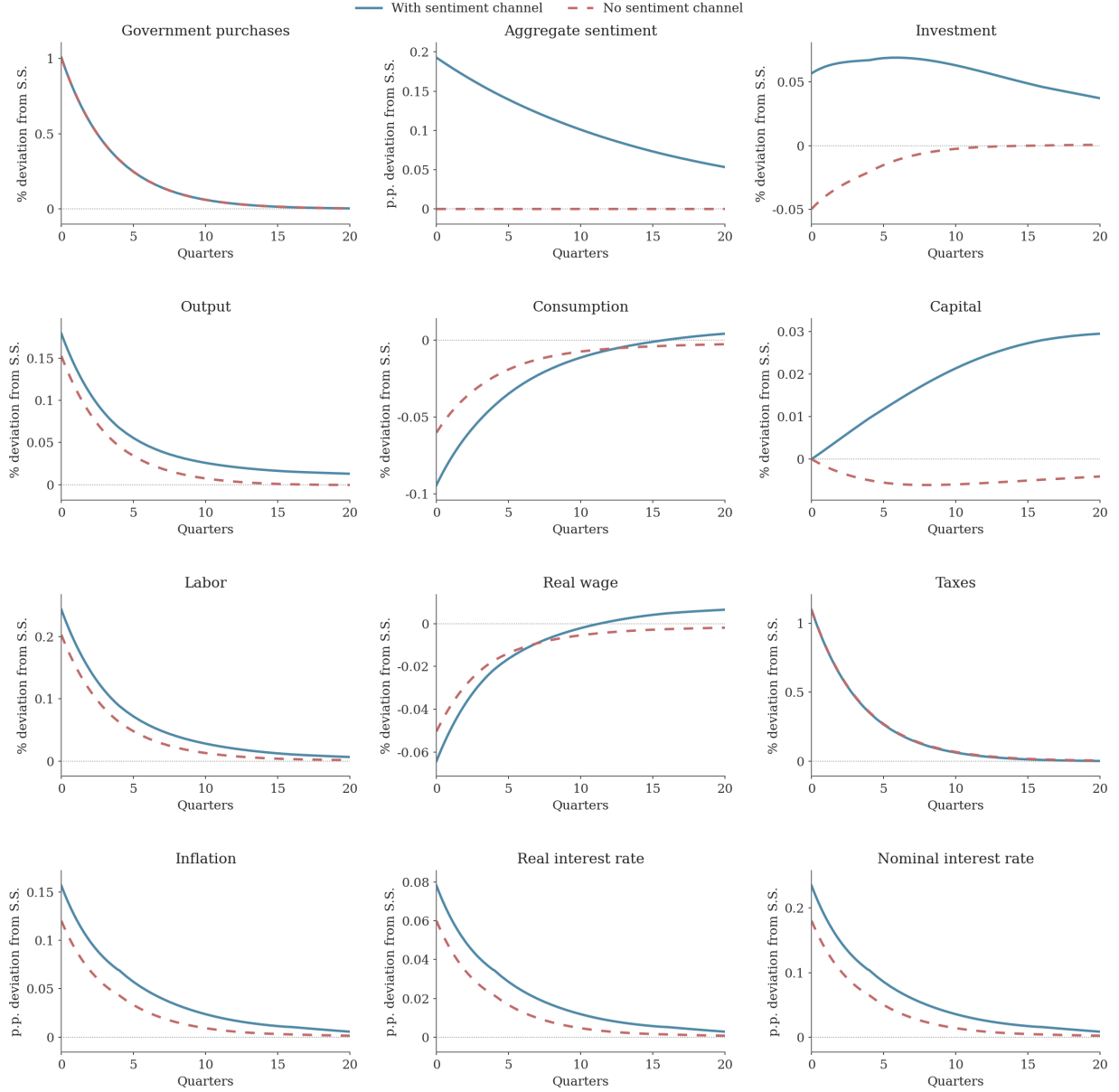
Note: Net investment at sentiment $S = 0$ and $S = \pm \frac{1}{2} \sigma_S$, where σ_S is the stationary cross-sectional standard deviation of sentiment. Private and public log demand are held at $z = g = 0$.

Figure D.3: Distribution of capital growth: model and data



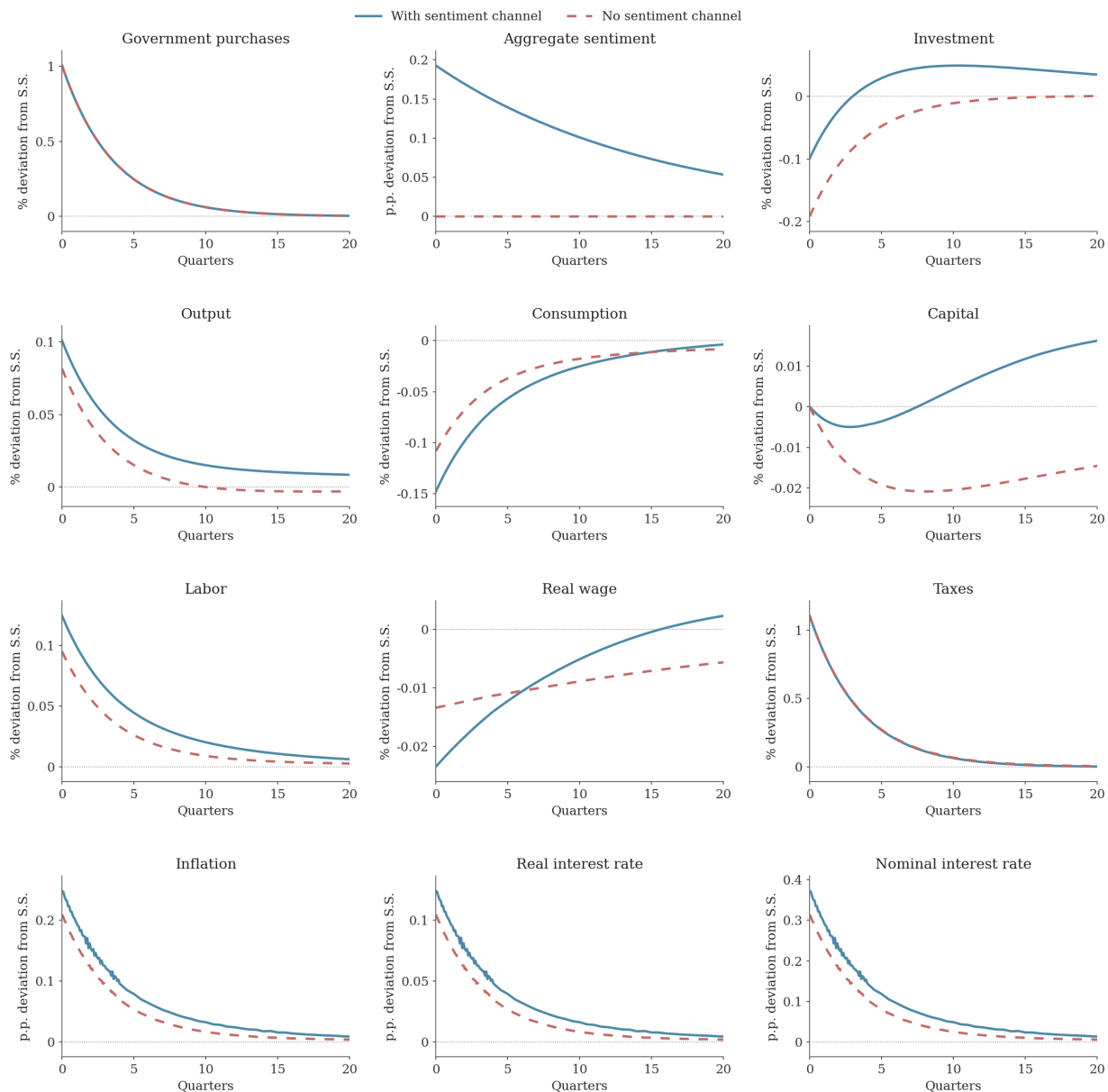
Note: Annual capital growth is $200(k_{t+1} - k_t) / (k_{t+1} + k_t)$. Blue bars include model exits (6.48% annually), assigning zero final capital and hence -200% growth to exiting firms. Red bars show growth in nominal tangible fixed assets for continuing firms in the sales-calibration sample, 2012–2017. Verified closures are unavailable in the data, so exits are included only in the model. Bars report bin probabilities.

Figure D.4: The sentiment channel of fiscal policy



Note: Responses to the same homogeneous procurement shock as in Figure 6. Blue solid lines include the aggregate sentiment channel; red dashed lines suppress it. Government expenditures denote real purchases G_t , while taxes finance nominal purchases $P_t^g G_t$. Aggregate sentiment is shown as 100 times the deviation of its cross-sectional mean from steady state; capital is shown as a percentage deviation from its steady-state aggregate. Interest rates and wage inflation are expressed in annual percentage points; other responses are percentage deviations from steady state. The horizontal axes measure quarters since the shock.

Figure D.5: The sentiment channel of fiscal policy without wage stickiness



Note: Responses to the same procurement shock as in Figure 6, using the baseline calibration with fully flexible wages ($\psi = 0$). Blue solid lines include the aggregate sentiment channel; red dashed lines suppress it. Government expenditures denote real purchases, while taxes finance nominal purchases. Sentiment is 100 times the change in its cross-sectional mean. Inflation and interest rates are in annual percentage points; other variables are percentage deviations from steady state. Horizontal axes measure quarters since the shock.

